Steeper Simplicity Priors in Solomonoff Function Induction:
Absorption, the Teacher's Mistakes and the Model–Randomness Split

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October 2026

Abstract

Solomonoff function induction (SFI) predicts labels from inputs with a universal prior over random functions. The author's notes proposed penalising program length more steeply than log-likelihood, by a factor $\lambda > 1$, with private randomness per input, so that learning selects a simple model plus noise, not a model of the teacher's errors. A standard objection: under a steeper penalty, a universal mixture taken as one hypothesis absorbs everything. We prove that it does in every class containing its own mixture, such as Solomonoff induction and shared-randomness SFI, while no private-randomness hypothesis is universal; the protection has a price on latent causes shared across inputs. For a random parity with a systematic error rule incompressible relative to it, we prove that for $\lambda = cn$, with c in an explicit window, every near-optimal private-randomness hypothesis knows the parity and gives the erroneous labels little probability. In the original example the window spans three orders of magnitude; it needs λ proportional to n and fails for some compressible error rules of the same density and description length. At $\lambda = 1$ a computable sampler and programs that fill in its sampled bits get similar posterior; for $\lambda > 1$ the sampler wins with private but not with shared randomness.

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*This paper was developed and written with extensive assistance from Claude (Anthropic), an AI system, which produced the proofs, the computations and the text under the author's direction. It was checked by independent AI referees, and no human has yet independently checked all of the mathematics. Companion paper: Hänni (2026a). Code and data: see the statement at the end of Section 9.

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1 Introduction

Solomonoff function induction (SFI) is Bayesian prediction of a label y from an input x with a universal prior over random partial functions (Hänni, 2026b,a). It is the supervised-learning counterpart of Solomonoff induction (SI), and like SI it learns every computable regularity of the labels, including the labeller's mistakes. Hänni (2026b) states the problem as follows: a human can understand "what someone means when using a word", "as opposed to becoming a predictor of what they would say", whereas "roughly speaking solomonoff would be making the same mistakes the person would make". His proposed remedy is "an even stronger simplicity prior than solomonoff": "if there are explainable mistakes on a simple model, you want the simple model that doesn't predict the mistakes", with a noisy version of the simple hypothesis and a likelihood term that is penalised less. In the form studied here, a hypothesis is a program π of length $\ell(\pi)$ with likelihood P_π, and one minimises

$$J_\lambda(\pi; D) = \lambda\, \ell(\pi) - \log P_\pi(D), \qquad \lambda > 1,$$

or uses the power prior $2^{-\lambda \ell(\pi)}$ (Definition 2.5). The proposal is made for function induction in which "each input gets its own private randomness" (Hänni and Wyeth, undated): a hypothesis is a probabilistic program that uses fresh coins at every input.

A standard objection, which we call the *absorption objection*, is that under a steeper penalty a hypothesis that is itself a universal mixture is short, penalises complexity inside itself only in the usual way, and therefore absorbs all the structure of the data, errors included. Hänni's note makes this point for sequence prediction, where "it'll be better to drop bits and take the likelihood loss", and suggests that the pathology is absent in the function case when "each input has to sample its own random bits" (Hänni, 2026b); with private randomness, "I don't see how to write down a universal hypothesis that gets good loss here" (Hänni and Wyeth, undated). He supports the proposal with an example: a 100-bit simple model, "systematic human error" on a 2^{-10} fraction of the inputs whose set takes 10,000 bits to specify, and three hypotheses (the simple model with noise, a model that learns the errors, and a fair coin), of which the first is selected for a range of λ/n. He leaves open whether another hypothesis does better ("Maybe I'm just failing to construct the right 'universal hypothesis' for this example?"), expects that "some other pathology" may be present, and calls the example "somewhat cherry-picked for the idea to work" (Hänni and Wyeth, undated). A second question concerns the split between model and randomness: whether a steeper prior gives "a preference for a simple sampler over a model where the sampled bits are filled in (solomonoff doesn't have a preference here, both get similar posterior)" (Hänni, 2026c), and whether the split shows as a change in slope when the entropy left given the hypothesis is plotted against its length, which in the sequence case is "pretty much just 1 bit paying for 1 bit" (Hänni, 2026b).

Contributions. We answer these questions with proofs; Table 1 lists Hänni's specific claims with their verdicts.

- *Absorption, and why private randomness blocks it* (Section 3). In every class that contains its own Bayes mixture, such as SI, SFI with shared randomness and history-aware SFI, every λ-MAP is shorter than a bound that does not depend on the data, and a computable deterministic teacher's labels, errors included, are eventually predicted (Theorem 3.2). No product hypothesis, with randomness private to each input, is universal: each loses $2^d - 1$ bits on some parity with a program of $d + 2\log d + O(1)$ bits (Corollary 3.5). The protection costs a factor $\min(\lambda, 2)$ on latent coins shared by pairs of inputs (Proposition 3.8) and ends

when inputs carry labelled history (Proposition 3.9); Remark 3.11 assembles the answer to the objection.

- *The teacher's-mistakes theorem* (Section 4). A teacher flips a random parity on $\{0,1\}^d$ on the inputs whose first m bits lie in a random error rule S of density p. If S is incompressible relative to the parity and $\lambda = cn$ with c in an explicit window $(2^{-m}, c_{\mathrm{hi}}]$, every near-optimal product hypothesis knows the parity, is short, holds little information about S and gives the erroneous labels small average probability (Theorem 4.7). At Hänni's sizes ($d = 100$, $m = 20$, $p = 2^{-10}$, 11,710 bits for S) the window is $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ with the machine constant set to 0 for illustration and $(9.54 \cdot 10^{-7}, 3.66 \cdot 10^{-3}]$ with 30 bits, against his three-hypothesis window $(1.12 \cdot 10^{-6}, 8.99 \cdot 10^{-3})$ (Table 2). Full truth tables, sampled data, his other version of the example and the power posterior are treated too (Section 4.5).
- *Limits* (Section 5). Plain Bayes, MDL, SI and every schedule with $\lambda_n/n \to 0$ learn the errors, so the trick needs $\lambda = \Theta(n)$ and is deliberately inconsistent for the teacher's labels (Proposition 5.1 and Remark 5.2). Incompressibility cannot be dropped: for an error rule with Hänni's density and about 10,400 bits, made of a cheap subcube and random points, no hypothesis that fits no better than the good model is near-optimal on almost a third of the window's logarithmic width (Proposition 5.3). A cheap, frequent error rule defeats every c once the simple model is long enough (Proposition 5.4 and Lemma 5.5); selection is by rate, not validity, cannot be tuned from imitation data and depends on the input law (Section 5.3).
- *The model–randomness split* (Sections 6 and 7). At $\lambda = 1$, for every computable sampler of length k, the posterior odds of the programs that fill in its sampled bits against the sampler lie in $[2^{-2\log(t_n+1)-\kappa}, 2^{k+\delta^*}]$ whatever the sampled randomness, with $2\log(t_n + 1)$ doubly logarithmic in the input length and δ^* a Ville deficiency; SI behaves similarly (Theorem 6.4 and Proposition 6.5). For fixed $\lambda > 1$ with private randomness every λ-MAP is at most $(k + \delta^*)/(\lambda - 1)$ bits longer than the sampler, so it is not a fill-in once the sampled randomness is larger, and the fill-ins' posterior is exponentially small in the sampled randomness (Theorem 6.6); with shared randomness, and in SI, the universal mixture wins (Theorem 6.8). For a noisy random parity, product sophistication exceeds set sophistication by $d - O(\log d)$ bits (Proposition 7.6).
- *Computable classes* (Section 8). The lower bounds and the private-randomness split hold in every c.e. prefix-free product class, with the class's own constants (Proposition 8.1).

Prior work. Two-part codes with a complexity weight above one appear in MDL theory, and the posteriors studied here are generalised, fractional or safe posteriors with learning rate $1/\lambda$ (Section 2.3). Closest to the teacher's-mistakes question is λ-weighted two-part MDL for noisy binary classification (Grünwald and Langford, 2007; Zhu and Srebro, 2025; Zhu et al., 2026); as their abstracts describe them, these works find that a bounded λ fits the noise and that λ must grow with the sample size. We do not claim this finding. Their target is the data distribution; ours deliberately differs from it (Remarks 5.2 and 8.5). In algorithmic statistics, restricting the model family is known to move the split between model and noise (Vereshchagin and Vitányi, 2010), and product models are such a restriction (Remark 7.7). An earlier report formalised Hänni's proposal for a general imitation channel, mostly by comparing a few hypotheses by rate (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation"); we re-derive what we

Claim	Verdict	Where
Sequence case: "it'll be better to drop bits and take the likelihood loss"	Confirmed, in every class that contains its own mixture	Thm. 3.2, Prop. 5.1(c)
The pathology is absent in the function case if "each input has to sample its own random bits"	Confirmed for absorption. Qualified: it returns with labelled history, and private randomness cannot express shared latent causes	Cor. 3.5 and Props. 3.8 and 3.9
In his example "the 'good model' could be a vertex of the convex hull"	Confirmed for error rules incompressible relative to the model; refuted for some error rules with the same density and description length	Thm. 4.7 and Prop. 5.3
"Maybe I'm just failing to construct the right 'universal hypothesis' for this example?"	Settled for incompressible error rules: the theorem covers all product hypotheses. For structured rules the danger is error envelopes, not universal hypotheses	Thm. 4.7 and Prop. 5.3
The arithmetic of both versions of the example	Confirmed. Version 2 is realised with 11,710 bits for the error rule; Version 1 in population form, with $\lambda \in (9.77, 58.0]$ at $n = 10^4$ ($\kappa = 0$) against his $(10, 100)$	Rem. 4.1, Thm. 4.7, and Ex. 4.12
"we will need to increase the hypothesis term coefficient ... as the data set size increases"	Confirmed: $\lambda = \Theta(n)$ is needed	Prop. 5.1 and Rem. 5.2
An absolute bound on model complexity is "in a sense ... equivalent" to the penalty	Confirmed at vertices of the hull only	Lem. 2.8
"some other pathology is present"	Yes: rate blindness, error envelopes, guard erosion, non-identifiability of λ, input-law dependence, labelled history, confident wrong extrapolation	§ 5, Prop. 3.9
Sampler versus filled-in bits: "both get similar posterior"	Confirmed with explicit factors, in SFI and in SI, for every computable sampler	Thm. 6.4 and Prop. 6.5
A steeper prior gives "a preference for a simple sampler"	Confirmed for fixed $\lambda > 1$ with private randomness, in the limited sense of Rem. 6.7; with shared randomness and in SI the universal mixture wins	Thms. 6.6 and 6.8
The slope of the remaining entropy changes at the model; in the sequence case "1 bit paying for 1 bit"	Confirmed: the product structure function has a kink at the parity for every flip rate below 1/4; the set version lies on the sufficiency line	Prop. 7.6
"have people not already considered this for solving the model + data split problem?"	Partly: λ-weighted MDL in noisy classification; restricted model families in algorithmic statistics	Rems. 7.7, 8.5 and 8.8

Table 1: Hänni's specific claims about the proposal (Hänni, 2026b; Hänni and Wyeth, undated; Hänni, 2026c), with verdicts. Numbers with $\kappa = 0$ set the machine constant to 0 and are illustrations.

use, with two corrections (Remark 8.7). Our literature search went only as far as abstracts, so the novelty claims of Remark 8.8 need a check against full texts.

Credits and scope. The notion of SFI, the proposal, the example with its numbers, the model–randomness question and the idea of reading the split from the slope of the remaining entropy come from the author's earlier notes and comments (Hänni, 2026b; Hänni and Wyeth, undated; Hänni, 2026c), which are cited in the third person, as Hänni's; Table 1 says where the analysis confirms, qualifies or corrects them. The samplers of Sam Eisenstat's coin example, as reported by Hänni (2026c), are covered by Theorem 6.4. The basic theory of SFI is in the companion paper (Hänni, 2026a). The positive result is proved only for parities, error sets of product form, uniform inputs, flip rates below 0.056 and incompressible error rules (Remark 4.9); for realistic, simple

error patterns we have a counterexample but no positive result. Numbers computed with the machine constant $\kappa = 0$ are illustrations. The finite-sample guarantee needs about 10^{11} samples at Hänni's sizes (Corollary 4.11), and with $\lambda = cn$ the learner is a family of batch objectives, not one Bayesian predictor (Remark 2.7).

Reader's guide. Section 2 fixes data, hypotheses, classes and objectives; Section 3 treats absorption; Sections 4 and 5 the teacher's mistakes; Sections 6 and 7 the model–randomness split; Section 8 computable classes and related work; Section 9 the discussion and open problems. Proofs are in the appendices, one per section. Notation follows the companion paper: logarithms are to base 2, n is the number of samples, $N = 2^d$, $c = \lambda/n$, and κ is a constant that depends only on the reference machine and fixed programs (Section 2.2).

2 Setting: hypothesis classes and steeper objectives

This section fixes the data, the two kinds of hypotheses, hypothesis classes indexed by programs, and the steeper objectives. Proofs and standard facts are in Appendix A.

Objects from the companion paper. We use the objects of Hänni (2026a). A *function machine* T reads, on input x, a prefix of one random tape $\omega \sim \mathbb{U}$ that all inputs share, and outputs a label; its random partial function is f_ω^T and its *machine law* is $\nu_T(D) = \mathbb{U}\{\omega : f_\omega^T \supseteq D\}$. Solomonoff function induction (SFI) is the random partial function of the universal function machine, with law M_F; $M_F(D) = \sum_i \mathbf{m}(i)\,\nu_{T_i}(D)$ for $D \neq \emptyset$. Product laws of lower-semicomputable (lsc) conditional semimeasures, Hänni's "computably samplable" x-only predictors (Hänni, 2026b), are machine laws; their mixture is M_{prod}. M^w is a universal lsc weak function semimeasure, and M_H, the law of SFI whose hypotheses see the earlier inputs (SFI^{hist}), is a universal lsc chronological semimeasure. M_{prod}, M_F and M^w agree up to $K(\operatorname{dom} D) + O(1)$ bits, but M_F/M_{prod} is unbounded. If labels are redrawn at every query (fresh noise) and inputs repeat, random-function SFI has infinite expected loss (Hänni, 2026a, the fresh-noise proposition); all noisy data in this paper draw each label once per distinct input (*frozen* noise).

2.1 Data and the two kinds of hypotheses

Definition 2.1 (Data)**.** steeper §1.1 The input set is $X = \{0,1\}^*$; statements about a fixed input length use $\{0,1\}^d \subseteq X$. The label set Y is *finite*, $|Y| \geq 2$. A *data set* is a finite partial function $D : X \rightharpoonup Y$ with $\operatorname{dom} D = \{x_1, \ldots, x_n\}$ (n distinct inputs) and $y_i = D(x_i)$. Noise is frozen: each distinct input has one label.

Definition 2.2 (Product hypothesis; private randomness)**.** steeper Def 1.1 A *product hypothesis* is an lsc conditional semimeasure $\nu(y \mid x)$ ($\sum_y \nu(y \mid x) \leq 1$). It is the law of a random partial function whose values $F(x) \sim \nu(\cdot \mid x)$ are independent across distinct x: a probabilistic machine that uses fresh coins of its own at each input. Its likelihood is the *product law* $\nu(D) := \prod_{x \in \operatorname{dom} D} \nu(D(x) \mid x)$, and its prediction at a new input x is $\nu(\cdot \mid x)$, whatever D is. It is *deterministic* if it is a partial computable function f; then $\nu(D) = \mathbf{1}[f \supseteq D]$. The randomness is private to each distinct input, not to each query; on data sets the two readings coincide (Hänni, 2026a, the per-query proposition).

Definition 2.3 (Shared hypothesis)**.** steeper Def 1.2 A *shared hypothesis* is an lsc weak function semimeasure P: $P(\emptyset) \leq 1$ and $\sum_y P(D + (x,y)) \leq P(D)$ for $x \notin \operatorname{dom} D$. Its prediction is $P(y \mid x, D) := P(D + (x,y))/P(D)$. Machine laws, Bayes mixtures $\sum_i w_i P_i$ with $\sum_i w_i \leq 1$, and

product laws are shared hypotheses. By telescoping, the cumulative sequential log loss of P on D_n is $-\log P(D_n)+\log P(\emptyset) \le -\log P(D_n)$ in every reveal order. Hypotheses of Solomonoff induction (SI), semimeasures on an interleaved string $x_1y_1x_2y_2\cdots$, are not weak function semimeasures: they depend on the order and assign probabilities to the inputs.

2.2 Hypothesis classes

A steeper penalty acts on single hypotheses, so its effect depends on what counts as one hypothesis and on its length. We therefore make the programs explicit.

Definition 2.4 (Hypothesis class)**.** steeper Def 1.3, §1.2 Fix a countable set $\mathcal{O}$ of *observations*: data sets for SFI, binary strings for SI, input–label sequences for history-aware prediction. A *hypothesis class* is a c.e. prefix-free set $\mathcal{H} \subseteq \{0,1\}^*$ of *programs* π, of length $\ell(\pi) := |\pi|$, with a family $(P_\pi)_{\pi\in\mathcal{H}}$ of maps $\mathcal{O} \to [0,1]$ that are lsc uniformly in π. Two programs for the same law are different hypotheses. By Kraft's inequality, $\sum_\pi 2^{-\ell(\pi)} \le 1$.

In a *canonical class*, $\mathcal{H} := \operatorname{dom} V$ for the fixed universal prefix machine V, and P_π is the member with index $V(\pi)$ of a fixed effective enumeration of a family, trimmed so that exactly the members of the family occur. The canonical classes and their mixtures $\xi_1 := \sum_\pi 2^{-\ell(\pi)}P_\pi$ are: $\mathcal{H}_\times$, lsc conditional semimeasures with their product laws ν_π, whose mixture is $\xi_\times = \sum_i \mathbf{m}(i)\nu_i = M_{\mathrm{prod}}$ exactly; $\mathcal{H}_F$, function machines with their machine laws, whose mixture is M_F on nonempty data sets; $\mathcal{H}_w$, lsc weak function semimeasures, with mixture M^w; $\mathcal{H}_{\mathrm{SI}}$, lsc semimeasures on binary strings, with the Solomonoff–Levin mixture M; $\mathcal{H}_{\mathrm{chr}}$, history-oracle function machines with their chronological semimeasures $\rho(y_{1:n}\|x_{1:n})$, with mixture M_H, itself an lsc chronological semimeasure that dominates every one (Hänni, 2026a, the corollary on $\mathrm{SFI}^{\mathrm{hist}}$). We use only this universality of M_H, not its identification with Hutter's universal chronological semimeasure. Canonical classes are *closed under prefix-free simulation*: for every prefix machine V' whose outputs are indices of the family there is a string r with $V(rw) = V'(w)$ for all $w \in \operatorname{dom} V'$, so distinct w give distinct programs rw.

Throughout, κ is a constant that depends only on the reference machine, the class and the fixed programs built in the proofs, never on the data or the hypotheses; different occurrences may differ. We use five standard facts (F1)–(F5) of algorithmic information theory, stated in Appendix A.1: $K(\pi) \le \ell(\pi) + \kappa$ in every class (F1), the conditional coding theorem (F2), the chain rule (F3), symmetry of information (F4), and $K(j) \le 2\log(j+1) + \kappa$ (F5).

2.3 Steeper objectives

Definition 2.5 (Steeper objectives)**.** steeper §1.3 Fix a class, an observation o and a *steepness* $\lambda \ge 1$. The *two-part objective* is $J_\lambda(\pi; o) := \lambda\,\ell(\pi) - \log P_\pi(o) \in [0,\infty]$; a minimiser is a λ-*MAP*. The *power-prior posterior* is $\Pi_\lambda(\pi \mid o) := 2^{-\lambda\ell(\pi)}P_\pi(o)/\xi_\lambda(o)$, with the *steeper mixture* $\xi_\lambda(o) := \sum_\pi 2^{-\lambda\ell(\pi)}P_\pi(o)$: Bayes' rule with the prior $2^{-\lambda\ell}$, whose Kraft sum is at most 1. We write $\xi_\lambda^\times$, ξ_λ^F for $\mathcal{H}_\times$, $\mathcal{H}_F$, and $\xi_\mathcal{C} := \sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)}P_\pi$ for a class $\mathcal{C}$. The *tempered posterior* with exponent $\eta \in (0,1]$ is $\Pi^\eta(\pi \mid o) \propto 2^{-\ell(\pi)}P_\pi(o)^\eta$.

These are the two forms of Hänni's proposal: "a higher coefficient on the length of the program than on the likelihood terms", and "penalize the likelihood term less" (Hänni, 2026b; Hänni and Wyeth, undated). For data-set classes, ξ_λ is a shared hypothesis. In statistics these objects are generalised, Gibbs, fractional or safe posteriors with learning rate $\eta = 1/\lambda$ (Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019; Grünwald, 2012; Grünwald and van Ommen, 2017), and two-part codes with a complexity weight above 1 are reported to appear in MDL convergence

theory (Barron and Cover, 1991; Grünwald, 2007); λ-weighted MDL for noisy classification is discussed in Remark 8.5. What we study is how $\lambda > 1$ interacts with universal hypotheses and with private versus shared randomness.

Lemma 2.6 (Basic properties)**.** [proved]
steeper Lemma 1.1 *(i) If $J_\lambda(\pi; o) < \infty$ for some π, the set of λ-MAPs is finite and nonempty. (ii)* $\arg\min_\pi J_\lambda(\pi; o) = \arg\max_\pi \Pi_\lambda(\pi \mid o) = \arg\max_\pi \Pi^{1/\lambda}(\pi \mid o)$. *(iii) On deterministic hypotheses $P_\pi(D)^\eta = P_\pi(D) \in \{0, 1\}$, so on any set of deterministic hypotheses the tempered posterior has the relative weights of the plain ($\lambda = 1$) posterior.*

Proof. (i) $J_\lambda \geq \lambda\ell$, and by Kraft only finitely many programs have $\ell(\pi) \leq J_\lambda(\pi_0; o)/\lambda$. (ii) $2^{-J_\lambda} = (2^{-\ell} P_\pi^{1/\lambda})^\lambda$. (iii) is immediate. □

Remark 2.7 (Per-sample form; $\lambda = cn$ gives batch objectives). [remark] steeper §1.3 For a product class and n data points put $c := \lambda/n$; then $J_\lambda/n = \hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \leq n} -\log \nu_\pi(y_i \mid x_i)$. For an input law μ and a fixed teacher $y^* : X \to Y$ the population versions are the risk $R(\pi) := \mathbb{E}_{x\sim\mu}[-\log \nu_\pi(y^*(x) \mid x)]$, $J_c(\pi) := c\,\ell(\pi) + R(\pi)$ and $\Phi(c) := \inf_\pi J_c(\pi)$. Hänni's remark that "we will need to increase the hypothesis term coefficient compared to the sum of log likelihoods term as the data set size increases" (Hänni and Wyeth, undated) is the choice $\lambda = cn$ with c fixed. For fixed λ, ξ_λ is one shared hypothesis and its sequential loss telescopes; with $\lambda = cn$ the prior exponent changes with n and this is not available. All statements with $\lambda = cn$ are therefore *batch* statements at a given n, not statements about one Bayesian predictor.

The next lemma is the geometry behind Hänni's observation that "the 'good model' could be a vertex of the convex hull of the set of attainable (hypothesis complexity, expected neg log likelihood) tuples" (Hänni and Wyeth, undated), and behind his alternative, an "abs bound on model complexity" (Hänni, 2026b).

Lemma 2.8 (Hull lemma)**.** [proved; computed; strengthening of (iv) refuted] steeper Lemma 1.2 (T5 Lemma 3.1); checks c5(a), c6(F) *Let $\mathcal{A} := \{(\ell(\pi), R(\pi))\}$ with $R \in [0, \infty]$ and finitely many points below each ℓ-level, let $C := \mathrm{conv}(\mathcal{A} \cap \{R < \infty\}) + \mathbb{R}^2_{\geq 0}$, and call the extreme points of C its* vertices*.*

(i) For $c > 0$, every minimiser of $c\,\ell + R$ lies on the boundary of C, on a supporting line with normal $(c, 1)$.

(ii) If $c < c'$ and h, h' minimise $J_c, J_{c'}$, then $\ell(h) \geq \ell(h')$ and $R(h) \leq R(h')$.

(iii) A point of $\mathcal{A}$ is the unique minimising point of $c\,\ell + R$ for every c in some nonempty open interval if and only if it is a vertex of C; the least-ℓ vertex and the least-R vertex, if it exists, count. A hypothesis is then the unique minimiser if and only if no other hypothesis has the same point.

(iv) (Absolute length bound.) Every minimiser h_c of $c\,\ell + R$ ($c > 0$) solves $\min\{R(h) : \ell(h) \leq \ell(h_c)\}$. Conversely, a solution of $\min\{R(h) : \ell(h) \leq L\}$ minimises $c\,\ell + R$ for some $c > 0$ if and only if its point lies on the boundary of C and not on the horizontal ray to the right of the least-R vertex.

The strengthening of the converse in (iv) with "on the boundary of C" alone is false: in $\mathcal{A} = \{(0, 1), (10, 0), (20, 0)\}$ the point $(20, 0)$ lies on the boundary and solves the constrained problem for $L \geq 20$, but $(10, 0)$ is strictly better for every $c > 0$.

So Hänni's two proposals, the penalty and the absolute bound, select the same hypotheses at the vertices; the absolute bound can also select points above the boundary, which the penalty never selects. Examples are in Appendix A.2.

The last lemma is the sufficiency line for data-set classes: one bit of description pays for at most one bit of likelihood.

Lemma 2.9 (Coding bound)**.** [proved] steeper Lemma 1.6 *In every data-set class (product or shared hypotheses), for every program π and data set D,*

$$\ell(\pi) - \log P_\pi(D) \;\geq\; K(y_D \mid \operatorname{dom} D) - \kappa,$$

where y_D lists the labels of D in the canonical order of $\operatorname{dom} D$*.*

Proof idea. Given π and $\operatorname{dom} D$, the labelling $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ is an lsc semimeasure on $Y^{\operatorname{dom} D}$; apply (F2), (F3) and (F1) (Appendix A.3). What separates the two kinds of class is whether one member attains this bound up to a constant on *every* data set. In a class that contains its own mixture, the mixture does (Section 3); in the product class nothing does (Corollary 3.5).

3 The absorption objection, and why private randomness blocks it

Under a steeper penalty, a hypothesis that is itself a universal mixture, and so penalises complexity only in the usual way inside itself, can absorb all the structure of the data. This is a standard point.[1] Hänni's note makes it for sequence prediction, where one has the "pathology of not specifying even the hypothesis in the seq prediction case (like it'll be better to drop bits and take the likelihood loss)", and suggests that the pathology is absent in the function case "if we make the randomness not shared between different inputs — each input has to sample its own random bits" (Hänni, 2026b). We call it the *absorption objection*. Theorem 3.2 makes it exact in every class that contains its own mixture; Proposition 3.4 and Corollary 3.5 show that the product class contains no such hypothesis; Propositions 3.8 and 3.9 give the price and the boundary of this protection; Remark 3.11 assembles the answer. Proofs are in Appendix B.

3.1 Shared randomness: a universal hypothesis absorbs everything

A class satisfies (Dom) if it contains a program π_U, with $U := P_{\pi_U}$ and $\ell_U := \ell(\pi_U)$, such that

$$P_\pi(o) \;\leq\; 2^{\ell(\pi)+\kappa_0}\, U(o) \qquad \text{for all programs } \pi \text{ and observations } o. \tag{Dom}$$

Remark 3.1 (Classes with an absorbing member). [proved] steeper §1.4, examples (1)–(5) (Dom) holds with $\kappa_0 = 0$, and $\xi_1 \leq U$, in every class that contains its own mixture as a member: in $\mathcal{H}_F$ with $U = M_F$, in $\mathcal{H}_w$ with $U = M^w$, in $\mathcal{H}_{\mathrm{SI}}$ with $U = M$, and in $\mathcal{H}_{\mathrm{chr}}$ with $U = M_H$ (only the universality of M_H is used). See Appendix B.1.

Let $\ell^*_\lambda := (\lambda \ell_U + \kappa_0)/(\lambda - 1)$.

Theorem 3.2 (Absorption)**.** [proved] steeper Thm 1 *Assume (Dom) and let $\lambda > 1$.*

(a) MAP. *For every observation o, every λ-MAP $\hat\pi$ has $\ell(\hat\pi) \leq \ell^*_\lambda$ and $-\log P_{\hat\pi}(o) \leq -\log U(o) + \lambda\ell_U - \lambda\ell(\hat\pi)$. So if $L > \ell_U$ and $\lambda > (L + \kappa_0)/(L - \ell_U)$, no hypothesis of length at least L is a λ-MAP for any observation; this holds for fixed λ, and for $\lambda = cn$ once cn exceeds the bound.*

[1] It was also raised in the comment thread in which the proposal was discussed (Hänni and Wyeth, undated).

(b) Power posterior. $2^{-\lambda \ell_U} U(o) \le \xi_\lambda(o) \le \xi_1(o)$. *If* $\xi_1 \le 2^{\kappa_1} U$, *then* $-\kappa_1 \le -\log \xi_\lambda(o) + \log U(o) \le \lambda \ell_U$ *for every* o.

(c) Deterministic teacher, errors included (data-set classes). *Let* $t : X \to Y$ *be total computable with a program* π_t *of length* ℓ_t *in the class, let* $x_1, x_2, \dots$ *be distinct and* $D_n := \{(x_i, t(x_i)) : i \le n\}$.

(c1) $\sum_{i \le n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) \le \min\{\lambda \ell_t,\ \lambda \ell_U + \kappa_0 + \ell_t\}$ *for every* n*; so at most that many inputs, over the whole sequence, get probability below* $1/2$ *for the teacher's label.*

(c2) Every λ*-MAP for* D_n *has* $-\log P_{\hat\pi}(D_n) \le \lambda \ell_U + \kappa_0 + \ell_t$.

(c3) There are n_0 *and a finite set* $\mathcal{H}_{\text{fin}}$ *such that every* λ*-MAP for* D_n, $n \ge n_0$, *lies in* $\mathcal{H}_{\text{fin}}$, *and at only finitely many* n *does a* λ*-MAP give* $t(x_{n+1})$ *probability below* $1/2$.

(d) SI with random inputs. *Let the inputs lie in* $\{0,1\}^d$, *encode* $(x_1, y_1, x_2, \dots)$ *as a binary string with fixed-length blocks, let the* x_i *be i.i.d. from a computable law and* $y_i = t(x_i)$, *and let* $\mathbb{P}$ *be the resulting computable measure, with a program* $\pi_{\mathbb{P}} \in \mathcal{H}_{\text{SI}}$. *Then* $\mathbb{E}_{\mathbb{P}} \sum_{i \le n} -\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i) \le \lambda \ell_U + \kappa_0 + \ell(\pi_{\mathbb{P}})$.

Proof idea. (a) A λ-MAP is at least as good as π_U, and (Dom) bounds its gain in likelihood by $\ell(\hat\pi) + \kappa_0$, so $(\lambda - 1)\ell(\hat\pi) \le \lambda \ell_U + \kappa_0$. (b) Keep one term of the mixture. (c) Telescope $\xi_\lambda(D_n) \ge 2^{-\lambda \ell_U} U(D_n)$; for (c3), each of the finitely many short programs either has bounded loss along the sequence or is eventually beaten by π_U. (d) is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions. See Appendix B.2.

Remark 3.3 (Reading). [remark; known]

steeper remarks after Thm 1 Part (a) is the absorption objection made exact, and the precise form of Hänni's "drop bits and take the likelihood loss"; $\ell^*_\lambda = \ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$ is the threshold of the earlier formalisation in Claude (Anthropic) (2026, section "Simplicity and normativity from imitation"). As $\lambda \to \infty$ it tends to ℓ_U, not 0; with $\lambda = cn$, every model longer than about ℓ_U is eventually excluded. Whether a given model is longer than ℓ_U depends on the reference machine; a dog classifier with millions of bits certainly is. For adaptive or non-computable inputs, (d) is replaced by $\mathcal{H}_{\text{chr}}$, where (a)–(b) hold. Part (c3) has no explicit rate; explicit online bounds for two-part MDL are exponentially worse than for Bayes mixtures in general (Poland and Hutter, 2005).

3.2 Private randomness: no hypothesis absorbs anything

In the product class, in Hänni's words, "you have to pay the likelihood term separately at every x" (Hänni, 2026b). For a finite set F of total functions $X \to Y$ and a distribution Q on F, let Q_x be the law of $f(x)$ for $f \sim Q$.

Proposition 3.4 (Per-input pricing)**.** [proved; computed] steeper Prop 1.3 (T5 Thm 2.1, Prop 2.2); check c2(a); referee r2 *Let* $x_1, \dots, x_n$ *be distinct.*

(i) For every product hypothesis ν *and every* Q, $\max_{f \in F} \sum_{i \le n} -\log \nu(f(x_i) \mid x_i) \ge \sum_{i \le n} H(Q_{x_i})$. *The value* $\inf_\nu \max_f \sum_i -\log \nu(f(x_i) \mid x_i)$ *equals* $\max_Q \sum_i H(Q_{x_i})$, *which can be strictly below* $\sum_i \log |\{f(x_i) : f \in F\}|$.

(ii) If F *is the set of parities on* $\{0,1\}^d$ *and the* x_i *are nonzero, every product hypothesis pays at least* n *bits on some parity, while the Bayes mixture over parities, a shared hypothesis, pays at most* d *bits on every parity.*

(iii) A per-input mixture $\nu(y \mid x) = \sum_q w_q\, q(y \mid x)$ is a product hypothesis and pays up to $\log(1/w_q)$ at every input: with the two constant functions and $w = 1/2$ it pays n bits on n ones, where the shared mixture pays 1 bit.

The bound in (i) is Gibbs' inequality applied to $\mathbb{E}_{f\sim Q}$, and the value follows from Sion's minimax theorem (Sion, 1958). For $F = \{(a,a),(a,b),(b,a),(c,a)\}$ on two inputs a direct computation (code in the accompanying repository) gives the value $2.5431 < \log 3 + \log 2 = 2.5850$.

Corollary 3.5 (No universal product hypothesis)**.** [proved]
steeper Cor 1.4 *For every product hypothesis ν_U and every $d \geq 1$ there are a data set D with $\operatorname{dom} D = \{0,1\}^d \setminus \{0^d\}$ and a deterministic $f \supseteq D$ with a program of length at most $d + 2\log d + \kappa$ such that $\nu_U(D) \leq 2^{-(2^d-1)}$. Hence for every C there are such D and f with $\nu_U(D) < 2^{-C} 2^{-\ell(\pi_f)} f(D)$, and (Dom) fails in every product class.*

Proof. Take the parity given by Proposition 3.4(ii) on which ν_U loses at least $2^d - 1$ bits; it is described by d and $a \in \{0,1\}^d$, and (F5) gives the length bound. Choose d with $2^d - 1 - d - 2\log d - \kappa > C$. □

Remark 3.6 (The universal conditional semimeasure never learns). [proved] steeper Rem 1.5 (T5 Thm 2.4) The universal lsc conditional semimeasure satisfies $\mathbf{m}(y \mid x) \geq 2^{-K(y|x)-\kappa}$. For $Y = \{0,1\}$ this gives $\mathbf{m}(y \mid x) \leq 1 - 2^{-\kappa'}$ for both labels, so it pays at least $\log \frac{1}{1-2^{-\kappa'}} > 0$ bits for every label on every data set. It dominates each product hypothesis input by input, $\mathbf{m} \geq 2^{-K(\nu)-\kappa}\nu$, but pays the factor $2^{K(\nu)}$ again at every input; its prediction does not depend on the data.

3.3 Why private randomness, and what it costs

Remark 3.7 (Why private randomness is what matters). [remark; cites proved results] steeper §1.6; foundations Thm 4.2, Prop 4.5, Rem 4.5a At $\lambda = 1$ the decomposition into hypotheses does not change the predictor ($\xi_\times = M_{\text{prod}}$ is one shared hypothesis); at $\lambda > 1$ it does. Over the class $\mathcal{H}_F$ of all machine laws, M_F would be a short member and absorption would return. Steeper SFI is therefore defined over $\mathcal{H}_\times$, whose $\lambda \to 1$ limit is M_{prod}, not M_F. At $\lambda = 1$ this costs at most $K(\operatorname{dom} D) + O(1)$ bits against M_F, but not a constant: M_F/M_{prod} is unbounded already against the $O(1)$-complexity law ν_{len} with one shared fair coin per input length; on data with inputs of length at most k the gap to ν_{len} is at most $K^{\text{pad}}(k+1) + O(1)$ bits, where $K^{\text{pad}}(N) := \min_{L \geq N} K(L)$ (Hänni, 2026a, the chain theorem, the shared-coin proposition and the remark after it). At $\lambda > 1$ the cost is a constant factor on shared latent causes (Proposition 3.8).

For $k \geq 1$ and $\beta \in \{0,1\}^k$ let $D_\beta := \{(0^i, \beta_i), (1^i, \beta_i) : i \leq k\}$: one fair coin per pair of inputs, shown on both.

Proposition 3.8 (The price of private randomness)**.** [proved; computed] steeper Prop 1.7; check c6(D) *Let $Y = \{0,1\}$. (i) For every $\lambda \geq 1$ and β: $-\log \xi^\times_\lambda(D_\beta) \geq \min(\lambda, 2)\,\big(K(\beta \mid k) - \kappa\big)$. (ii) Hence $J_\lambda(\pi; D_\beta) \geq \min(\lambda,2)\big(K(\beta \mid k) - \kappa\big)$ for every $\pi \in \mathcal{H}_\times$, and in every product class with a prefix-free code, with κ depending on the class. (iii) ν_{len} is a machine law of complexity $O(1)$ with $\nu_{\text{len}}(D_\beta) = 2^{-k}$, so $-\log M_F(D_\beta) \leq k + \kappa$ and $-\log \xi^F_\lambda(D_\beta) \leq k + \lambda\kappa$. (iv) For uniform β and $j \geq 0$, with probability at least $1 - 2^{-j}$ the steeper product predictor pays at least $(\min(\lambda,2) - 1)k - 2j - (\lambda+2)\kappa$ bits more than M_F and than ξ^F_λ.*

Proof idea. By the inequality of arithmetic and geometric means, $\nu(y \mid 0^i)\nu(y \mid 1^i)$ is at most the square of their average, so every product law on D_β is at most the square of a semimeasure

on β; mixing and the coding theorem give (i) (Appendix B.4). A product class must treat a coin shared by several inputs either as *model*, at λ bits per coin, or as *noise*, paid again at every input that shows it (2 bits here); a shared class pays 1 bit. Persistent latent causes in real labelling, such as a labeller's mood or a per-batch coin, are of this kind.

The protection also depends on what an input is.

Proposition 3.9 (Labelled history restores universality)**.** [proved; computed]
steeper Prop 4.1 (T5 Prop 2.6); check c5(d) *Suppose a hypothesis's input at position i of a block contains the block's earlier labelled pairs. Let $(\nu_\theta)_{\theta\in\Theta}$, Θ countable, be uniformly computable history-dependent conditional measures with positive values, and w computable positive weights with a computable sum $\sum_\theta w_\theta \le 1$. With $Z_i := \sum_\theta w_\theta \prod_{k<i} \nu_\theta(y_k \mid \cdot)$, the in-context Bayes mixture $h_w(y_i \mid \cdot) := Z_{i+1}/Z_i$ is a single product hypothesis on the extended inputs, with a program of length at most $K(w,(\nu_\theta)) + \kappa$, and on every block*

$$\textstyle\sum_i -\log h_w(y_i \mid \cdot) \;\le\; \min_\theta \big[\log(1/w_\theta) + \sum_i -\log \nu_\theta(y_i \mid \cdot)\big].$$

So it pays $\log(1/w_\theta)$ once per block, not once per input.

Proposition 3.10 (A history of inputs only)**.** [proved]
steeper Prop 4.2 *A hypothesis that sees the earlier inputs but not the earlier labels, and draws fresh randomness for each extended input $(x_{<i}, x_i)$, is a product hypothesis on the extended inputs, which are pairwise distinct; so Proposition 3.4 and Corollary 3.5 apply and per-input pricing persists. With shared randomness instead, as in* $\mathrm{SFI}^{\mathrm{hist}}$*, $\mathcal{H}_{\mathrm{chr}}$ has the universal member M_H and Theorem 3.2 applies.*

So the property that makes $\mathrm{SFI}^{\mathrm{hist}}$ as strong as a universal chronological predictor, shared randomness, is the property that removes the protection against absorption.

3.4 Answer to the absorption objection

Remark 3.11 (Answer to the absorption objection). [remark; cites proved results] steeper §1.8

(i) *The objection is correct wherever a universal hypothesis is one member of the class:* in SI ($U = M$), in SFI with shared randomness ($U = M_F$), in the weak class ($U = M^w$) and in history-aware SFI ($U = M_H$). There every λ-MAP has length at most ℓ^*_λ and fits almost as well as U, so it reproduces a deterministic teacher's errors (Theorem 3.2(c)), and samplers longer than about ℓ_U are never selected (Theorem 6.8).

(ii) *The construction is unavailable in the product class.* No product hypothesis is universal (Corollary 3.5); the universal conditional semimeasure never learns (Remark 3.6); seeing earlier inputs does not help while the randomness stays private (Proposition 3.10). This is Hänni's reply made exact: "if each input gets its own private randomness, then I don't see how to write down a universal hypothesis that gets good loss here" (Hänni and Wyeth, undated).

(iii) *The restriction has a price and a boundary.* The product class cannot represent latent randomness shared across inputs: at $\lambda = 1$ this costs at most $K(\mathrm{dom}\, D) + O(1)$ bits against M_F, unboundedly in the worst case (Remark 3.7); at $\lambda > 1$ a factor $\min(\lambda, 2)$ on shared coins (Proposition 3.8). The protection ends where hypotheses see labelled history (Proposition 3.9). So "nothing is lost at $\lambda = 1$" holds only relative to the product class, not relative to M_F.

(iv) *The algorithmic-statistics reading.* For set models, which share their randomness, the minimal sufficient statistic of a noisy random parity treats the parity's d-bit parameter as noise (sophistication $O(\log d)$); for product models the parameter is model (product sophistication $d \pm O(\log d)$; Proposition 7.6). Restricting the model family to "parameter plus private noise per input" is what moves the split (Remark 7.7).

The restriction is a modelling choice, and we present it as one. Two reasons support it. Hänni's dominance template for SFI is stated against x-only samplers that "use different bits of randomness from the universal semimeasure's infinite read tape for sampling on different inputs" (Hänni, 2026b), which are exactly the product hypotheses. And in supervised learning "the model" is naturally the part of the description that is reused across inputs; noise private to each input is, by construction, not part of it. The price in (iii) is what the choice gives up.

Open question. Which hypotheses should count as shared? Whether every lsc law of a random partial function is dominated by M_F up to a constant is open (Hänni, 2026a). Theorem 3.2 holds for $\mathcal{H}_w$ and for $\mathcal{H}_F$ either way.

4 The teacher's-mistakes theorem

Hänni asks how a learner can grasp what a teacher means rather than become a predictor of what the teacher would say; Solomonoff induction "would be making the same mistakes the person would make" (Hänni, 2026b). His remedy is the steeper objective of Definition 2.5 over product hypotheses: "if there are explainable mistakes on a simple model, you want the simple model that doesn't predict the mistakes" (Hänni, 2026b). This section makes his example rigorous. Theorem 4.7 quantifies over *all* product hypotheses and shows that, for error rules that are incompressible relative to the model, every near-optimal hypothesis is, up to explicit slack, the simple model with noise, on a window of $c = \lambda/n$ of more than three orders of magnitude at his sizes. Section 5 shows where this stops. Proofs are in Appendix C.

4.1 Hänni's calculation

Remark 4.1 (Hänni's calculation). [computed] steeper §2.1; checks c1, T5 re-run Hänni gives two versions of his example (Hänni, 2026b; Hänni and Wyeth, undated).

(1) *Version 1* (total bits): "a simple model of 100 bits which predicts 100 bits per specification bit, on top of which there is a systematic error model of 100 bits which predict 10 bits per specification bit". Coin, simple model, and simple model plus errors lie on a hull with slopes 100 and 10, so the simple model alone is selected exactly for $\lambda \in (10, 100)$.

(2) *Version 2* (per sample): a 100-bit "simple model" with "systematic human error" on "a 2^{-10} fraction of the inputs, with the set of those inputs taking 10000 bits to specify". Against the "model that learns the errors" (10,100 bits, loss 0) and the "50/50 random distribution", the "good model" (the simple model with flip probability $p = 2^{-10}$, 10 bits for p) has rate $(1 - H(p))/110 = 8.989 \cdot 10^{-3}$ and the errors have rate $H(p)/9990 = 1.1185 \cdot 10^{-6}$. So the good model is selected among the three if and only if $c \in (1.12 \cdot 10^{-6}, 8.99 \cdot 10^{-3})$, a ratio of about 8000; plain MDL ($\lambda = 1$) switches to the memoriser at $n \approx 8.94 \cdot 10^5$.

A direct computation (code in the accompanying repository) confirms this arithmetic; his "$\approx p \log_2(1/p) \approx 1/100$" is 0.0098 against the exact $H(2^{-10}) = 0.0112$, which does not matter. The calculation compares three hypotheses only. It does not exclude partial models, partial memorisers

or mixtures, and Hänni asks: "Maybe I'm just failing to construct the right 'universal hypothesis' for this example?" (Hänni and Wyeth, undated).

4.2 The instance

Definition 4.2 (The teacher's-mistakes instance)**.** steeper Setting 2.0 Fix integers $1 \le m \le d$ and $2 \le r \le m$, and put $p := 2^{-r}$ and $s_0 := p2^m$.

(a) *Model, errors, teacher.* The *simple model* is the parity $f_a(x) = \langle a, x\rangle \bmod 2$ on $\{0,1\}^d$, for some $a \in \{0,1\}^d$ with $K(a) \ge d$. The *error rule* is a set $S \subseteq \{0,1\}^m$ with $|S| = s_0$, and the *error set* is $E := S \times \{0,1\}^{d-m}$, the inputs whose first m bits lie in S. The *teacher* is $y^* := f_a \oplus \mathbf{1}_E$. For $m < d$ the errors are a systematic rule (a random junta) that generalises to new inputs; for $m = d$ they are an idiosyncratic random set.

(b) *Description lengths.* $L_S := \log \binom{2^m}{s_0}$, $B_S := K(S \mid a, K(a))$, and the *deficiency* $\delta_S := L_S - B_S$. The *information* a program π holds about the rule beyond the parity is $I(\pi) := B_S - K(S \mid a, \pi)$, so $L_S - K(S \mid a, \pi) = \delta_S + I(\pi)$. One has $2^m H(p) - \log(2^m + 1) \le L_S \le 2^m H(p)$ (Cover and Thomas, 2006); $\delta_S \ge -2\log m - 2\log s_0 - \kappa$, since S can be given by its index among the s_0-subsets; and $\Pr_S[\delta_S > j] < 2^{-j}$ for a uniformly random rule, since fewer than $2^{L_S - j}$ programs are shorter than $L_S - j$.

(c) *Population quantities.* Inputs are uniform on $\{0,1\}^d$, so $\Pr[x \in E] = p$. For a product program π and a truncation level $B \in [\log(1/p), \infty]$: $R(\pi) := \mathbb{E}_x[-\log \nu_\pi(y^*(x) \mid x)]$, $R_B(\pi) := \mathbb{E}_x \min\{B, -\log \nu_\pi(y^*(x) \mid x)\} \le R(\pi)$, $J_c := c\,\ell + R$, $J_c^B := c\,\ell + R_B$, $\Phi(c) := \inf_\pi J_c(\pi)$ and $\Phi^B(c) := \inf_\pi J_c^B(\pi) \le \Phi(c)$, the infima over the class. Put $\tau_B := 2^{1-B}$ and $\beta_B := 2^m \log(1 + \tau_B)$, both 0 if $B = \infty$.

(d) *Agreement and level.* $u(\pi) := \mathbb{E}_x \nu_\pi(f_a(x) \mid x)$. With $\theta_t := \frac{1}{2} + 2^{-t-1}$, the *level* is $\mathrm{lev}(\pi) := \min\{t \ge 1 : u(\pi) > \theta_t\}$, and $\mathrm{lev}(\pi) := \infty$ if $u(\pi) \le \frac{1}{2}$; so $\mathrm{lev}(\pi) = 1$ if and only if $u(\pi) > \frac{3}{4}$. Put $s(t) := 2t + 2\log(t+1) \le 4t$ and $G_j := 1 - \log(1 + 2^{-j} + 2p + \tau_B)$ for $j \in [0, \infty]$; G_j is increasing and concave in j.

(e) *Reference programs* in $\mathcal{H}_\times$: the *coin* π_{coin}, with $\ell \le \kappa$ and $R = 1$; the *good model* π_g, which gives $f_a(x)$ probability $1 - p$, with $\ell(\pi_g) \le d + \kappa_g$, $\kappa_g := 2\log d + 2\log r + \kappa$, and $R(\pi_g) = R_B(\pi_g) = H(p)$; the *memoriser* π_b, the deterministic y^*, with $\ell(\pi_b) \le d + \lceil L_S \rceil + \kappa_b$, $\kappa_b := 2\log d + 2\log m + 2\log s_0 + \kappa$, and $R(\pi_b) = 0$.

(f) *Slack constant.* $\Lambda_1 := 2\log d + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \beta_B + \kappa$.

Hänni's sizes are $d = 100$, $m = 20$, $p = 2^{-10}$; then $s_0 = 1024$ and $L_S = 11{,}710.27$ bits, his "10000 bits".

The objective is the population objective J_c; finite samples are treated in Corollary 4.11. The deficiency δ_S enters every slack term below, and every number we give states the δ_S it assumes; numbers with $\delta_S = 0$ concern error rules that are incompressible given the model, and say nothing about compressible ones (Proposition 5.3).

4.3 Four lemmas

The first lemma is the reason for choosing parities: a hypothesis that agrees with f_a noticeably more often than a coin determines a up to a few bits.

Lemma 4.3 (List decoding for semimeasures)**.** [proved; computed] steeper Lemma 2.1 (T5 Lemma 2.5a, generalised); checks c2(b), referee r3 *If* $\mathrm{lev}(\pi) = t < \infty$*, then* $K(a \mid \pi) \le s(t) + 2\log d + \kappa$.

Proof idea. The agreement of ν_π with a parity $f_{a'}$ is $\frac{1}{2}$ plus half a Fourier coefficient of $\nu_\pi(0 \mid x) - \nu_\pi(1 \mid x)$, and by Parseval fewer than 2^{2t} parities have agreement above θ_t (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The agreement is lower semicomputable, so this list can be enumerated from π (Appendix C.1).

Lemma 4.4 (Risk floor)**.** [proved; computed]
steeper Lemma 2.2 (T5 Lemma 2.5b); check c2(b) $R_B(\pi) \ge -\log\left(u(\pi) + p + 2^{-B}\right)$*, with* $2^{-\infty} = 0$*. Hence* $\mathrm{lev}(\pi) = t \ge 2$ *implies* $R_B(\pi) \ge G_{t-1}$*, and* $\mathrm{lev}(\pi) = \infty$ *implies* $R_B(\pi) \ge G_\infty$.

Lemma 4.5 (Coding the error rule with a hypothesis)**.** [proved] steeper Lemma 2.3; referee r3, q2(c) *For integer or infinite* B*,* $K(S \mid a, \pi) \le 2^m R_B(\pi) + \beta_B + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$.

Lemma 4.6 (Two-part lower bounds)**.** [proved] steeper Lemma 2.4 (T5 Thm 2.5(c)) *If* $\mathrm{lev}(\pi) = t < \infty$*, then*

$$\textit{(D1)}\quad \ell(\pi) \ge d - s(t) - \Lambda_1, \qquad\qquad \textit{(D2)}\quad \ell(\pi) + 2^m R_B(\pi) \ge d + B_S - s(t) - \Lambda_1.$$

Proof ideas. Lemma 4.4 is Jensen's inequality; a hypothesis can be right on E for free only on a p-fraction of the inputs. Lemma 4.5 turns the per-z average likelihoods into a semimeasure on subsets of $\{0,1\}^m$; the product form of E makes the cost $2^m R_B$, not $2^d R_B$. (D1) is $d \le K(a) \le \ell(\pi) + K(a \mid \pi) + \kappa$; (D2) adds Lemma 4.5 and symmetry of information. See Appendix C.2, which also proves a sharper floor (Lemma C.1).

4.4 The theorem

Define

$$\psi_c(R) := c\max\{0, B_S - 2^m R\} + R, \qquad \Phi_1(c) := c(d - 4 - \Lambda_1) + B_S/2^m, \qquad t^* := (d - \Lambda_1)/4,$$

$$\Phi_2(c) := \begin{cases} \min\{c(d - 8 - \Lambda_1) + G_1,\ G_{t^*-1}\} & t^* \ge 2, \\ G_1 & t^* < 2, \end{cases}$$

$$J_g(c) := c(d + \kappa_g) + H(p) \ge J_c(\pi_g), \qquad J_b(c) := c(d + \lceil L_S \rceil + \kappa_b) \ge J_c(\pi_b),$$

$$\Delta(c) := \max\{0,\ c(4 + \Lambda_1 + \kappa_g) + H(p) - B_S/2^m\}.$$

The pieces of Φ_2 are nondecreasing in c and J_g is linear, so $\{c > 0 : J_g(c) \le \Phi_2(c)\} = (0, c_{\mathrm{hi}}]$ with

$$c_{\mathrm{hi}} := \min\left\{\frac{G_1 - H(p)}{8 + \Lambda_1 + \kappa_g},\ \frac{G_{t^*-1} - H(p)}{d + \kappa_g}\right\}\ \ (t^* \ge 2), \qquad c_{\mathrm{hi}} := \frac{G_1 - H(p)}{d + \kappa_g}\ \ (t^* < 2).$$

$c_{\mathrm{hi}} > 0$ requires $G_1 > H(p)$; with $B = \infty$ this holds if and only if $p < 0.0560$.

Theorem 4.7 (The teacher's-mistakes theorem)**.** [proved]
steeper Thm 2 *In the setting of Definition 4.2, with* $B \in [\log(1/p), \infty]$*, the following hold for every program* π *of* $\mathcal{H}_\times$*, and of every product class covered by Proposition 8.1.*

- *()* Level bounds. *If* $\mathrm{lev}(\pi) = 1$*, then* $J_c^B(\pi) \ge c(d - 4 - \Lambda_1) + \psi_c(R_B(\pi))$*. If* $\mathrm{lev}(\pi) \ge 2$*, including* ∞*, then* $J_c^B(\pi) \ge \Phi_2(c)$.
- *(i)* Window. *If* $c\,2^m > 1$ *and* $c \le c_{\mathrm{hi}}$*, then* $\Phi(c) \ge \Phi^B(c) \ge \min\{\Phi_1(c), \Phi_2(c)\}$ *and* $J_c(\pi_g) \le \Phi(c) + \Delta(c)$*, where* $\Delta(c) \le c(4 + \Lambda_1 + \kappa_g) + (\log(2^m + 1) + \delta_S)/2^m$.

(ii) Identification. *Let* $c\,2^m > 1$, $\epsilon \geq 0$ *and* $J_g(c) + \epsilon < \Phi_2(c)$, *and let* $J_c^B(\pi) \leq J_g(c) + \epsilon$; *this includes every* π *with* $J_c(\pi) \leq \Phi(c) + \epsilon$. *Then* $\mathrm{lev}(\pi) = 1$. *Put* $\Lambda_2 := 2^m H(p) - B_S + 4 + \Lambda_1 + \kappa_g \leq \log(2^m + 1) + \delta_S + 4 + \Lambda_1 + \kappa_g$, $\Lambda_3 := 4 + 2\log d + \kappa$ *and*

$$\Delta_\ell := \kappa_g + \frac{\epsilon}{c} + \frac{\Lambda_2 + \epsilon/c}{c2^m - 1}.$$

(a) It knows the simple model: $u(\pi) \geq \max\{\frac{3}{4},\ 2^{-(J_g(c)+\epsilon)} - p - 2^{-B}\}$.

(b) It fits no better than noise: $R_B(\pi) \geq H(p) - \dfrac{c\Lambda_2 + \epsilon}{c2^m - 1}$.

(c) It is short: $\ell(\pi) \leq d + \Delta_\ell$.

(d) It holds little information about the error rule: $I(\pi) \leq \Delta_\ell + \Lambda_3$, *so* $L_S - K(S \mid a, \pi) \leq \delta_S + \Delta_\ell + \Lambda_3$.

(e) It does not predict the errors: *for every integer* $g \geq 1$,

$$e(\pi) := \mathbb{E}\big[\nu_\pi(y^*(x) \mid x) \,\big|\, x \in E\big] \ \leq\ 1 - u(\pi) + \gamma_g + 2^{-g},$$

where $\gamma_g := \sqrt{j_g \ln 2/(2s_0)}$, j_g *is the least integer* $j \geq 0$ *with* $j - 2\log(j+1) > \Gamma_g$, *and* $\Gamma_g := \delta_S + \Delta_\ell + \Lambda_3 + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$.

(iii) Memoriser regime. *Let* $c\,2^m < 1$, *and let* $J_c^B(\pi) \leq J_b(c) + \epsilon$ *with* $J_b(c) + \epsilon < \Phi_2(c)$. *Then* $\mathrm{lev}(\pi) = 1$ *and* $(1 - c2^m)\min\{R_B(\pi), B_S/2^m\} \leq \varrho := c(\lceil L_S \rceil - L_S + \delta_S + 4 + \Lambda_1 + \kappa_b) + \epsilon$. *If* $\varrho < (1 - c2^m)B_S/2^m$, *then* $R_B(\pi) \leq \varrho/(1 - c2^m)$ *and* $e(\pi) \geq 2^{-R_B(\pi)/p} - 2^{-B}$.

(iv) The lower end is sharp. *If* $c < H(p)/(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$, *then* $J_c(\pi_b) < J_c(\pi)$ *for every* π *with* $\mathrm{lev}(\pi) = 1$ *and* $R(\pi) \geq H(p)$. *Since* $L_S = 2^m H(p) - O(m)$, *this threshold is* $2^{-m}\big(1 + O((m + \log d + \kappa)/L_S)\big)$.

Proof idea. For a program of level 1, (D1) and (D2) give $J_c^B \geq c(d - 4 - \Lambda_1) + \psi_c(R_B)$, and when $c2^m > 1$ the function ψ_c is minimised at $R = B_S/2^m$: each bit of rule information costs c per sample in length and saves only 2^{-m} in risk. A program of level $t \geq 2$ pays in length through (D1) or in risk through Lemma 4.4; concavity of G reduces the t-dependence to two endpoints. Part (ii) reads these inequalities backwards. For (e), if a near-optimal program gave the erroneous labels much probability, S would lie in a small set enumerable from the program (Hoeffding's inequality for sampling without replacement (Hoeffding, 1963)), contradicting (d). Parts (iii) and (iv) follow from (D2) and (D1). See Appendix C.3.

So in the window $2^{-m} < c \leq c_{\mathrm{hi}}$, for error rules that are incompressible relative to the model, no product hypothesis of any kind beats the good model by more than $\Delta(c)$, and every near-optimal hypothesis knows the parity, is at most Δ_ℓ bits longer than d, holds little information about the rule and gives the erroneous labels little probability. Below 2^{-m}, by a relative margin, near-optimal hypotheses reproduce the errors. This answers Hänni's question about a missing "universal hypothesis" for such rules.

Remark 4.8 (Numbers at Hänni's sizes). [computed] steeper §2.4 "Numbers"; checks c3, c6(B), c7(D); referee r1, q4(B) At $d = 100$, $m = 20$, $p = 2^{-10}$, $B = \infty$ and $\epsilon = 0$, a direct computation (code in the accompanying repository) gives Table 2. Rows with $\kappa = 0$ are illustrations, not results about any particular reference machine; rows with $\kappa = 30$ show how the bounds degrade, and every bound is explicit in κ and δ_S. The window is $(9.54 \cdot 10^{-7},\ 8.06 \cdot 10^{-3}]$ at $\kappa = 0$ (a ratio of 8453) and $(9.54 \cdot 10^{-7},\ 3.66 \cdot 10^{-3}]$ at $\kappa = 30$ (ratio 3837), against Hänni's three-hypothesis window

(a) Identification at $c = 10^{-4}$ (Theorem 4.7(ii); g optimised)

κ	δ_S	$u \geq$	$\ell \leq$	$I \leq$	$L_S - K(S \mid a, \pi) \leq$	$e \leq$
0	0	0.983	$d + 20.4$	37.7	37.7	0.199
0	20	0.983	$d + 20.6$	37.9	57.9	0.218
30	0	0.981	$d + 51.0$	98.3	98.3	0.274
30	20	0.981	$d + 51.2$	98.5	118.5	0.287

(b) Memoriser regime (Theorem 4.7(iii), $\delta_S = 0$): lower bound on e

$c\,2^m$	0.5	0.8	0.9	0.94	0.97	0.99
$\kappa = 0$	0.955	0.831	0.658	0.483	0.223	0.010
$\kappa = 30$	0.917	0.706	0.457	0.256	0.060	0.000

Table 2: The teacher's-mistakes theorem at Hänni's sizes $d = 100$, $m = 20$, $p = 2^{-10}$ ($L_S = 11{,}710$ bits), with $B = \infty$ and $\epsilon = 0$. Rows with $\kappa = 0$ are illustrations. Information bounds are in bits, out of L_S.

$(1.12 \cdot 10^{-6}, 8.99 \cdot 10^{-3})$. The memoriser has $e = 1$ and the good model $e = 2^{-10}$; for $m = 24$ ($L_S = 187{,}457$) the bound at $c = 10^{-4}$ is $e \leq 0.065$ ($\kappa = 0$). Near 2^{-m} the memoriser regime is only partly covered: at $\kappa = 0$, $\delta_S = 0$, (iii) applies only for $c < 0.99418 \cdot 2^{-m}$, and $e \geq \frac{1}{2}$ needs $c \leq 0.937 \cdot 2^{-m}$; in a band of relative width about 6% below 2^{-m} nothing useful is proved.

Remark 4.9 (What the theorem relies on). [remark] steeper §2.4 "The hypotheses Theorem 2 relies on" Without one of the first four of the following hypotheses a step of the proof fails as written; the fifth enters every bound. (1) *A list-decodable model class*: parities admit Lemma 4.3, while simple models with short approximations, such as majority-like functions, do not. (2) *The product form* $E = S \times \{0,1\}^{d-m}$: Lemma 4.5 averages each $z \in \{0,1\}^m$ over 2^{d-m} inputs; for an unstructured $E \subseteq \{0,1\}^d$ it would give $2^d R_B$ bits, useless at $d = 100$. (3) *Uniform inputs*: rates, and with them the window, scale with the input law (Proposition 5.9). (4) $p < 0.056$, so that $G_1 > H(p)$ (Remark 5.11). (5) *A small deficiency* δ_S: every slack term contains it, and Proposition 5.3 shows that this is not an artefact of the proof.

4.5 Idiosyncratic errors, finite samples and Version 1

With idiosyncratic errors and the whole truth table, the window becomes a window in λ itself. Write L_E, B_E, δ_E for L_S, B_S, δ_S when $m = d$.

Corollary 4.10 (Idiosyncratic errors, full truth table)**.** [proved; computed] steeper Cor 2.5 (T5 Thm 3.5, Cor 3.5a); checks c3, c7(H) *Take $m = d$, $N = 2^d$, $B = \infty$, and let the data be the whole truth table of y^*, so $n = N$. A λ-MAP minimises $\lambda\ell + NR = N\,J_{\lambda/N}$.*

(i) For every $\lambda > 1$ with $J_g(\lambda/N) < \Phi_2(\lambda/N)$, in particular every $\lambda \in (1, Nc_{\mathrm{hi}})$, every λ-MAP π satisfies Theorem 4.7(ii) with $c = \lambda/N$, $\epsilon = 0$: $u(\pi) \geq \max\{\frac{3}{4}, 2^{-J_g(\lambda/N)} - p\}$, $NR(\pi) \geq NH(p) - \lambda\Lambda_2/(\lambda - 1)$, $\ell(\pi) \leq d + \kappa_g + \Lambda_2/(\lambda - 1)$, $I(\pi) \leq \kappa_g + \Lambda_2/(\lambda - 1) + \Lambda_3$, and (e).

(ii) At $\lambda = 1$ the optimum, $J_1(\pi_g)$ and $J_1(\pi_b)$ lie within $O(\log N) + \delta_E$ of each other, so $\lambda = 1$ does not decide whether the errors are memorised; lower-order terms and the reference machine do.

At $d = 20$, $p = 2^{-10}$ ($L_E = 11{,}710$ bits) a direct computation gives the window $\lambda \in (1, 1.19 \cdot 10^4]$ at $\kappa = 0$ and $(1, 6456]$ at $\kappa = 30$ (bounds in Appendix C.4).

The theorem is a population statement. For sampled data, let $n \le 2^d$ inputs be drawn uniformly without replacement, so that the data form a partial function, and labelled by y^*; let $B < \infty$, $\delta \in (0,1)$, $\epsilon_1 := \log(1/p)\sqrt{\ln(2/\delta)/(2n)}$, $\hat{L} := (J_c(\pi_g) + \epsilon_1)/c$ and $\epsilon_n := \epsilon_1 + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$.

Corollary 4.11 (Hänni's numbers with sampled data)**.** [proved; computed] steeper Cor 2.7, Lemma 2.6; check c3 *With probability at least $1-\delta$, every λ-MAP with $\lambda = cn$ satisfies $J_c^B \le J_g(c) + \epsilon_n$, and hence the conclusions of Theorem 4.7(ii) with $\epsilon = \epsilon_n$, whenever $c2^m > 1$ and $J_g(c) + \epsilon_n < \Phi_2(c)$.*

The proof is uniform convergence over all programs with Kraft weights (Lemma C.2), which is pessimistic: at Hänni's sizes, with $B = 30$, $\delta = 0.01$, $\kappa = 0$, $\delta_S = 0$ and $c = 10^{-4}$, the excess is $\epsilon_n/c = 9.2$ bits at $n = 10^{11}$ and 0.9 bits at $n = 10^{13}$ (Appendix C.5).

Example 4.12 (Hänni's Version 1, in population form)**.** [proved (instance of Theorem 4.7); computed] steeper Ex 2.7a; checks c6(C), c6(H) Take $d = 100$, $m = 10$, $p = 2^{-6}$, so $s_0 = 16$ and $L_S = 115.6$ bits. At $n = 10^4$ the model saves $n(1 - H(p)) = 8839$ bits for its 100 bits (88 bits per bit) and the error rule saves $nH(p) = 1161$ bits for its 116 bits (10.0 bits per bit): Version 1's sizes and rates. The window is $c \in (2^{-10}, c_{\mathrm{hi}}]$ with $c_{\mathrm{hi}} = 5.80 \cdot 10^{-3}$ at $\kappa = 0$ and $2.53 \cdot 10^{-3}$ at $\kappa = 30$; at $n = 10^4$ and $\kappa = 0$ this is $\lambda \in (9.77, 58.0]$, against Hänni's $(10, 100)$. Only coarse identification is proved at this size: near-optimal hypotheses have $u > \frac{3}{4}$, and at $\kappa = 0$, $\delta_S = 0$, $c = 5.5 \cdot 10^{-3}$ they have $\ell \le d + 28.3$ and $I \le 45.6$ bits out of 115.6; near the lower end the information bound exceeds L_S, and (e) is vacuous because $s_0 = 16$ is too small. At $n = 10^4$ samples Corollary 4.11 gives nothing (Appendix C.6).

Finally, the power posterior with $\lambda = cn$, not only its maximiser, concentrates on near-optimal programs. Let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \le n} -\log \nu_\pi(y_i \mid x_i)$ (Remark 2.7) and $\hat{J}_g := \hat{J}_c(\pi_g)$.

Proposition 4.13 (The power posterior concentrates on near-minimisers)**.** [proved; computed] steeper Prop 2.8; checks c5(e), c7(F); referee q1(F) *Let $\lambda = cn$ and let D be any data set of size n. For every $\epsilon > 0$,*

$$\frac{\Pi_\lambda(\{\pi : \hat{J}_c(\pi) \ge \hat{J}_g + \epsilon\} \mid D)}{\Pi_\lambda(\pi_g \mid D)} \;\le\; 2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}.$$

Consequently, for data sampled as in Corollary 4.11, with probability at least $1-\delta$ all but this fraction of the posterior mass lies on programs that satisfy Theorem 4.7(ii) with $\epsilon' := \epsilon + \epsilon_1 + B\sqrt{(\hat{L}'\ln 2 + \ln(2/\delta))/(2n)}$ in place of ϵ, where $\hat{L}' := (J_g(c) + \epsilon_1 + \epsilon)/c$, provided $J_g(c) + \epsilon' < \Phi_2(c)$.

Proof idea. $2^{-\lambda\ell(\pi)}\nu_\pi(D) = 2^{-n\hat{J}_c(\pi)}$ and $n\hat{J}_c(\pi) - \ell(\pi) \ge (1 - 1/\lambda)\, n\hat{J}_c(\pi)$; sum over the bad set with Kraft's inequality (Appendix C.7). With $\hat{J}_g$ replaced by its population value 0.0232 ($\kappa = 0$), $c = 10^{-4}$, $\epsilon = 10^{-3}$ and $n = 10^6$, the ratio is at most 2^{-758}.

Open question. Which simple-model classes give a window? Parities do, through list decoding; majority-like functions have short approximations and presumably do not. Can the window method reach beyond $p < 0.056$, towards $p < 1/4$ where the kink exists (Proposition 7.6, Remark 5.11), with sharper constants? And what can be proved at realistic sample sizes, far below the $n \approx 10^{11}$ that uniform convergence needs here?

5 Limits of the trick

Theorem 4.7 needs λ to grow linearly in n and the error rule to be incompressible relative to the model. Neither requirement can be dropped, and steeper selection by description length has further failure modes. The setting is Definition 4.2 unless stated otherwise; proofs are in Appendix D.

5.1 Why λ must be proportional to n

Proposition 5.1 (Plain Bayes, MDL and SI learn the errors)**.** [proved; computed] steeper Props 2.9, 2.10; checks c3, c6(G), c7(G)

(a) Let P be a shared hypothesis with $P \geq 2^{-\ell(\pi)-\kappa}\nu_\pi$ for every $\pi \in \mathcal{H}_\times$, such as M_{prod} or M_F. For every sequence of distinct inputs and every n, $\sum_{i\leq n} -\log P(y^*(x_i) \mid x_i, D_{i-1}) \leq \ell(\pi_b)+\kappa \leq d+\lceil L_S\rceil+\kappa_b+\kappa$. So at most $d+L_S+O(\log dm)$ inputs in total get probability below $\frac{1}{2}$ for the teacher's label (semimeasure convention, hence also normalised), and every other revealed error input gets the erroneous label with probability at least $\frac{1}{2}$; if all inputs are revealed, this is a fraction $1-9.6\cdot10^{-24}$ of the error inputs at Hänni's sizes ($\kappa=0$).

(b) Let $B<\infty$, let $\lambda=\lambda_n\geq 1$ be fixed or depend on n, and draw $n\leq 2^d$ inputs uniformly without replacement. With probability at least $1-\delta$, every λ-MAP $\hat\pi$ has $R_B(\hat\pi)\leq \lambda\hat L/n + B\sqrt{(\hat L\ln 2+\ln(2/\delta))/(2n)}$ with $\hat L := d+\lceil L_S\rceil+\kappa_b$, hence $e(\hat\pi)\geq 2^{-R_B(\hat\pi)/p}-2^{-B}$. If $\lambda_n/n\to 0$, this tends to $1-2^{-B}$ as n grows (within $n\leq 2^d$).

(c) On the interleaved string with i.i.d. uniform inputs, Solomonoff's M has expected total label loss at most $d+L_S+O(\log dm)$, and the steeper SI predictor pays at most $\lambda\ell_U$ more (Theorem 3.2(d)).

Proof idea. (a) Telescope $P(D_n)\geq 2^{-\ell(\pi_b)-\kappa}$; for M_F this is the mistake bound of Hänni (2026a). (b) is Lemma C.2 with the memoriser as reference. At $\kappa=0$, $B=30$, $\delta=0.01$, a direct computation (code in the accompanying repository) gives $e(\hat\pi)\geq 0.649$ at $n=10^{13}$ for $\lambda=1$, and $e(\hat\pi)\geq 0.734$ at $n=10^{15}$ for $\lambda_n=\sqrt{n}$, whatever δ_S (Appendix D.1).

Remark 5.2 (Fixed λ versus $\lambda\propto n$). [remark (from proved results); known] steeper end of §2.5, §4.4 So the trick needs $\lambda=\Theta(n)$: λ/n must stay above 2^{-m} (Theorem 4.7(iii),(iv)), and every schedule with $\lambda_n/n\to 0$ eventually reproduces the errors. This confirms Hänni's remark that "we will need to increase the hypothesis term coefficient compared to the sum of log likelihoods term as the data set size increases" (Hänni and Wyeth, undated). A fixed $\lambda>1$ removes noise (Theorem 6.6), not finite-complexity errors. With $\lambda=cn$ the learner discards every component whose rate is below c, valid or not (Proposition 5.6): it is deliberately inconsistent for the teacher's label distribution. Its price is a batch excess: $-\log\xi_\lambda(D_n)\leq n\,(c\,\ell(\pi)+\hat R(\pi))$ for every π, with $\hat R$ the empirical risk, so the code length exceeds π's own loss by $c\,\ell(\pi)$ bits per sample, against $\ell(\pi)/n$ at $\lambda=1$. This is not a sequential regret, since $\lambda=cn$ is not one Bayesian predictor (Remark 2.7). That a bounded λ overfits and λ must grow with the sample size is known for λ-weighted two-part MDL in noisy classification (Zhu and Srebro, 2025; Zhu et al., 2026) and is not claimed here (Remark 8.5); there, consistency means converging to the data distribution, which here contains the teacher's errors, and by (b) every schedule with $\lambda_n/n\to 0$ learns them.

5.2 Which error rules defeat the trick

Hänni's example fixes only a density and a description length, and realistic systematic errors are often low-complexity patterns. Split the rule as $S = S_1 \sqcup S_2$, $|S| = s_0 = p2^m$, with $s_i := |S_i|$, $p_1 := s_1/2^m$ and $p_2 := s_2/(2^m - s_1)$, where S_1 is *cheap*, $K(S_1 \mid m) \le k_1$; every input of $E_1 := S_1 \times \{0,1\}^{d-m}$ is an error input. The *envelope* π_e predicts $f_a \oplus 1$ with probability 1 on E_1 and f_a with flip probability p_2 elsewhere; it has risk $R_e = (1-p_1)H(p_2) < H(p)$ and length at most $d + k_1 + \kappa_e$, $\kappa_e := 2\log d + 2\log m + 2\log(s_2+1) + \kappa$. Put $J_e(c) := c(d + k_1 + \kappa_e) + R_e$, $\zeta := c(k_1 + \kappa_e + 4 + \Lambda_1) + \epsilon$ and $c_{\mathrm{env}} := (H(p) - R_e)/(k_1 + \kappa_e + 4 + \Lambda_1)$.

Proposition 5.3 (Error envelopes)**.** [proved; computed] steeper Prop 2.17 (T5 Prop 4.5, all-hypothesis form); check c7(B); referee q4(C) *Let $B = \infty$, and let $J_c(\pi) \le J_e(c) + \epsilon$ with $J_e(c) + \epsilon < \Phi_2(c)$.*

(i) $\mathrm{lev}(\pi) = 1$ *and $R(\pi) \le R_e + \zeta$. So for $c < c_{\mathrm{env}}$ and ϵ small enough that $\zeta < H(p) - R_e$, no program with $R(\pi) \ge H(p)$, the good model included, is ϵ-near-optimal.*

(ii) With $B_2 := K(S_2 \mid a, K(a))$, the average probability of reproducing the cheap errors is

$$e_1(\pi) := \mathbb{E}\big[\nu_\pi(y^*(x) \mid x) \,\big|\, x \in E_1\big] \;\ge\; 2^{-\bar{R}_1},$$
$$\bar{R}_1 := \tfrac{1}{p_1}\big(\zeta + (1-p_1)H(p_2) - \min(c, 2^{-m})\,(B_2 - K(S_1))^+\big).$$

Proof idea. (i) is Theorem 4.7(o) with (D1). For (ii), code S_2 given S_1 with the hypothesis, as in Lemma 4.5; this bounds its risk off E_1 from below, so its risk on E_1 is small (Appendix D.4).

At Hänni's sizes take $S_1 = \{z : z_1 = \cdots = z_j = 0\}$ and S_2 uniformly random. For $j = 13$ (128 cheap and 896 random points; $\log\binom{2^m - s_1}{s_2} = 10{,}418$ bits), a direct computation gives $c_{\mathrm{env}} = 1.64 \cdot 10^{-5}$, which is 31% of the logarithmic width of the window $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ of Theorem 4.7 at $\kappa = 0$ (25% at $\kappa = 30$), and $e_1 \ge 0.545$ at $c = 1.01 \cdot 2^{-m}$ ($\kappa = 0$, B_2 typical). For $j = 11$ (512+512 points, 6,364 bits): 47% of the log-width and $e_1 \ge 0.862$ (table in Appendix D.4).

So *the extension of Hänni's example to every error rule with his density and description length is false*: the rule with $j = 13$ has density 2^{-10} and about 10,400 bits, close to his "10000 bits", yet on almost a third of the window's log-width no hypothesis that fits no better than the good model is near-optimal, and near the lower end near-optimal hypotheses reproduce the cheap errors. (Its deficiency is about 1,300 bits, so Theorem 4.7 does not cover it; above c_{env} nothing is proved for it.) Hänni called the example "definitely somewhat cherry-picked for the idea to work, though I also don't consider it completely contrived" (Hänni and Wyeth, undated). What is confirmed is the example for error rules incompressible relative to the model. For structured rules the dangerous hypotheses are envelopes, not universal ones: a cheap, dense error region has a high rate and is learned like any valid component. The extreme case is a cheap, frequent rule.

Proposition 5.4 (Rate blindness)**.** [proved; computed] steeper Prop 2.11; checks c3, c6(A) *Let $B = \infty$ and $\ell(\pi_{\mathrm{coin}}) \le \kappa$. (i) If $H(p)/(\ell(\pi_b) - \ell(\pi_g)) \ge (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\mathrm{coin}}))$, then at every c the good model is beaten by the memoriser or beaten or tied by the coin. (ii) If*

$$\frac{H(p)}{\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1} \;\ge\; \frac{1 - H(p)}{d - 4 - \Lambda_1 - \kappa} \quad \text{and} \quad d - 4 - \Lambda_1 - \kappa > 0, \tag{$\star$}$$

then at every c every program with $u > \frac{3}{4}$ and $R \ge H(p)$ is beaten by the memoriser or beaten or tied by the coin. (iii) For $c < 2^{-m}$, near-optimal programs reproduce the errors, under the condition of Theorem 4.7(iii).

Lemma 5.5 (The criterion is robust in the machine constant)**.** [proved; computed] steeper Lemma 2.11a; check c6(A) *Fix m and r, let κ bound the machine constants in Λ_1, κ_b and $\ell(\pi_{\text{coin}})$, and put $H := H(2^{-r})$, $q := (1-H)/H$, $a := 2+4q$, $b := 4+2\log m+2\kappa+q(\lceil L_S\rceil+4\log m+2\log s_0+2\kappa+4)$ and $F(d) := d - a\log d - b$. (i) For $d \geq 1$, $(\star)$ holds if and only if $F(d) \geq 0$. (ii) If $b \geq a/\ln 2$, then $\{d \geq 1 : F(d) \geq 0\} = [d_0, \infty)$ for a unique real $d_0 = d_0(\kappa)$. (iii) $d_0(\kappa)$ grows linearly in κ, with slope $(2+2q)(1+o(1))$.*

For a single 3-bit prefix as error rule ($m = 3$, $p = 1/8$), a direct computation gives $d_0 = 49$ at $\kappa = 0$ and 169 at $\kappa = 30$; for a single 6-bit prefix, 450 and 1005 (Appendix D.2). So for each fixed cheap rule and each κ, once the simple model is long enough, no program that knows the parity and fits no better than noise is optimal at any c. Hänni's 11,710-bit rule, by contrast, leaves a window of a factor of about 8000 ($\kappa = 0$) or 3800 ($\kappa = 30$).

5.3 Other pathologies

Proposition 5.6 (Separable components)**.** [proved; computed]
steeper Prop 2.12 (T5 Thm 3.2, Thm 4.1); check c5(a) *Let the inputs split into regions $X_1, \ldots, X_k$ and the hypotheses be products of per-region choices with additive code lengths. Each region selects a vertex of its own hull for slope c (Lemma 2.8). If region j offers a default or a component j of extra length k_j and population risk reduction r_j, component j is selected if and only if $r_j/k_j > c$ (ties arbitrary). So a valid component whose rate is at most an error component's rate cannot be kept without the error, except at a common tie value. Two worlds with the same teacher channel and input law, one where a component is an error and one where it is a rare valid rule, get the same selection at every c.*

The steeper penalty sorts components by *rate*, risk reduction per bit, not by validity. Three consequences follow (proofs in Appendix D.3).

Proposition 5.7 (Guard erosion)**.** [proved; computed] steeper Prop 2.13 (T5 Prop 4.4); checks c5(b), c6(G), c7(E); referee q1(I) *In a region of contexts a schema σ is one of n_{st} possible steps, and a guard G splits the region into G-contexts (input mass $\mu(G)$), where the teacher plays σ with probability $1 - \epsilon_G + \epsilon_G/n_{\text{st}}$ and every other step with probability ϵ_G/n_{st}, and $\neg G$-contexts (mass $\mu(\neg G)$), where it plays one fixed step other than σ. Over a common base, "none" is uniform (length 0), "unguarded σ" predicts the G-context law everywhere (length k_σ), and "guarded σ" does so in G-contexts and is uniform over the other $n_{\text{st}} - 1$ steps elsewhere (length $k_\sigma + k_G$). With H_G the entropy of the G-context law, $g := \log n_{\text{st}} - H_G$ and $\Delta_G := \log(n_{\text{st}}/\epsilon_G) - \log(n_{\text{st}} - 1)$, put $\rho_\sigma := (\mu(G)g - \mu(\neg G)\log(1/\epsilon_G))/k_\sigma$ and $\rho_G := \mu(\neg G)\Delta_G/k_G$. If $\rho_G < \rho_\sigma$, then for every $c \in (\rho_G, \rho_\sigma)$ the unguarded rule is the unique selection, and it applies σ where the teacher never did. Otherwise the selection goes from guarded to none.*

With $n_{\text{st}} = 64$, $\epsilon_G = 0.05$, $\mu(G) = 0.05$, $\mu(\neg G) = 5 \cdot 10^{-4}$, $k_\sigma = 22$ and $k_G = 12$, a direct computation gives $\rho_\sigma = 1.2226 \cdot 10^{-2}$ and $\rho_G = 1.8103 \cdot 10^{-4}$: the unguarded rule wins on a factor of 67 in c.

Proposition 5.8 (λ cannot be chosen from imitation data)**.** [proved; computed; (i) known] steeper Prop 2.14 (T5 Thm 4.3); check c5(c) *(i) Every strictly proper scoring rule has expected score on the teacher's data uniquely maximised by the teacher's channel $P^*(\cdot \mid x)$, errors included (Gneiting and Raftery, 2007), so validation by such a score prefers the teacher's channel to the intended model. (ii) Along the population selection path the held-out log loss $R(h_c)$ is nonincreasing as c*

decreases (Lemma 2.8(ii)), so log-loss validation over a grid of values of c is minimised at the smallest, error-memorising, value; in Hänni's example the losses are 1.0, 0.0112, 0.0112, 0 *at* $c = 10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}$. *(iii) In the twin worlds of Proposition 5.6, every rule that chooses c from imitation data alone is wrong in one of them.*

Proposition 5.9 (The selection depends on the input law)**.** [proved] steeper Prop 2.15 *In the setting of Proposition 5.6, let the inputs have law μ and let Δ_j be component j's expected log-loss reduction per input of X_j. Then component j is selected if and only if $\mu(X_j)\Delta_j > c\,k_j$. So the window that keeps a valid component j and drops an error component j' is $c \in (\mu(X_{j'})\Delta_{j'}/k_{j'},\ \mu(X_j)\Delta_j/k_j)$; it moves with μ and is empty once $\mu(X_{j'})/\mu(X_j) \geq \Delta_j k_{j'}/(\Delta_{j'} k_j)$, and after a shift to μ' a c chosen in μ's window keeps the error if and only if $\mu'(X_{j'})\Delta_{j'} > c\,k_{j'}$.*

Theorem 4.7's window depends on the input law by the same mechanism; this is proved here only in the separable case.

5.4 Unseen properties

For M_F, a simple property absent from the training data gets a constant probability of bad behaviour, $2^{-K(Q,b)-O(1)}$, where $K(Q,b)$ is the length of a shortest self-delimiting code for a decider of a set Q and a program for a behaviour b (Hänni, 2026a, Hänni's proposition, with no log term). The steeper counterpart is as follows (proof in Appendix D.5).

Proposition 5.10 (Steeper simplicity and unseen properties)**.** [proved] steeper Prop 2.16 *In $\mathcal{H}_\times$, let $Q \subseteq X$ be decidable, b partial computable, D a data set with* $\operatorname{dom} D \cap Q = \emptyset$, *and $Z \subseteq Q \cap \operatorname{dom} b$ finite. For every $\lambda \geq 1$: (i) the programs that answer $b(x)$ on Q and run another program elsewhere carry posterior mass exactly $2^{-\lambda(K(Q,b)+\kappa)}$, so $\xi_\lambda(D \cup b|_Z) \geq 2^{-\lambda(K(Q,b)+\kappa)}\xi_\lambda(D)$; (ii) if $\nu_{\pi_1}(D) = \nu_{\pi_2}(D) > 0$, then $\Pi_\lambda(\pi_1 \mid D)/\Pi_\lambda(\pi_2 \mid D) = 2^{-\lambda(\ell(\pi_1)-\ell(\pi_2))}$.*

At $\lambda = 1$, (i) is Hänni's proposition for M_{prod}. With $\lambda = cn$ the bound vanishes, so steeper simplicity removes his constant lower bound, but only through (ii): the posterior concentrates on the shortest hypotheses that fit the training data, whatever they do on Q. If the shortest one extrapolates well, steeper SFI is more reliable out of distribution than SFI; if a shorter one fits the training data and is wrong on Q, it is more confidently wrong. No guarantee about unseen properties follows.

Remark 5.11 (A limitation of the method)*.* [remark; computed] steeper end of §2.6; check c7(C) The window needs $G_1 > H(p)$, that is, $p < 0.056$. The sharper floor of Lemma C.1 gives the kink for every $p < 1/4$ (Proposition 7.6), but does not extend the window as the argument stands: with it and no deviation term the level-2 floor $1 - \log(3/2 - p)$ exceeds $H(p)$ only for $p < 0.123$; and a program of level at least 2 has $I(\pi)$ bounded only through its length, which near $c = 2^{-m}$ allows $I(\pi) \approx L_S$ and a deviation term of order $\sqrt{H(p)/p}$. A window for $p < 0.123$ and c well above 2^{-m} looks within reach but is not proved.

Open question. Which structured error rules leave a window? For envelopes with error density $\theta < 1$ in the cheap region only part (i) of Proposition 5.3 carries over; whether near-optimal hypotheses memorise the random part S_2, and what is selected above c_{env}, are open. A theory for realistic, low-complexity human errors is the main gap between the trick and Hänni's motivation. Is there a principled, data-free choice of c? By Propositions 5.6 and 5.8 imitation data alone cannot tell errors from rare valid rules; what extra information would suffice is open.

6 The model–randomness split

Hänni has suggested that a steeper prior might give "a preference for a simple sampler over a model where the sampled bits are filled in (solomonoff doesn't have a preference here, both get similar posterior)" (Hänni, 2026c). This section makes both halves precise. At $\lambda = 1$ the sampler and the programs that fill in its sampled bits get similar posterior, up to explicit factors that do not depend on how much randomness was sampled (Theorem 6.4 and Proposition 6.5). At $\lambda > 1$ with private randomness the sampler wins in every countable product class (Theorem 6.6). With shared randomness, or in SI, the universal mixture wins instead and no split is made (Theorem 6.8). Proofs are in Appendix E.

6.1 Setting

Definition 6.1 (Samplers and fill-ins)**.** steeper §3.1 A *sampler* is a computable conditional measure ν ($\nu(y \mid x)$ computable uniformly in (x, y), $\sum_y \nu(y \mid x) = 1$) with a program π_ν of length k in the class considered. Distinct inputs $x_1, x_2, \ldots$ are chosen by any rule, possibly adaptive, randomised or non-computable, and then $y_i \sim \nu(\cdot \mid x_i)$; $\mathbb{P}$ is the joint law and D_n the first n pairs. The *sampled randomness* is $-\log \nu(D_n)$. *Fill-ins* are deterministic programs consistent with D_n; in a product class their total weight is $W_{\mathrm{det}}(D) := \sum\{2^{-\ell(\pi)} : \pi \text{ deterministic}, \ \nu_\pi \supseteq D\}$.

The first lemma bounds how much any program, or the whole mixture, can gain over the sampler on its own data.

Lemma 6.2 (Ville deficiency)**.** [proved; computed; Ville's inequality known] steeper Lemma 3.1; check c4(A); referee q3(B) *Let ξ be a shared hypothesis with $\xi(\emptyset) \le 1$, for example the mixture $\xi_{\mathcal{C}}$ of a countable product class $\mathcal{C}$ with $\sum_{\pi \in \mathcal{C}} 2^{-\ell(\pi)} \le 1$ (for $\mathcal{H}_\times$ this is M_{prod}). Let $\delta^* := \sup_n \log\big(\xi(D_n)/\nu(D_n)\big)$. Then $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$ for all $j \ge 0$. Consequently, for $\xi = \xi_{\mathcal{C}}$, every $\pi \in \mathcal{C}$ and every n,*

$$\nu_\pi(D_n) \le 2^{\ell(\pi)+\delta^*}\, \nu(D_n).$$

Adaptive inputs are allowed.

Proof idea. $\xi(D_n)/\nu(D_n)$ is a nonnegative supermartingale with initial value at most 1, by the one-point inequality; apply Ville's maximal inequality (Ville, 1939) (Appendix E.1).

The second lemma shows that fill-ins are cheap: a deterministic program can store a finite piece of each input's private random stream. This adapts the tape-length bound of Hänni (2026a) to product classes and to every computable sampler.

Lemma 6.3 (Fill-ins from stored stream prefixes)**.** [proved; computed]
steeper Lemma 3.2′ (adapts foundations Prop 4.4); check c7(A) *Let the class be $\mathcal{H}_\times$, let ν be a sampler with a program π_ν of length k, and let D be a data set with n inputs, all of length at most d, with $\beta_D := \min_{(x,y)\in D} \nu(y \mid x) > 0$. Put $i_D := \lceil \log(6n/\beta_D) \rceil$ and $t_D := \lceil \log(2d + 2 + 2\log(i_D + 1)) \rceil$. Then*

$$W_{\mathrm{det}}(D) \ \ge \ 2^{-k - 2\log(t_D+1) - \kappa}\, \nu(D).$$

Since $\beta_D \ge \nu(D)$, $i_D \le \log(6n) + \log(1/\nu(D)) + 1$; so the cost $2\log(t_D + 1)$ is doubly logarithmic in d and triply logarithmic in n and in the sampled randomness $-\log \nu(D)$.

Proof idea. Give each input its own stream of tape positions and read a label from a stream prefix as soon as its dyadic interval certifiably lies inside one label's interval of $\nu(\cdot \mid x)$. Each stored tape w of length 2^{2^t} gives one deterministic program; for uniform w the streams are independent

and certify the right labels with probability at least $\nu(D)/2$, and the only cost beyond π_ν is a code of t (Appendix E.2). A direct computation (code in the accompanying repository) gives $2\log(t_D+1) = 5.6$ bits at $d=20$, $n=10^3$, and 6.3 bits at $d=100$, $n=10^6$. The lemma needs a computable *measure*: for a semimeasure the label boundaries are not computable. For dyadic samplers a hashing argument gives a weaker bound (Lemma E.1).

6.2 $\lambda = 1$: the sampler and its fill-ins get similar posterior

For a run of the data process put $d_n := \max_{i\le n}|x_i|$, $\beta_n := \min_{i\le n}\nu(y_i \mid x_i)$, $i_n := \lceil \log(6n/\beta_n)\rceil$ and $t_n := \lceil \log(2d_n + 2 + 2\log(i_n+1))\rceil$. Let $\Pi_1(\det \mid D_n)$ be the posterior mass of all deterministic programs.

Theorem 6.4 (SFI at $\lambda = 1$)**.** [proved] steeper Thm 3 *In the setting of Definition 6.1 with the class $\mathcal{H}_\times$, for every sampler ν with a program of length k, simultaneously for all n:*

(i) $\Pi_1(\pi_\nu \mid D_n) \ge 2^{-k-\delta^*}$;

(ii) $\Pi_1(\det \mid D_n) = W_{\det}(D_n)/\xi_\times(D_n) \ge 2^{-k-2\log(t_n+1)-\kappa-\delta^*}$;

(iii) the posterior odds of all fill-ins against the sampler program lie in $\left[2^{-2\log(t_n+1)-\kappa},\, 2^{k+\delta^*}\right]$, *whatever the sampled randomness* $-\log\nu(D_n)$;

(iv) for a sampler whose values are multiples of 2^{-r}, *Lemma E.1 gives the alternative lower ends* $2^{-k-2\log(n+1)-2\log(d_n+1)-2\log(r+1)-\kappa-\delta^*}$ *in (ii) and* $2^{-2\log(n+1)-2\log(d_n+1)-2\log(r+1)-\kappa}$ *in (iii).*

Proof. $\Pi_1(\pi_\nu \mid D_n) = 2^{-k}\nu(D_n)/\xi_\times(D_n)$ and $\Pi_1(\det \mid D_n) = W_{\det}(D_n)/\xi_\times(D_n)$. Bound $\xi_\times(D_n) \le 2^{\delta^*}\nu(D_n)$ by Lemma 6.2, $W_{\det}(D_n)$ from below by Lemma 6.3 (or Lemma E.1 for (iv)) and from above by $\xi_\times(D_n)$. □

The lower ends of (ii) and (iii) hold for every data set, not only for samples from ν, and cover non-dyadic samplers: one that gives every label probability at least $1/3$, such as the Bern(1/3) threshold hypotheses in Sam Eisenstat's coin example (Hänni, 2026c,a), has $i_n \le \lceil\log(18n)\rceil$. The hashing bound of (iv) is weaker at realistic sizes (32 to 143 bits; Appendix E.3).

Proposition 6.5 (SI at $\lambda = 1$)**.** [proved; (iii) known]
steeper Prop 3.3 *Let a string $z_{1:n}$ be drawn from a computable measure ν with a program $\pi_\nu \in \mathcal{H}_{\mathrm{SI}}$ of length k, and let $\delta^* := \sup_n \log\left(M(z_{1:n})/\nu(z_{1:n})\right)$, where $M = \xi_1$ is the mixture of $\mathcal{H}_{\mathrm{SI}}$. (i) $\Pi_1(\pi_\nu \mid z_{1:n}) \ge 2^{-k-\delta^*}$ for all n, and $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$. (ii) The deterministic hypotheses consistent with $z_{1:n}$ (point masses on computable infinite extensions) have total weight at least $2^{-k-K(n)-\kappa}\nu(z_{1:n})$. (iii) In the deterministic-program representation of M every hypothesis is a fill-in; the sampler appears only as the programs "decoder for ν, then a code of z", of total weight at least $2^{-k-\kappa}\nu(z)$.*

For (ii) the point mass on $z_{1:n}0^\infty$ suffices; (iii) is the usual proof that $M \ge 2^{-K(\nu)-O(1)}\nu$ (Li and Vitányi, 2008) (Appendix E.4). So Hänni's "both get similar posterior" holds in SFI and in SI, up to factors 2^k, $2^{2\log(t_n+1)+\kappa}$ (SFI) or $2^{O(k+K(n))}$ (SI), and the Ville deficiency, whatever the amount of sampled randomness.

6.3 $\lambda > 1$ with private randomness: the sampler wins

Call a program π *ν-equivalent on the run* if $\sum_n d_{\mathrm{H}}^2(\nu(\cdot \mid x_n), \nu_\pi(\cdot \mid x_n)) < \infty$, where $d_{\mathrm{H}}^2(P,Q) := 1 - \sum_y \sqrt{P(y)Q(y)} \in [0,1]$, also for a sub-probability Q.

Theorem 6.6 (Private randomness, $\lambda > 1$)**.** [proved; computed; Kakutani-type argument known] steeper Thm 4; check c4(C); referee r4, q3(C) *Let $\mathcal{C}$ be a countable class of product programs with $\sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} \le 1$ that contains π_ν (for example $\mathcal{H}_\times$, or a time-bounded or circuit subclass). Fix $\lambda > 1$ and define δ^* from $\xi_{\mathcal{C}}$. Simultaneously for all n:*

(i) MAP. *Every λ-MAP $\hat\pi_n$ has $\ell(\hat\pi_n) \le (\lambda k+\delta^*)/(\lambda-1) = k+(k+\delta^*)/(\lambda-1)$ and $-\log\nu(D_n) - \ell(\hat\pi_n) - \delta^* \le -\log\nu_{\hat\pi_n}(D_n) \le -\log\nu(D_n) + \lambda k$.*

(ii) Length tail. $\Pi_\lambda(\{\pi : \ell(\pi) > L\} \mid D_n) \le 2^{\lambda k+\delta^*-(\lambda-1)L}$ *for every L.*

(iii) Fill-ins. *Every deterministic program consistent with D_n has $\ell \ge -\log\nu(D_n) - \delta^*$; $\Pi_\lambda(\det \mid D_n) \le 2^{\lambda k+\lambda\delta^*-(\lambda-1)(-\log\nu(D_n))}$; and for the tempered posterior with $\eta = 1/\lambda$, $\Pi^\eta(\det \mid D_n) \le 2^{k+\delta^*-(1-\eta)(-\log\nu(D_n))}$.*

(iv) Identification. *Suppose the input sequence is infinite. Then, almost surely, from some n on every λ-MAP is ν-equivalent on the run, and $\Pi_\lambda(\nu\text{-equivalent on the run} \mid D_n) \to 1$.*

Proof idea. (i)–(iii) compare with the sampler program, $J_\lambda(\pi_\nu) = \lambda k - \log\nu(D_n)$, and bound every program's likelihood by Lemma 6.2. For (iv), the likelihood ratio $\Lambda_n := \nu_\pi(D_n)/\nu(D_n)$ of a program that is not ν-equivalent on the run tends to 0 almost surely, by the martingale argument behind Kakutani's dichotomy (Kakutani, 1948), which holds for adaptive inputs (Appendix E.5).

Remark 6.7 (What the theorem says, and what it does not). [remark] steeper §3.3 This is the split Hänni hoped for, for every fixed $\lambda > 1$, in a precise and limited sense. The selected model is a sampler, not a fill-in, but not necessarily ν: it may carry up to $(k + \delta^*)/(\lambda - 1)$ bits beyond the sampler, for example memorised noise. The fill-ins' posterior is exponentially small in the sampled randomness, but positive. "ν-equivalent on the run" is not equality of predictive laws and says nothing about unseen inputs. On a finite domain such as $\{0, 1\}^d$, (iv) is vacuous.

6.4 $\lambda > 1$ with shared randomness: no split

Theorem 6.8 (SI, or SFI with shared randomness, $\lambda > 1$)**.** [proved; computed] steeper Thm 5; check c4(C); referee r4, q4(A) *Assume (Dom) and $\xi_1 \le 2^{\kappa_1}U$, let $\lambda > 1$, and let π_ν be a program of length k for a sampler in the class. For every observation o:*

(i) if $(\lambda-1)k > \lambda\ell_U + \kappa_0$, then $J_\lambda(\pi_U; o) < J_\lambda(\pi_\nu; o)$: the sampler program is never a λ-MAP, and every λ-MAP has length at most $\ell^*_\lambda < k$;

(ii) for every set A of programs, $\Pi_\lambda(A \mid o) \le 2^{-(\lambda-1)\min_{\pi\in A}\ell(\pi)+\lambda\ell_U+\kappa_1}$. In particular the power posterior puts at most $2^{-(\lambda-1)L+\lambda\ell_U+\kappa_1}$ on all programs of length at least L.

For SI with adaptive or non-computable inputs, (Dom) in the needed form is supplied by $\mathcal{H}_{\mathrm{chr}}$ with $U = M_H$, not by $\mathcal{H}_{\mathrm{SI}}$; with i.i.d. computable inputs $\mathcal{H}_{\mathrm{SI}}$ with $U = M$ works.

Proof idea. (i) compares π_U with π_ν using (Dom); (ii) bounds the numerator by $2^{-(\lambda-1)\min_A\ell}\xi_1(o)$ and the normaliser by $2^{-\lambda\ell_U}U(o)$ (Appendix E.6).

Compare Theorem 6.6. In a shared class the steeper posterior lives, up to mass 2^{-s}, on programs of length at most $(\lambda\ell_U + \kappa_1 + s)/(\lambda - 1)$, *whatever the data*; in the product class it concentrates on samplers ν-equivalent on the run, whose length is unbounded over ν. Predictively both converge to ν's predictions (in $\mathcal{H}_F$ by the merging theorem of Hänni, 2026a). The difference is *what the selected hypothesis is*: a model plus private noise, or a short universal learner that contains no model of ν.

Example 6.9 (Noisy parity, full table: an illustration)**.** [computed; illustration] steeper Ex 3.4; check c4(C); referee q1(J) Let $d = 12$ and let the data be the full table ($N = 4096$ inputs) of a random parity with i.i.d. flips at rate 1/8 (509 flips). A small product class contains the coin, parity-plus-noise samplers ($\ell = 24.2$ bits) and the memoriser (parity plus flip set); a "shared class" adds the finite Bayes mixture U over all parities and a grid of noise rates, with an *assigned* length $\ell_U = 12$. A direct computation (code in the accompanying repository) gives: the product-class MAP is the sampler at $\lambda = 1, 1.1, 1.5, 2, 3, 6$, and at $\lambda = 1$ the memoriser is only 8.2 bits worse, a tie up to $O(\log N)$; the shared-class MAP is the sampler for $\lambda < 1.315$ and U above, the threshold being $(-\log U(D) + \log \nu(D))/(k - \ell_U)$ (the rounded values 2234.0, 2218.0, 24.2, 12 give 1.311). Since $\ell_U = 12$ is assigned and this finite class does not satisfy (Dom), the example illustrates the mechanism of Theorem 6.8; it is not an instance of it.

Remark 6.10 (No split in SI's program representation). [proved] steeper Rem 4.3 Solomonoff's M is a mixture over deterministic programs. Tempering does not change deterministic hypotheses (Lemma 2.6(iii)), and a power prior $2^{-\lambda|p|}$ charges sampled bits and model bits alike, so it is indifferent between "decoder for ν plus a code of the randomness" and a literal fill-in of the same length. The split needs a representation in which the randomness is not part of the program: a product hypothesis with private coins.

7 Relation to algorithmic statistics

Hänni's note proposes a second way to find the split between model and noise: "look at how many bits of entropy are left given the hypothesis, like imagine this graphed as a function of hypothesis length, and like see some point at which the derivative changes or sth. (this doesn't show up in the seq case because there it's pretty much just 1 bit paying for 1 bit (until you specify it in full if it's finite complexity))" (Hänni, 2026b). For set models this graph is Kolmogorov's structure function. We define a product version and show, for a noisy random parity, that the two place the split about d bits apart. Proofs are in Appendix F.

7.1 Set models and steeper two-part codes

Definition 7.1 (Set models)**.** steeper §3.5, Definitions For a string z, a *set model* is a finite set $A \ni z$, with two-part length $K(A) + \log|A|$. The *structure function* is $h_z(\alpha) := \min\{\log|A| : A \ni z,\ K(A) \le \alpha\}$. The *sufficiency line* is the bound $K(A) + \log|A| \ge K(z) - \kappa$, valid for every $A \ni z$ since $K(z \mid A) \le \log|A| + \kappa$. A is *b-sufficient* if $K(A) + \log|A| \le K(z) + b$; a *minimal sufficient statistic* minimises $K(A)$ among b-sufficient models; and the *set sophistication* is $\mathrm{soph}_b^{\mathrm{set}}(z) := \min\{K(A) : A \ni z,\ K(A) + \log|A| \le K(z) + b\}$. Conditional versions given a string w (for example a domain) replace K by $K(\cdot \mid w)$.

These notions are standard (Gács et al., 2001; Vereshchagin and Vitányi, 2004; Vereshchagin and Shen, 2017); we use them as stated. Sophistication goes back to Koppel (1987); Koppel and Atlan (1991), with total programs as models; see also Vitányi (2006) and Antunes and Fortnow (2009). We do not use the equivalences between the versions.

Proposition 7.2 (Steeper two-part codes over sets)**.** [proved; a natural reading refuted] steeper Prop 3.5 *Let $\lambda > 1$ and let A_λ minimise $\lambda K(A) + \log|A|$ over $A \ni z$. Then, for every $b \ge 0$,*

$$K(A_\lambda) \le \mathrm{soph}_b^{\mathrm{set}}(z) + \frac{b + \kappa}{\lambda - 1},$$

and A_λ is $\big(b+(\lambda-1)\operatorname{soph}_b^{\mathrm{set}}(z)\big)$-sufficient. The two bounds trade off: as $\lambda \to 1$ the first becomes vacuous, and as λ grows the sufficiency deficiency grows. So the natural reading "λ slightly above 1 selects an approximately minimal sufficient statistic" does not follow: near $\lambda = 1$ the length bound says nothing about minimality. The proposition is useful at $\lambda - 1 = \Theta(1)$ when the sophistication is small.

Proof. Let A_b attain $\operatorname{soph}_b^{\mathrm{set}}(z)$. Then $\lambda K(A_\lambda)+\log|A_\lambda| \le \lambda K(A_b)+\log|A_b| \le (\lambda-1)\operatorname{soph}_b^{\mathrm{set}}(z)+K(z)+b$. Subtract the sufficiency line for the first claim; use $K(A_\lambda) \le \lambda K(A_\lambda)$ for the second. A minimiser exists because only finitely many A have $K(A) \le L$. □

7.2 Product models

Definition 7.3 (Product structure function and sophistication)**.** steeper §3.5, product versions For a data set D and the class $\mathcal{H}_\times$, the *product structure function* is $\rho_D^\times(\alpha) := \min\{-\log \nu_\pi(D) : \ell(\pi) \le \alpha\}$; π is *product-b-sufficient* if $\ell(\pi) - \log \nu_\pi(D) \le K(y_D \mid \operatorname{dom} D) + b$; and the *product sophistication* is $\operatorname{soph}_b^\times(D) := \min\{\ell(\pi) : \pi \text{ product-}b\text{-sufficient}\}$. Lemma 2.9 is the product sufficiency line. Product notions are conditional on the domain by construction, since product hypotheses do not model the inputs.

Proposition 7.4 (Steeper two-part codes over products)**.** [proved] steeper Prop 3.6 *Every λ-MAP over $\mathcal{H}_\times$ for D has length at most $\operatorname{soph}_b^\times(D) + (b+\kappa)/(\lambda-1)$ and is $\big(b+(\lambda-1)\operatorname{soph}_b^\times(D)\big)$-product-sufficient.*

Proof. As for Proposition 7.2, with Lemma 2.9 as the sufficiency line. The same trade-off applies. □

Set models are at least as coarse as product models, at least for computable hypotheses.

Proposition 7.5 (Sets from products)**.** [proved for computable hypotheses; that it is the standard set–distribution translation is believed (known; unverified)]
steeper Prop 3.7 *Let π be product-b-sufficient for D, and let ν_π be a computable conditional measure with rational values. Put $\rho := \lceil -\log \nu_\pi(D) \rceil$ (of order n) and $A := \{y' \in Y^{\operatorname{dom} D} : \nu_\pi(\operatorname{dom} D \mapsto y') \ge 2^{-\rho}\}$. Then $y_D \in A$, $K(A \mid \operatorname{dom} D) \le \ell(\pi) + 2\log(\rho+1) + \kappa$, and A is $(b+2\log(\rho+1)+\kappa)$-sufficient conditionally on $\operatorname{dom} D$: $K(A \mid \operatorname{dom} D) + \log|A| \le K(y_D \mid \operatorname{dom} D) + b + 2\log(\rho+1) + \kappa$. Consequently $\operatorname{soph}_{b'}^{\mathrm{set}}(y_D \mid \operatorname{dom} D) \le \operatorname{soph}_b^\times(D) + 2\log(\rho+1) + \kappa$ with $b' = b + 2\log(\rho+1) + \kappa$, provided a computable rational product hypothesis attains $\operatorname{soph}_b^\times(D)$. Against the unconditional set sophistication of the string y_D, add $K(\operatorname{dom} D)$.*

A is computable from $(\pi, \operatorname{dom} D, \rho)$ and has at most 2^ρ elements (Appendix F.1). Vereshchagin and Shen (2017) state that set and distribution models are equivalent up to logarithmic terms; the thresholding above is presumably that translation.

Open question. Does Proposition 7.5 hold for lower-semicomputable product hypotheses? For them the set A is only c.e.

7.3 A noisy random parity

We use the teacher of Definition 4.2 with $m = d$ (the whole truth table, idiosyncratic errors), and restate what is needed. Let $N = 2^d$, $p = 2^{-r}$ with $2 \le r \le d$, $a \in \{0,1\}^d$ with $K(a) \ge d$, $f_a(x) = \langle a, x\rangle \bmod 2$, and let $E \subseteq \{0,1\}^d$ have $|E| = pN$. The teacher is $y^* = f_a \oplus \mathbf{1}_E$; let

$z = y^* \in \{0,1\}^N$ be its table and D the full data set, so $\operatorname{dom} D = \{0,1\}^d$ and $K(y_D \mid \operatorname{dom} D) = K(z \mid d) \pm \kappa$. Put $L_E := \log \binom{N}{pN}$ and $\delta_E := L_E - K(E \mid a, K(a))$. For a product program π let $R(\pi) := -\frac{1}{N} \log \nu_\pi(D)$ (the risk under uniform inputs), $u(\pi) := \mathbb{E}_x \nu_\pi(f_a(x) \mid x)$, and let $\operatorname{lev}(\pi) := \min\{t \geq 1 : u(\pi) > \frac{1}{2} + 2^{-t-1}\}$ (∞ if $u(\pi) \leq \frac{1}{2}$). Put $s(t) := 2t + 2\log(t+1)$, $G_1 := 1 - \log(\frac{3}{2} + 2p)$, $\Lambda_1 := 4\log d + \kappa$, $\kappa_g := 2\log d + 2\log r + \kappa$ and $\kappa_b := 4\log d + 2\log(pN) + \kappa$; the good model π_g (the parity with flip probability p) has $\ell(\pi_g) \leq d + \kappa_g$, and the memoriser π_b (the table itself) has $\ell(\pi_b) \leq d + \lceil L_E \rceil + \kappa_b$. We use three facts from Section 4: list decoding (Lemma 4.3), the risk floor (Lemma 4.4: $\operatorname{lev}(\pi) \geq 2$ implies $R(\pi) \geq G_1$) and the length bound (D1) of Lemma 4.6: $\operatorname{lev}(\pi) = t < \infty$ implies $\ell(\pi) \geq d - s(t) - \Lambda_1$. Part (iv) also uses the sharper floor Lemma C.1.

Proposition 7.6 (The two splits differ by about d bits)**.** [proved; computed] steeper Prop 3.8 (T5 Thm 2.5); checks c2(c), c4(C), c7(C) *Let $p < 1/4$.*

(i) Sets: the parity is noise. *$A_0 := \{f_{a'} \oplus \mathbf{1}_{E'} : a' \in \{0,1\}^d,\ |E'| = pN\}$ has $|A_0| = 2^{d+L_E}$ and $K(A_0) \leq 2\log d + 2\log r + \kappa$, and $K(z) \geq d + L_E - \delta_E - \kappa$. So A_0 is $(\delta_E + 2\log d + 2\log r + \kappa)$-sufficient, and $\operatorname{soph}_b^{\mathrm{set}}(z) \leq 2\log d + 2\log r + \kappa$ for every such b. The structure function lies on the sufficiency line from $\alpha = O(\log d)$ on, within $\delta_E + O(\log d)$. Every λ-optimal set model A_λ with $\lambda > 1$ holds at most $O(\log d) + (\delta_E + O(\log d))/(\lambda - 1)$ bits of information about a.*

(ii) Products: the parity is model. *Assume also $G_1 > H(p)$ (true for $p < 0.056$). For $\delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa \leq b < N(G_1 - H(p)) - d - 2\log N - \kappa$,*

$$d - 4 - \Lambda_1 \ \leq\ \operatorname{soph}_b^{\times}(D) \ \leq\ d + \kappa_g,$$

and $\rho_D^{\times}(\alpha) \geq NG_1$ for every $\alpha < d - 4 - \Lambda_1$: the product structure function has a kink at $\alpha \approx d$.

(iii) The gap $\operatorname{soph}^{\times} - \operatorname{soph}^{\mathrm{set}} \geq d - O(\log d)$ is unbounded in d.

(iv) Every flip rate below 1/4. *Drop the condition $G_1 > H(p)$; let $p = |E|/N < 1/4$ have small complexity $K(p)$, with $\kappa_g := 2\log d + K(p) + \kappa$. For integers $t, g \geq 1$ let $\gamma_d := \sqrt{j_d \ln 2/(2pN)}$, where j_d is the least integer with $j_d - 2\log(j_d + 1) > \delta_E + d + 1 + 2\log d + 2\log(pN) + g + 2\log g + \kappa$, and*

$$F_t := 1 - \log\Big(1 + (1-2p)2^{-t} + 2p\big(2\gamma_d + 2^{1-g}\big)\Big).$$

Every product program with $\ell(\pi) < d - s(t) - \Lambda_1$ has $R(\pi) \geq F_t$, so $\rho_D^{\times}(\alpha) \geq NF_t$ for $\alpha < d - s(t) - \Lambda_1$. Moreover $d - s(t) - \Lambda_1 \leq \operatorname{soph}_b^{\times}(D) \leq d + \kappa_g$ for every b with

$$\delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa \ \leq\ b \ <\ NF_t - d - \lceil L_E \rceil - \kappa_b - 2\kappa.$$

For fixed $p < 1/4$ and $\delta_E = O(d)$, $\gamma_d \to 0$ as $d \to \infty$, and with g growing slowly $F_t \to 1 - \log(1 + (1-2p)2^{-t})$, which exceeds $H(p)$ for some fixed $t = t_p$. So the admissible range of b has length of order $N(F_{t_p} - H(p))$, product sophistication is $d \pm O(\log d)$, and the gap of (iii) holds for every $p < 1/4$.

Proof idea. (i) Minimum-distance decoding recovers (a, E) from z, since two parities differ on $N/2 > 2pN$ inputs, so $K(z) \geq d + L_E - \delta_E - \kappa$, while A_0 is described by d and r. (ii) The good model is product-sufficient; a program with $\operatorname{lev}(\pi) \geq 2$ pays NG_1, more than the memoriser's two-part length; so a product-sufficient program has $\operatorname{lev}(\pi) = 1$, and (D1) bounds its length.

(iv) uses the sharper floor (Appendix F.2). A direct computation (code in the accompanying repository; $\kappa = 0$ as an illustration, $\delta_E = 0$) gives, at $d = 100$ and $p \in \{1/8, \dots, 15/64\}$, $\mathrm{soph}_b^{\times}(D) \in [d - 33.7,\ d + 19.3]$ to $[d - 33.7,\ d + 26.9]$ for b from about 140 up to about 10^{28} (Table 3).

Remark 7.7 (Restricted model families). [known; unverified (content per abstracts); remark]
steeper §3.5, relation to restricted model families That restricting the model family moves the split is known. Vereshchagin and Vitányi (2010) study Hamming balls and other distortion balls containing the data, and Milovanov (2016) restricts models to Vereshchagin's strong models. For the full-table parity the Hamming ball of radius pN around f_a is a set model of complexity about d and log-size about $NH(p)$, "parity plus i.i.d. flips" as a set; so the kink of $\rho_D^{\times}$ at $\alpha \approx d$ is the same phenomenon as a kink of the Hamming-ball-restricted curve at the complexity of the centre. As far as we can tell from the abstracts, what the product view adds is (a) a family defined by an operational constraint, private randomness per input, for general lsc product models on partial data sets; (b) its role as the hypothesis class of steeper SFI (Theorem 3.2 versus Corollary 3.5; Theorems 4.7 and 6.6); and (c) structured, non-i.i.d. noise with identification bounds (Theorem 4.7). A comparison with the full texts is left open.

Remark 7.8 (Reading: one bit paying for one bit). [remark]
steeper §3.5, Reading With shared randomness (a set model, or any shared model) a uniformly random d-bit parameter costs the same in the model as in the data-to-model code: Hänni's "1 bit paying for 1 bit". So the minimal sufficient statistic treats the parity's parameter as noise. With private randomness, not knowing the parameter costs about one bit *per input*, so the product split keeps it in the model, and the derivative of $\rho_D^{\times}$ changes at $\alpha \approx d$, as Hänni expected. For supervised learning the product split is the relevant one, because "the model" is the part of the description reused across inputs. This is also why steeper simplicity is useless over set models in the parity example (Proposition 7.6(i)) and decisive over product models (Theorem 4.7 and Corollary 4.10).

8 Computable versions and related work

8.1 Computable classes

The lower bounds of Section 4 use only coding properties of the class, so they survive in resource-bounded classes (proofs in Appendix G).

Proposition 8.1 (Computable product classes)**.** [proved; the examples are remarks] steeper Prop 4.4 *Let $\mathcal{H}'$ be a hypothesis class of product hypotheses (Definition 2.4). (a) Lemmas 4.3 to 4.6 and the lower-bound parts of Theorem 4.7 ((o), the bound on Φ in (i), (ii), and the level and R_B statements of (iii)) hold for every $\pi \in \mathcal{H}'$, with κ replaced by a constant that depends on $\mathcal{H}'$. The parts that use reference programs ($\Delta(c)$, J_g, J_b, (iv)) hold if $\mathcal{H}'$ contains a good model π'_g and a memoriser π'_b, with $\kappa_g := \ell(\pi'_g) - d$ and $\kappa_b := \ell(\pi'_b) - d - \lceil L_S \rceil$. (b) Theorem 6.6 holds in every countable product class with Kraft weights that contains a program for the sampler. (c) Proposition 3.8(ii) holds in every product class with a prefix-free code.*

Examples are probabilistic programs clocked at $t(|x|)$ steps and weighted by code length, as in time-bounded SFI (Hänni, 2026a, the time-bounded theorem), where generous time bounds change κ_g, κ_b by logarithmic terms; and circuits with private output noise, where a parity costs about $d \log d$ bits, so κ_g is of order $d \log d$ and c_{hi} shrinks by a factor of order $\log d$ (constants not computed). Theorem 3.2 does not transfer: it needs a member that is the class's own mixture,

and a mixture of all time-t programs is not a time-t program. So the contrast between product and shared classes is established only for the unbounded classes.

Proposition 8.2 (Computability of the steeper predictor)**.** [proved] steeper Prop 4.5 *Let $\lambda \geq 1$ be computable, the class $\mathcal{H}_\times$ or $\mathcal{H}_F$, X infinite and $\{0,1\} \subseteq Y$. (a) $\xi_\lambda(D)$ is lower semicomputable uniformly in D. (b) Neither the raw conditional $\xi_\lambda(D+(x,y))/\xi_\lambda(D)$ nor its normalised version is computable. (c) Both are limit-computable.*

Part (b) is the diagonal argument for M_F (Hänni, 2026a, the computability proposition): a computable conditional yields a computable labelling on which the predictor loses a constant per input, against the weight $2^{-\lambda(K(f)+\kappa)}$ of a program for it.

Remark 8.3 (Cold posteriors are a different direction)*.* [known; remark]
steeper Rem 4.6 Wenzel et al. (2020) report that "cold" posteriors $\propto \exp(-U(\theta)/T)$ with $T < 1$, where $U(\theta) = -\sum_i \log p(y_i \mid x_i, \theta) - \log p(\theta)$, improve Bayesian neural networks. Prior and likelihood get the same exponent, so their ratio stays at $\lambda = 1$. The $\lambda > 1$ direction is that of fractional posteriors, with likelihood exponent $\eta = 1/\lambda < 1$ (Bhattacharya et al., 2019).

Remark 8.4 (Neural-network Bayes)*.* [remark]
steeper Rem 4.7 With labels independent given the parameters θ, a network's Bayes posterior is a mixture of product laws, a M_{prod}-type object (Hänni, 2026a, NN Bayes is function-space Bayes). Theorem 6.6 applies to countable discretisations with weights $p(\theta)$: under the power prior $p(\theta)^\lambda$, $\lambda > 1$, parameters that memorise the label noise get posterior mass exponentially small in the sampled randomness, if the noise model is in the family. Whether a parameter prior behaves like a description-length prior, as Theorem 4.7 needs, is the bridge question raised by the Vaintrob–Hänni result on neural-network Bayes (Vaintrob and Hänni, 2026); nothing here settles it.

8.2 Related work

Remark 8.5 (λ-weighted two-part MDL in classification)*.* [known; unverified] steeper §1.3 Three papers study $\lambda\ell + \sum -\log \nu$ over essentially product hypotheses, a classifier plus a code for the label noise. As their abstracts describe them, Grünwald and Langford (2007) show that at $\lambda = 1$ MDL and Bayes classifiers can stay bounded away from the best achievable error under misspecification, Zhu and Srebro (2025) characterise the limiting error along the regularisation path of two-part-code MDL as a function of λ and the noise level, and Zhu et al. (2026) show for a PAC-Bayesian rule that $\lambda = 1$ can overfit while a much larger, sample-size-dependent λ gives vanishing excess loss. That a bounded λ fits the noise and λ must grow with the sample size is their finding (and Hänni's remark), not ours. Their target is the data distribution; Hänni's is deliberately not (Remark 5.2). Their abstracts do not mention universal hypotheses, the shared/private contrast, or Kolmogorov-complexity classes with lower bounds over all hypotheses; whether their analysis covers some of this can only be settled from the full texts.

Remark 8.6 (Tempered and safe posteriors)*.* [known; unverified; remark] steeper §1.3, remark after Prop 2.14 The posteriors of Definition 2.5 are generalised, Gibbs, fractional or safe posteriors with learning rate $\eta = 1/\lambda$ (Bissiri et al., 2016; Bhattacharya et al., 2019; Grünwald, 2012; Grünwald and van Ommen, 2017; Zhang, 2006), whose consistency theory concerns the data distribution. SafeBayes chooses η from data by a sequential randomised log loss (Grünwald, 2012); by Proposition 5.8(i) such a criterion has no reason to leave $\eta = 1$ in well-specified imitation. This is our reading, not checked against the papers' precise criteria.

Remark 8.7 (Relation to the earlier formalisation)*.* [remark]
steeper §6 An earlier report (Claude (Anthropic), 2026, section "Simplicity and normativity from

imitation") formalised Hänni's proposal for a general imitation channel. We re-derive, rather than copy, the results we use from it: per-input and per-block pricing, the sequence-case threshold, the lemmas behind its kink theorem (here for semimeasures), the hull lemma, rate thresholds, rate blindness, non-identifiability, guard erosion and error envelopes (Appendix G.1). Two statements change. Its full-table window at $d = 20$ has upper end about $(1 - H(p))/d \approx 0.049$ when logarithmic terms are ignored; with the $2 \log d$ terms, as large as d there, the rigorous value is $c_{\mathrm{hi}} = 0.01139$ at $\kappa = 0$ (Corollary 4.10). And its lemma that the randomness of the error set removes the slack $2p$, stated for computable hypotheses, holds for semimeasures (Lemma C.1).

Remark 8.8 (What is new). [remark] steeper §3.6 Known: two-part codes and MDL with complexity weights above 1 (Section 2.3); λ-weighted MDL in noisy classification (Remark 8.5); the structure function, sophistication and restricted model families (Section 7); tempered posteriors; and our tools (dominance bounds, Kakutani's dichotomy, Ville's inequality, hashing). We have not found the following in the literature, though our search was limited to abstracts: the contrast under a steeper prior between classes with a universal member and the private-randomness class, with its price (Section 3); the analysis over all hypotheses of Kolmogorov-complexity classes through list decoding (Theorem 4.7, Propositions 5.3 and 5.4); the observation that consistent schedules learn the teacher's errors; the model–randomness theorems (Section 6); and product sophistication with its gap to set sophistication (Proposition 7.6). We do not claim that a fixed λ eventually learns the errors and λ must grow with n.

Open question. Does a time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$? If not, steeper simplicity could work in time-bounded SI too. Is the λ-MAP computable from the data? It presumably is not, but we have no proof. And the novelty claims of Remark 8.8 remain to be checked against the full texts of Zhu and Srebro (2025); Zhu et al. (2026); Grünwald and Langford (2007) and of the literature on restricted model families.

9 Discussion

What counts as one hypothesis. Whether a steeper simplicity prior can separate a model from its noise depends on what counts as one hypothesis. When a universal mixture is a member of the class, as in SI and in SFI with shared randomness, it cannot: selected hypotheses are shorter than a bound that does not depend on the data, and a computable teacher's labels, errors included, are eventually predicted (Theorem 3.2). With private randomness per input no member is universal, and steeper selection can work. The same contrast appears in the split between a sampler and its filled-in bits (Theorem 6.6 against Theorem 6.8) and in algorithmic statistics (Proposition 7.6): the product view counts as model what is reused across inputs and as noise what is private to each input. This is a modelling choice with a price on latent causes shared by several inputs (Remark 3.7 and Proposition 3.8), and the protection ends when inputs carry labelled history (Proposition 3.9).

The window and its price. For error rules incompressible relative to the model, near-optimal product hypotheses are the simple model with noise for $2^{-m} < c \le c_{\mathrm{hi}}$ (Theorem 4.7). The lower end is sharp because a bit of information about the error rule costs c per sample in length and saves about 2^{-m} per sample in risk. So λ must grow linearly with n, and the learner is then deliberately inconsistent for the teacher's labels: it discards every component whose rate is below c, valid or not, and its guaranteed code length exceeds a program's own loss by $c\,\ell(\pi)$ bits per

sample (Remark 5.2). Imitation data cannot justify this price, because strictly proper scores prefer the teacher's channel, errors included (Proposition 5.8).

The gap to realistic errors. Human labelling errors are often cheap, frequent patterns rather than incompressible sets, and for such errors the trick provably fails (Propositions 5.3 and 5.4): steeper simplicity sorts components by rate, not by validity (Proposition 5.6), so by itself it does not tell errors from rare valid rules. This agrees with Hänni's assessment that "it's very unlikely this steeper penalization is anywhere close to a full solution to the philosophical problem here", with "some hope that it works in some specific toy cases" (Hänni and Wyeth, undated): it provably works in one such case and fails in others.

The model–randomness split. At $\lambda = 1$ a Bayes mixture is indifferent, up to bounded factors, between a sampler and the programs that fill in its sampled bits (Theorem 6.4). A steeper prior over product hypotheses breaks the tie in favour of samplers, in the limited sense of Remark 6.7; in SI's representation by deterministic programs no form of steepness does (Remark 6.10).

Relation to other work. The objective is that of λ-weighted two-part MDL in noisy classification, and the posteriors are generalised, fractional or safe posteriors; those works aim at the data distribution, while the target here differs from it by design (Remarks 8.5 and 8.6). Cold posteriors are a different direction (Remark 8.3). A neural network whose labels are independent given its parameters has a Bayes posterior that mixes product laws, so Theorem 6.6 applies to countable discretisations; whether a parameter prior behaves like a description-length prior, as Theorem 4.7 would need, is the bridge question raised by Vaintrob and Hänni (2026) and is not settled here (Remark 8.4).

9.1 Open problems

The open questions of the sections, collected:

(1) *Structured error rules.* For envelopes whose cheap region has error density below one, only part (i) of Proposition 5.3 carries over; whether near-optimal hypotheses memorise the random part, and what is selected above c_{env}, are open (Section 5.4). A theory for realistic, low-complexity errors is the main gap between the trick and its motivation.

(2) *Other model classes and flip rates.* Which simple-model classes give a window (parities do; majority-like functions presumably do not)? Can the window method reach flip rates up to $1/4$, where the kink exists (Section 4.5, Remark 5.11)?

(3) *Non-uniform inputs and finite samples.* A version of Theorem 4.7 for non-uniform input laws (only the separable case is proved, Proposition 5.9), and guarantees at realistic sample sizes, far below the 10^{11} samples that uniform convergence needs (Section 4.5).

(4) *Choosing c.* A principled choice of c that does not use the imitation data, and the extra information that would tell errors from rare valid rules (Section 5.4).

(5) *Shared hypotheses.* Is every lower-semicomputable law of a random partial function dominated by M_F up to a constant (Section 3.4)?

(6) *Resource bounds and computability.* Does a time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$? Is the λ-MAP computable from the data (Section 8.2)?

(7) *Lower-semicomputable product models.* Does Proposition 7.5 hold for them (Section 7.2)?

(8) *Literature.* The novelty claims of Remark 8.8 need a comparison with the full texts of the classification papers and of the work on restricted model families (Section 8.2).

Acknowledgements

This paper was developed and written with extensive assistance from Claude (Anthropic), an AI system. Under the author's direction, separate instances of Claude wrote the proofs and the computations, refereed them adversarially in two rounds with independently written code, revised them, and wrote the text. The earlier report on the proposal (Claude (Anthropic), 2026), whose results are re-derived here, was also prepared with Claude. No human has yet independently checked all of the mathematics, and the comparison with the literature rests on abstracts and search snippets. The author thanks Sam Eisenstat for the coin example discussed in the companion paper and in Section 6.2.

Code and data

Every numerical statement comes from a deterministic or seeded Python script (Python 3.11 with NumPy and SciPy) whose output is saved next to it. The scripts named in the appendices, independent re-implementations of the main computations, and the research notes with full proofs from which the paper was written are in the folder `solomonoff-function-induction/research/tracks/steeper/` of the repository `https://github.com/kaarelh/AI-works`; the companion paper's material is in the same project folder.

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A Proofs for the setting

A.1 Standard facts

The following facts are standard (Li and Vitányi, 2008); κ is as in Section 2.2.

(F1) $K(\pi) \le \ell(\pi) + \kappa$ for every program π of a class $\mathcal{H}$. The identity map on the c.e. prefix-free set $\mathcal{H}$ is a prefix machine, which the universal machine simulates with a fixed prefix; for canonical classes, $\mathcal{H} = \operatorname{dom} V$ and the fact is immediate.

(F2) *Conditional coding theorem.* If a fixed program lower-semicomputes from z a semimeasure Q_z on a countable set, then $K(w \mid z) \le -\log Q_z(w) + \kappa$.

(F3) $K(x) \le K(y) + K(x \mid y) + \kappa$, and $K(x, y \mid z) \le K(x \mid z) + K(y \mid x, z) + \kappa$.

(F4) *Symmetry of information.* $K(x, y) = K(x) + K(y \mid x, K(x)) + O(1)$.

(F5) $K(j) \le 2\log(j+1) + \kappa$ for every integer $j \ge 0$, by a doubling code.

A.2 Proof of the hull lemma

Examples. In $\{(0,1),(10,0.5),(20,0)\}$ the middle point lies on the segment between the other two, so it is never the unique minimiser; the end points are the unique minimisers on $(0.05, \infty)$ and on $(0, 0.05)$. In $\{(0,1),(5,0.9),(10,0)\}$ the point $(5, 0.9)$ lies above the boundary of C; it solves the constrained problem for $L \in [5, 10)$ and minimises no penalised objective.

We prove Lemma 2.8. Write $\mathcal{A}' := \mathcal{A} \cap \{R < \infty\}$ and $\langle (c,1), (\ell, R) \rangle = c\,\ell + R$.

(i). Since $(c, 1) \ge 0$, the infimum of $c\,\ell + R$ over $C = \operatorname{conv}(\mathcal{A}') + \mathbb{R}^2_{\ge 0}$ equals its infimum over $\mathcal{A}'$. A minimiser a over $\mathcal{A}$ therefore minimises a nonzero linear functional over C. So the line $\{c\,\ell + R = c\,\ell(a) + R(a)\}$ supports C at a, and a lies on the boundary of C.

Structure of the boundary. Fix $a \in \mathcal{A}'$ and $s > 0$. A point $b \in \mathcal{A}'$ with $\ell(b) > \ell(a)$ and $R(b) \le R(a) - s(\ell(b) - \ell(a))$ has $\ell(b) \le \ell(a) + R(a)/s$, because $R(b) \ge 0$; only finitely many points satisfy this. So from each point the steepest descending direction to the right is attained, and the lower-left boundary of C is a polygonal line. It consists of a vertical ray above the least-ℓ vertex, segments between consecutive vertices with slopes $-s$, $s \in (0, \infty)$, strictly increasing from left to right, and, if a point of least R exists, a horizontal ray to the right of the least-R vertex. Its corners are extreme points of C, and every extreme point of C lies in $\mathcal{A}'$, because the extreme points of a convex hull belong to the generating set and adding the orthant creates none.

(iii). Let a be a vertex. Its two incident boundary pieces have slopes $-s_- < -s_+ \le 0$, with $s_- = \infty$ at the least-ℓ vertex and $s_+ = 0$ at the least-R vertex. For $c \in (s_+, s_-)$ the line with normal $(c, 1)$ through a meets C only at a, so a is the unique minimising point. Conversely, if a is the unique minimising point for every c in an open interval, then a is an exposed point of C, hence an extreme point, hence a vertex. A boundary point that is not a vertex lies in the relative interior of a segment between two vertices, or on one of the two rays. In the first case, whenever it minimises $c\,\ell + R$, so do both endpoints of its segment. A point on the vertical ray is beaten by the vertex below it (smaller R), and a point on the horizontal ray by the least-R vertex (smaller ℓ, same R). So no such point is ever the unique minimising point. The statement about hypotheses is immediate: a hypothesis is the unique minimiser exactly when its point is the unique minimising point and no other hypothesis has that point.

(ii). Add $c\,\ell(h) + R(h) \le c\,\ell(h') + R(h')$ and $c'\ell(h') + R(h') \le c'\ell(h) + R(h)$ to get $(c' - c)(\ell(h') - \ell(h)) \le 0$, so $\ell(h') \le \ell(h)$. Then the first inequality gives $R(h) \le R(h') + c(\ell(h') - \ell(h)) \le R(h')$.

(iv). If $\ell(h) \le \ell(h_c)$, then $R(h) \ge R(h_c) + c(\ell(h_c) - \ell(h)) \ge R(h_c)$, so h_c solves the constrained problem with $L = \ell(h_c)$. For the converse, by the description of the boundary: a vertex minimises $c\,\ell + R$ for every c in its interval (s_+, s_-), which contains positive values; a point on a segment between two vertices, of slope $-s$ with $s > 0$, lies on the supporting line with normal $(s, 1)$ and so minimises $s\,\ell + R$. A point on the open horizontal ray is beaten, for every $c > 0$, by the least-R vertex. A point on the vertical ray above the least-ℓ vertex is never a constrained solution, since that vertex is feasible whenever the point is and has smaller R. A point not on the boundary minimises no linear functional with normal $(c, 1)$ over $\mathcal{A}$, by (i).

The failed strengthening. In $\mathcal{A} = \{(0,1), (10,0), (20,0)\}$ the point $(20,0)$ lies on the horizontal ray to the right of the least-R vertex $(10,0)$. It solves $\min\{R : \ell \le L\}$ for $L \ge 20$, since it ties with $(10,0)$, but $c \cdot 20 < c \cdot 10$ fails for every $c > 0$.

Computation. A direct computation (code in the accompanying repository, seed 6) checked the equivalence in (iii) on 401 random point sets (5061 points, duplicates and collinear points included) and found no mismatch; a second computation (seed 5) checked the minimisers on 300 further point sets.

A.3 Proof of the coding bound

We prove Lemma 2.9. Given π and the finite set $\operatorname{dom} D$, the map $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ on $Y^{\operatorname{dom} D}$ is lower semicomputable, uniformly in $(\pi, \operatorname{dom} D)$. It is a semimeasure. For a product hypothesis its total mass is $\prod_{x \in \operatorname{dom} D} \sum_y \nu_\pi(y \mid x) \le 1$. For a shared hypothesis, repeated use of the one-point inequality gives $\sum_y P_\pi(\operatorname{dom} D \mapsto y) \le P_\pi(\emptyset) \le 1$. By (F2), $K(y_D \mid \pi, \operatorname{dom} D) \le -\log P_\pi(D) + \kappa$. By (F3) and (F1),

$$K(y_D \mid \operatorname{dom} D) \le K(\pi) + K(y_D \mid \pi, \operatorname{dom} D) + \kappa \le \ell(\pi) - \log P_\pi(D) + \kappa.$$

□

B Proofs for the absorption section

B.1 Classes with an absorbing member

We prove Remark 3.1. Suppose first that the mixture $\xi_1 = \sum_\pi 2^{-\ell(\pi)} P_\pi$ is itself a member, $\xi_1 = P_{\pi_U}$. Then $P_\pi \le 2^{\ell(\pi)} \xi_1 = 2^{\ell(\pi)} U$ for every π, which is (Dom) with $\kappa_0 = 0$, and $\xi_1 = U$. This covers $\mathcal{H}_w$ ($M^w = \sum_i \mathbf{m}(i)\nu_i$ is an lsc weak function semimeasure, hence occurs in the trimmed enumeration), $\mathcal{H}_{\mathrm{SI}}$ (M is an lsc semimeasure) and $\mathcal{H}_{\mathrm{chr}}$ (M_H is an lsc chronological semimeasure and, by the representation theorem for chronological semimeasures, the law of a history-oracle function machine at every nonempty history; at the empty history both sides of (Dom) are at most 1 and $P_{\pi_U} = 1$) (Hänni, 2026a). For $\mathcal{H}_F$: M_F is a machine law, so it has an index; let π_U be a shortest program for it. For $D \ne \emptyset$, $P_\pi(D) = \nu_{T_{V(\pi)}}(D) \le M_F(D)/\mathbf{m}(V(\pi)) \le 2^{\ell(\pi)} M_F(D)$, and at $\emptyset$ both sides equal 1. Here $\xi_1 = M_F$ on nonempty data sets and $\xi_1(\emptyset) \le 1 = M_F(\emptyset)$, so $\xi_1 \le U$. □

B.2 Proof of the absorption theorem

We prove Theorem 3.2. Throughout, (Dom) holds and $\lambda > 1$.

(a). Let $\hat\pi$ be a λ-MAP for o. Then $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_U; o) = \lambda \ell_U - \log U(o)$. By (Dom), $-\log U(o) \le \ell(\hat\pi) + \kappa_0 - \log P_{\hat\pi}(o)$. Substituting,

$$\lambda \ell(\hat\pi) - \log P_{\hat\pi}(o) \le \lambda \ell_U + \ell(\hat\pi) + \kappa_0 - \log P_{\hat\pi}(o),$$

so $(\lambda-1)\ell(\hat{\pi}) \le \lambda\ell_U + \kappa_0$, which is the length bound. (If $P_{\hat{\pi}}(o) = 0$ then $J_\lambda(\hat{\pi}; o) = \infty$, and $\hat{\pi}$ is a minimiser only if every program has infinite objective; we exclude this case, which does not occur when $U(o) > 0$.) The loss bound is $J_\lambda(\hat{\pi}; o) \le J_\lambda(\pi_U; o)$ rearranged. Finally, $\ell^*_\lambda < L$ if and only if $\lambda(L - \ell_U) > L + \kappa_0$.

(b). The first inequality keeps the single term of π_U in $\xi_\lambda(o) = \sum_\pi 2^{-\lambda\ell(\pi)} P_\pi(o)$. The second holds because $2^{-\lambda\ell} \le 2^{-\ell}$. If $\xi_1 \le 2^{\kappa_1} U$, then $-\log \xi_\lambda(o) \ge -\log \xi_1(o) \ge -\log U(o) - \kappa_1$, and the first inequality gives $-\log \xi_\lambda(o) \le \lambda\ell_U - \log U(o)$. For a data-set class, ξ_λ is a weak function semimeasure with $\xi_\lambda(\emptyset) \le \sum_\pi 2^{-\lambda\ell(\pi)} \le 1$, because each P_π is one and the weak inequalities are linear. Its sequential log loss on D_n is $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset)$ in every order (Definition 2.3), and the same holds for U.

(c1). Since π_t is deterministic and consistent with D_n, $P_{\pi_t}(D_n) = 1$, so $\xi_\lambda(D_n) \ge 2^{-\lambda\ell_t}$. Also, by (Dom), $U(D_n) \ge 2^{-\ell_t - \kappa_0} P_{\pi_t}(D_n) = 2^{-\ell_t - \kappa_0}$, so $\xi_\lambda(D_n) \ge 2^{-\lambda\ell_U} U(D_n) \ge 2^{-\lambda\ell_U - \kappa_0 - \ell_t}$. By telescoping, $\sum_{i \le n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) = -\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset) \le -\log \xi_\lambda(D_n)$. Each conditional is at most 1 by the one-point inequality, so each term is nonnegative, and a label with probability below $1/2$ adds more than one bit.

(c2). Combine (a) with $-\log U(D_n) \le \ell_t + \kappa_0$.

(c3). Let $\mathcal{H}_{\le \ell^*_\lambda} := \{\pi : \ell(\pi) \le \ell^*_\lambda\}$. It is finite by Kraft, and by (a) it contains every λ-MAP for every D_n. For $\pi \in \mathcal{H}_{\le \ell^*_\lambda}$ let $L_\pi(n) := -\log P_\pi(D_n)$. It is nondecreasing in n, because weak function semimeasures are monotone, so it has a limit $L_\pi(\infty) \in [0, \infty]$. Let $\mathcal{H}_{\text{fin}} := \{\pi \in \mathcal{H}_{\le \ell^*_\lambda} : L_\pi(\infty) < \infty\}$. It contains π_U, because $\ell_U \le \ell^*_\lambda$ and $L_{\pi_U}(\infty) \le \ell_t + \kappa_0$. For $\pi \in \mathcal{H}_{\le \ell^*_\lambda} \setminus \mathcal{H}_{\text{fin}}$, $J_\lambda(\pi; D_n) \to \infty$, while $J_\lambda(\pi_U; D_n) \le \lambda\ell_U + \ell_t + \kappa_0$ for every n. Since there are finitely many such π, there is n_0 beyond which none of them is a λ-MAP. For $\pi \in \mathcal{H}_{\text{fin}}$, $\sum_n [L_\pi(n+1) - L_\pi(n)] \le L_\pi(\infty)$, and each increment equals $-\log\big(P_\pi(D_{n+1})/P_\pi(D_n)\big)$, the loss of π's prediction of the next label. So π gives the next label probability below $1/2$ fewer than $L_\pi(\infty) + 1$ times. Taking the union over the finite set $\mathcal{H}_{\text{fin}}$ bounds the number of $n \ge n_0$ at which some λ-MAP does so.

(d). By (Dom), $\xi_\lambda \ge 2^{-\lambda\ell_U} U \ge 2^{-\lambda\ell_U - \kappa_0 - \ell(\pi_\mathbb{P})} \mathbb{P}$ on every string. Hence, for every length L and $\omega_{1:L}$ drawn from $\mathbb{P}$,

$$\mathbb{E}_\mathbb{P} \log \frac{\mathbb{P}(\omega_{1:L})}{\xi_\lambda(\omega_{1:L})} \le \lambda\ell_U + \kappa_0 + \ell(\pi_\mathbb{P}).$$

Write $\xi_\lambda(b \mid h) := \xi_\lambda(hb)/\xi_\lambda(h)$ and expand by the chain rule over bit positions: $\log(\mathbb{P}/\xi_\lambda)(\omega_{1:L}) = \sum_{j \le L} \log\big(\mathbb{P}(\omega_j \mid \omega_{<j})/\xi_\lambda(\omega_j \mid \omega_{<j})\big) - \log \xi_\lambda(\varepsilon)$, and $-\log \xi_\lambda(\varepsilon) \ge 0$. At each position the conditional expectation of the summand is $\mathrm{KL}\big(\mathbb{P}(\cdot \mid h) \,\|\, \xi_\lambda(\cdot \mid h)\big)$, which is at least $-\log \sum_b \xi_\lambda(b \mid h) \ge 0$ because $\xi_\lambda(\cdot \mid h)$ is a sub-probability. With fixed-length blocks the label positions are fixed. At a label position $\mathbb{P}(\cdot \mid h)$ is a point mass, because $y_i = t(x_i)$, so the summand equals $-\log \xi_\lambda(\omega_j \mid \omega_{<j})$; the sum of these over the bits of one label block is $-\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i)$. Dropping the nonnegative terms of the input positions gives the claim. This is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions. □

B.3 Proof of per-input pricing

We prove Proposition 3.4.

(i), the bound. $\max_f \ge \mathbb{E}_{f \sim Q}$. For each i,

$$\mathbb{E}_{f \sim Q}\big[-\log \nu(f(x_i) \mid x_i)\big] = H(Q_{x_i}) + \sum_y Q_{x_i}(y) \log \frac{Q_{x_i}(y)}{\nu(y \mid x_i)},$$

and the last sum is at least $-\log \sum_y \nu(y \mid x_i) \ge 0$, by Gibbs' inequality with a sub-probability.

(i), the value. For $\varepsilon \in (0, 1/|Y|)$ let Δ_ε be the set of conditional sub-probabilities on $\{x_1, \dots, x_n\}$ with $\nu(y \mid x_i) \geq \varepsilon$ for all y, i. It is compact and convex, and $L(\nu, Q) := \mathbb{E}_{f\sim Q} \sum_i -\log \nu(f(x_i) \mid x_i)$ is continuous on $\Delta_\varepsilon \times \Delta(F)$, convex in ν and linear in Q. By Sion's minimax theorem (Sion, 1958), $\min_{\Delta_\varepsilon} \max_Q L = \max_Q \min_{\Delta_\varepsilon} L$. Since L is linear in Q, $\max_Q L(\nu, Q) = \max_{f\in F} \sum_i -\log \nu(f(x_i) \mid x_i)$. For each Q, $\min_{\Delta_\varepsilon} L(\cdot, Q)$ lies between $\sum_i H(Q_{x_i})$, by the bound, and $\sum_i H(Q_{x_i}) - n \log(1 - |Y|\varepsilon)$, attained up to that error by $\nu = (1 - |Y|\varepsilon)Q_x + \varepsilon$, which lies in Δ_ε and sums to 1. The error term is uniform in Q. Hence

$$\max_Q \sum_i H(Q_{x_i}) \leq \inf_\nu \max_f \sum_i -\log \nu(f(x_i) \mid x_i) \leq \max_Q \sum_i H(Q_{x_i}) - n \log(1 - |Y|\varepsilon),$$

where the first inequality is the bound, applied to every ν. Let $\varepsilon \to 0$.

The example. For $F = \{(a, a), (a, b), (b, a), (c, a)\}$ a direct computation (code in the accompanying repository, seed 2) gives $\max_Q \sum_i H(Q_{x_i}) = 2.5431$ and the same value for $\min_\nu \max_f$ by constrained optimisation over ν; an independent computation by grid search, mirror ascent and a primal certificate agrees. The naive bound is $\log 3 + \log 2 = 2.5850$.

(ii). For $x \neq 0$ exactly half of the parities give each label at x, so the uniform Q has $H(Q_x) = 1$, and (i) gives a parity with loss at least n. The Bayes mixture with weight 2^{-d} on each parity gives every parity's data probability at least 2^{-d}.

(iii). Each term of the per-input mixture assigns $1/2$ to the observed label 1, at every input; the shared mixture gives the constant-1 function weight $1/2$. □

B.4 Proof of the price of private randomness

We prove Proposition 3.8.

Write $a_i := 0^i$ and $b_i := 1^i$, so $D_\beta = \{(a_i, \beta_i), (b_i, \beta_i) : i \leq k\}$.

(i). Fix $\pi \in \mathcal{H}_\times$ with conditional semimeasure $\nu = \nu_\pi$. Then $\nu(D_\beta) = \prod_{i\leq k} s_i(\beta_i)$ with $s_i(y) := \nu(y \mid a_i)\, \nu(y \mid b_i)$. By the inequality of arithmetic and geometric means, $s_i(y) \leq q_i(y)^2$ with $q_i(y) := \frac{1}{2}(\nu(y \mid a_i) + \nu(y \mid b_i))$, and each q_i is a sub-probability on $\{0, 1\}$. So $\nu(D_\beta) \leq Q_\pi(\beta)^2$ with $Q_\pi(\beta) := \prod_i q_i(\beta_i) \leq 1$. Since $\sum_\beta Q_\pi(\beta) = \prod_i \sum_y q_i(y) \leq 1$, Q_π is a semimeasure on $\{0, 1\}^k$, lower semicomputable uniformly in (π, k). Let $\tilde{R}(\beta) := \sum_{\pi\in\mathcal{H}_\times} 2^{-\ell(\pi)} Q_\pi(\beta)$. It is an lsc semimeasure on $\{0, 1\}^k$ given k, so $\tilde{R}(\beta) \leq 2^{-K(\beta|k)+\kappa}$ by (F2). Now let $\gamma := \min(\lambda, 2)$. If $1 \leq \lambda \leq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 = (2^{-\ell(\pi)} Q_\pi)^\lambda Q_\pi^{2-\lambda} \leq (2^{-\ell(\pi)} Q_\pi)^\lambda$. If $\lambda \geq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 \leq (2^{-\ell(\pi)} Q_\pi)^2$. For nonnegative t_π and $\gamma \geq 1$, $\sum_\pi t_\pi^\gamma \leq (\sum_\pi t_\pi)^\gamma$. Therefore

$$\xi_\lambda^\times(D_\beta) = \sum_\pi 2^{-\lambda\ell(\pi)} \nu_\pi(D_\beta) \leq \sum_\pi 2^{-\lambda\ell(\pi)} Q_\pi(\beta)^2 \leq \tilde{R}(\beta)^\gamma \leq 2^{-\gamma(K(\beta|k)-\kappa)}.$$

(ii). $2^{-J_\lambda(\pi; D_\beta)}$ is one term of $\xi_\lambda^\times(D_\beta)$. For a product class with a prefix-free code, define $\tilde{R}$ over that class; (F2) then holds with a constant for the class.

(iii). ν_{len} is the law of the function machine $T = \{(x, w, w_{|x|}) : x \neq \varepsilon,\ w \in \{0, 1\}^{|x|}\}$, where $w_{|x|}$ is the last bit of the tape prefix w. Inputs a_i, b_i have the same length i, so they receive the same bit, and distinct i receive independent bits; so $\nu_{\text{len}}(D_\beta) = 2^{-k}$. Universality gives $M_F \geq 2^{-K(\nu_{\text{len}})} \nu_{\text{len}}$ (Hänni, 2026a), so $-\log M_F(D_\beta) \leq k + \kappa$. For ξ_λ^F keep the single term of a shortest program for ν_{len}.

(iv). Fewer than 2^{k-j} strings β have $K(\beta \mid k) < k - j$, so $\Pr[K(\beta \mid k) < k - j] < 2^{-j}$. On the complement, (i) gives $-\log \xi_\lambda^\times(D_\beta) \geq \gamma(k - j - \kappa)$, while (iii) gives $-\log M_F(D_\beta) \leq k + \kappa$ and $-\log \xi_\lambda^F(D_\beta) \leq k + \lambda\kappa$. Subtracting, and using $\gamma \leq 2$ and $\gamma \leq \lambda$, the excess is at least $(\gamma - 1)k - 2j - (\lambda + 2)\kappa$ in both cases. □

Computation. A direct computation (code in the accompanying repository, seed 6) verified the key inequality $\xi_\lambda^\times(D_\beta) \le \tilde{R}(\beta)^{\min(\lambda,2)}$ in all 24,732 cases on random finite product classes with Kraft weights and $\lambda \in \{1, 1.3, 1.7, 2, 2.5, 4\}$; equality occurs.

Reading. At $\lambda = 1$ the first option of the product class, writing the coins into the program, costs the same as the shared class up to lower-order terms, which is why M_{prod} and M_F are close; at $\lambda > 1$ the product class pays $\min(\lambda, 2)$ bits per coin, against 1 for the shared class.

B.5 Proofs for the two history propositions

We prove Proposition 3.9. Each Z_i is a positive real, computable uniformly in the block's history: its terms $w_\theta \prod_{k<i} \nu_\theta(y_k \mid \cdot)$ are computable and positive, and the tail $\sum_{\theta>T} w_\theta \prod_{k<i} \nu_\theta(\cdot)$ is at most $\sum_{\theta>T} w_\theta$, which is computable from T because w and $\sum_\theta w_\theta$ are computable. So $h_w = Z_{i+1}/Z_i$ is computable. It sums to 1 over y_i, because the ν_θ are measures. So h_w is a computable conditional measure on extended inputs; a program for it is obtained from a description of w and (ν_θ), which gives the length bound in the canonical class. The product of the predictive probabilities over a block of length n' telescopes to $Z_{n'+1}/Z_1$, and $Z_1 = \sum_\theta w_\theta \le 1$, so

$$-\log \frac{Z_{n'+1}}{Z_1} \le -\log \frac{w_\theta \prod_i \nu_\theta(y_i \mid \cdot)}{\sum_{\theta'} w_{\theta'}} \le \log \frac{1}{w_\theta} + \sum_i -\log \nu_\theta(y_i \mid \cdot)$$

for every θ. □

Without the computability assumptions, Z_{i+1}/Z_i is a ratio of lsc sums, which need not be lsc; this is why they are made. A finite family with rational values and rational weights also satisfies them. A direct computation (code in the accompanying repository, seed 5) checked the block identity to $3.6 \cdot 10^{-15}$ on 2000 random instances with history-dependent inputs.

We prove Proposition 3.10. A hypothesis of the first kind is an lsc conditional semimeasure on the set of extended inputs, with private randomness per extended input, so it is a product hypothesis there. Extended inputs $(x_{<i}, x_i)$ at different positions are distinct. In Proposition 3.4(ii) take F to be the functions $(x_{<i}, x_i) \mapsto \langle a, x_i \rangle \bmod 2$ with nonzero x_i; under uniform a each label is a fair coin, and the argument is unchanged, as is that of Corollary 3.5. The second claim is Remark 3.1 for $\mathcal{H}_{\mathrm{chr}}$. □

C Proofs for the teacher's-mistakes theorem

Throughout, the setting is Definition 4.2, and (F1)–(F5) are the standard facts of Appendix A.1. For $z \in \{0,1\}^m$ and $w \in \{0,1\}^{d-m}$ we write (z, w) for the input with prefix z; w is uniform on $\{0,1\}^{d-m}$ in every expectation $\mathbb{E}_w$. Since $a \in \{0,1\}^d$, the length d can be computed from a.

C.1 List decoding

Proof of Lemma 4.3. Write $h_y(x) := \nu_\pi(y \mid x)$, $s_h := h_0 - h_1 \in [-1, 1]$, $\chi_{a'}(x) := (-1)^{\langle a', x \rangle}$ and $\hat{s}(a') := \mathbb{E}_x[s_h \chi_{a'}]$. Since $\nu_\pi(f_{a'}(x) \mid x) = \frac{1}{2}(h_0 + h_1) + \frac{1}{2}\chi_{a'}(x) s_h(x)$, the agreement with $f_{a'}$ is

$$u(\pi; a') := \mathbb{E}_x \nu_\pi(f_{a'}(x) \mid x) = \tfrac{1}{2}\big(\mathbb{E}_x[h_0 + h_1] + \hat{s}(a')\big) \le \tfrac{1}{2}\big(1 + \hat{s}(a')\big).$$

The characters are orthonormal under the uniform law on $\{0,1\}^d$, so Parseval's identity gives $\sum_{a'} \hat{s}(a')^2 = \mathbb{E}_x s_h^2 \le 1$. Hence $A_t := \{a' : u(\pi; a') > \theta_t\} \subseteq \{a' : \hat{s}(a') > 2^{-t}\}$ has fewer than 2^{2t} elements. This is the list-size bound for the Hadamard code (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The value $u(\pi; a')$ is a finite average of lower semicomputable

values, so it is lower semicomputable uniformly in (π, a', d), and A_t is computably enumerable given (π, d, t). If $\mathrm{lev}(\pi) = t$, then $a \in A_t$. So a is described, given π, by d, by t, and by its index in the enumeration of A_t, which takes $2t$ bits given t. By (F5), $K(a \mid \pi) \le 2t + 2\log(t+1) + 2\log d + \kappa = s(t) + 2\log d + \kappa$. □

The argument uses the lower-semicomputable agreement u rather than the correlation $\hat{s}$, which is not computable for semimeasures; this is what extends the earlier, computable-hypothesis version of the lemma (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation") to all product hypotheses. A direct computation on 400 random semimeasure hypotheses at $d = 7$ and $d = 8$ found no violation of the identity for u, of the list bound or of the risk floor (script `c2_pricing_kink.py`, seed 2); an independent computation on adversarial hypotheses agrees.

C.2 The risk floors, the coding lemma and the two-part bounds

Proof of Lemma 4.4. Since $\min\{B, -\log v\} = -\log\max\{v, 2^{-B}\}$ and $-\log$ is convex, Jensen's inequality gives

$$R_B(\pi) \ge -\log \mathbb{E}_x \max\{\nu_\pi(y^*(x) \mid x), 2^{-B}\} \ge -\log\left(\mathbb{E}_x \nu_\pi(y^*(x) \mid x) + 2^{-B}\right).$$

Off E, $y^* = f_a$; on E, $\nu_\pi(y^* \mid x) \le 1$, and $\Pr[E] = p$. So $\mathbb{E}_x \nu_\pi(y^*(x) \mid x) \le u(\pi) + p$. If $\mathrm{lev}(\pi) = t \ge 2$, then $u(\pi) \le \theta_{t-1} = \frac{1}{2} + 2^{-t}$, and $\frac{1}{2} + 2^{-t} + p + 2^{-B} = \frac{1}{2}(1 + 2^{-(t-1)} + 2p + \tau_B)$, which gives $R_B \ge G_{t-1}$. If $\mathrm{lev}(\pi) = \infty$, then $u \le \frac{1}{2}$ and $R_B \ge G_\infty$. □

Lemma 4.4 is tight as a function of u alone: a hypothesis that gives the teacher's label probability 1 on E and is uniform elsewhere has $u = \frac{1-p}{2}$ and $\mathbb{E}_x \nu(y^* \mid x) = \frac{1}{2} + \frac{p}{2} = u + p$. Such a hypothesis knows S. Removing the additive p therefore needs the randomness of S relative to the hypothesis, which the next lemma uses. It is the semimeasure version of a lemma of the earlier formalisation (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation"), which handled computable hypotheses through their exact correlation; for semimeasures only lower-semicomputable quantities are available. The lemma is used for the kink of Proposition 7.6 at every flip rate below 1/4, not for Theorem 4.7 (see Remark 5.11).

Lemma C.1 (Randomness of the error rule sharpens the floor)**.** [proved; computed] steeper Lemma 2.2c (T5 Lemma 2.5c for semimeasures); check c7(C) *For every program π, every integer $g \ge 1$ and every $B \in [\log(1/p), \infty]$,*

$$R_B(\pi) \;\ge\; -\log\Big(p + (1-2p)\,u(\pi) + p\big(2\gamma(\pi) + 2^{1-g}\big) + 2^{-B}\Big),$$

where $\gamma(\pi) := \sqrt{j \ln 2/(2s_0)}$ and j is the least integer with $j - 2\log(j+1) > \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. Always $I(\pi) \le \ell(\pi) + \kappa$, so the bound also holds with $\ell(\pi)$ in place of $I(\pi)$. The proof does not use $p = 2^{-r}$; it holds for every rule size s_0, with $p := s_0/2^m$.

Proof. For $z \in \{0,1\}^m$ let $A(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \mid (z,w))$ and $V(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \oplus 1 \mid (z,w))$. Then $A + V \le 1$, $u = 2^{-m}\sum_z A(z)$, and $\bar{v} := 2^{-m}\sum_z V(z) \le 1 - u$. Off E the teacher's label is f_a and on E it is $f_a \oplus 1$, so

$$\mathbb{E}_x \nu_\pi(y^*(x) \mid x) = 2^{-m}\Big[\sum_{z \notin S} A(z) + \sum_{z \in S} V(z)\Big] = u - p\,\mathrm{avg}_S A + p\,\mathrm{avg}_S V. \tag{1}$$

Information budget. By definition $K(S \mid a,\pi) = B_S - I(\pi) = L_S - \delta_S - I(\pi)$. By (F3), $B_S = K(S \mid a, K(a)) \le K(\pi \mid a, K(a)) + K(S \mid a,\pi) + \kappa$, so $I(\pi) \le K(\pi \mid a, K(a)) + \kappa \le \ell(\pi) + \kappa$ by (F1).

Coding step. Suppose $\mathcal{B}$ is a set of s_0-subsets of $\{0,1\}^m$ with $|\mathcal{B}| \le 2^{L_S - j}$ that contains S and is computably enumerable given (π, a, m, s_0) and a parameter θ with $K(\theta) \le g + 2\log g + \kappa$. Then S is described by j, m, s_0, θ and its index among the first $2^{\lceil L_S \rceil - j}$ enumerated elements, so $K(S \mid a,\pi) \le \lceil L_S \rceil - j + 2\log(j+1) + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. With the budget this gives $j - 2\log(j+1) \le \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$, which the choice of j excludes. So no such $\mathcal{B}$ exists.

The V side. If $1 - u + \gamma \ge 1$, then $\operatorname{avg}_S V \le 1 \le 1 - u + \gamma$. Otherwise let $\theta \le 1$ be the least multiple of 2^{-g} with $\theta \ge 1 - u + \gamma$; then $K(\theta) \le g + 2\log g + \kappa$ and $\theta \ge \bar{v} + \gamma$. If $\operatorname{avg}_S V > \theta$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \operatorname{avg}_{S'} V > \theta\}$ contains S and is computably enumerable, because V is lower semicomputable. For S' uniform among the s_0-subsets, Hoeffding's inequality for sampling without replacement (Hoeffding, 1963) gives $\Pr[\operatorname{avg}_{S'} V - \bar{v} \ge \gamma] \le e^{-2 s_0 \gamma^2} = 2^{-j}$, so $|\mathcal{B}| \le 2^{L_S - j}$, which the coding step excludes. Hence $\operatorname{avg}_S V \le 1 - u + \gamma + 2^{-g}$.

The A side. A is only lower semicomputable, so a set of the form "$\operatorname{avg}_{S'} A$ is small" need not be enumerable; we use a finite stage. Let $\nu_s \uparrow \nu_\pi$ be the stage-s lower approximations, $A_s(z) := \mathbb{E}_w \nu_s(f_a(z,w) \mid (z,w))$ and $u_s := 2^{-m} \sum_z A_s(z) \uparrow u$. If $u \le 2^{-g}$, then $\operatorname{avg}_S A \ge 0 > u - 2^{-g} - \gamma$. Otherwise let θ' be the largest multiple of 2^{-g} strictly below u, so $\theta' \ge u - 2^{-g}$ and $K(\theta') \le g + 2\log g + \kappa$, and let s^* be the first stage with $u_{s^*} > \theta'$. Then A_{s^*} is computable from (π, a, m, θ'). If $\operatorname{avg}_S A_{s^*} < u_{s^*} - \gamma$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \operatorname{avg}_{S'} A_{s^*} < u_{s^*} - \gamma\}$ is decidable from these data and contains S; the lower-tail form of the same inequality gives $|\mathcal{B}| \le 2^{L_S - j}$, which the coding step excludes. Hence $\operatorname{avg}_S A \ge \operatorname{avg}_S A_{s^*} \ge u_{s^*} - \gamma > u - 2^{-g} - \gamma$.

Combination. By (1), $\mathbb{E}_x \nu_\pi(y^* \mid x) \le u - p(u - 2^{-g} - \gamma) + p(1 - u + \gamma + 2^{-g}) = p + (1-2p)u + p(2\gamma + 2^{1-g})$. Conclude with Jensen's inequality as in Lemma 4.4. □

A direct computation on 400 random semimeasure hypotheses at $d = 8$, $m \in \{6, 8\}$, a third of them correlated with S, confirms (1) to $2.2 \cdot 10^{-16}$, and the deterministic form $\mathbb{E}\nu(y^*) \le p + (1-2p)u + p(\mathrm{dev}_A + \mathrm{dev}_V)$ with the actual deviations is never violated (script `c7_run2.py`, part (C), seed 7).

Proof of Lemma 4.5. For $z \in \{0,1\}^m$ and $b \in \{0,1\}$ let

$$q_z(b) := 2^{-\mathbb{E}_w \min\{B,\, -\log \nu_\pi(f_a(z,w) \oplus b \mid (z,w))\}}.$$

By Jensen's inequality for the convex function $t \mapsto 2^{-t}$, $q_z(b) \le \mathbb{E}_w \max\{\nu_\pi(f_a(z,w) \oplus b \mid (z,w)), 2^{-B}\}$, so $\sum_b q_z(b) \le 1 + 2 \cdot 2^{-B} = 1 + \tau_B$. Each $q_z(b)$ is lower semicomputable from (π, a, m, B), because the exponent is an average of upper semicomputable values. Define, on subsets $S' \subseteq \{0,1\}^m$,

$$Q(S') := \prod_{z \in \{0,1\}^m} \frac{q_z(\mathbf{1}_{S'}(z))}{1 + \tau_B},$$

normalised factor by factor. Its total mass is $\prod_z \sum_b q_z(b)/(1+\tau_B) \le 1$, so Q is a lower-semicomputable semimeasure given (π, a, m, B). Since $y^*(z,w) = f_a(z,w) \oplus \mathbf{1}_S(z)$,

$$-\log Q(S) = \sum_z \mathbb{E}_w \min\{B, -\log \nu_\pi(y^*(z,w) \mid (z,w))\} + 2^m \log(1 + \tau_B) = 2^m R_B(\pi) + \beta_B.$$

By (F2), $K(S \mid \pi, a, m, B) \le 2^m R_B(\pi) + \beta_B + \kappa$, and by (F3) and (F5) the descriptions of m and of the integer B cost at most $2\log m + 2\log B + \kappa$ more. For $B = \infty$ the terms in B vanish. □

The normalisation must be per factor: with a single division by $1+\tau_B$, Q is in general not a semimeasure (for a deterministic hypothesis at the smallest admissible B its total mass is 1.119). With per-factor normalisation an exhaustive computation over subsets confirms the identity for $-\log Q(S)$ to $2\cdot 10^{-15}$ and a total mass at most 1.

Proof of Lemma 4.6. (D1). By (F3), (F1) and Lemma 4.3, $d \le K(a) \le K(\pi) + K(a \mid \pi) + \kappa \le \ell(\pi) + s(t) + 2\log d + \kappa$. All terms of Λ_1 other than $2\log d + \kappa$ are nonnegative.

(D2). By symmetry of information (F4), $K(a,S) = K(a) + K(S \mid a, K(a)) + O(1) \ge d + B_S - \kappa$. By (F3), $K(a,S) \le K(\pi) + K(a \mid \pi) + K(S \mid a, \pi) + \kappa$. Insert (F1), Lemma 4.3 and Lemma 4.5: $d + B_S \le \ell(\pi) + s(t) + 2\log d + 2^m R_B(\pi) + \beta_B + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$. □

C.3 Proof of the theorem

Proof of Theorem 4.7. (o). If $\mathrm{lev}(\pi) = 1$, then (D1) and (D2) with $s(1) = 4$ give $\ell(\pi) \ge d - 4 - \Lambda_1 + \max\{0, B_S - 2^m R_B(\pi)\}$; multiply by c and add $R_B(\pi)$. If $\mathrm{lev}(\pi) = t \in [2,\infty)$, then (D1), $\ell \ge 0$ and Lemma 4.4 give

$$J_c^B(\pi) \ge c\,(d - s(t) - \Lambda_1)^+ + G_{t-1} \ge \varphi(t) := c\,(d - 4t - \Lambda_1)^+ + G_{t-1},$$

using $s(t) \le 4t$. Extend φ to real $t \ge 2$. If $t^* \ge 2$: on $[2, t^*]$ the first term is linear and G_{t-1} is concave, so $\varphi \ge \min\{\varphi(2), \varphi(t^*)\} = \min\{c(d - 8 - \Lambda_1) + G_1,\, G_{t^*-1}\}$; for $t > t^*$, $\varphi(t) \ge G_{t-1} \ge G_{t^*-1}$ because G is increasing. If $t^* < 2$: $\varphi(t) \ge G_{t-1} \ge G_1$. If $\mathrm{lev}(\pi) = \infty$: $J_c^B \ge R_B \ge G_\infty \ge G_j$ for every j, so $J_c^B \ge \Phi_2(c)$ in both cases. (G_j is increasing and concave: with $v = 2^{-j}$, $\mathrm{d}G/\mathrm{d}j = v/(1 + v + 2p + \tau_B) > 0$ and $\mathrm{d}^2G/\mathrm{d}j^2 = -\ln 2 \cdot v(1 + 2p + \tau_B)/(1 + v + 2p + \tau_B)^2 < 0$.)

(i). When $c2^m > 1$, ψ_c has slope $1 - c2^m < 0$ on $[0, B_S/2^m]$ and slope 1 after it, so $\min_{R \ge 0} \psi_c(R) = \psi_c(B_S/2^m) = B_S/2^m$, and every program of level 1 has $J_c^B \ge \Phi_1(c)$. With (o), $\Phi^B(c) \ge \min\{\Phi_1(c), \Phi_2(c)\}$, and $\Phi \ge \Phi^B$ because $R \ge R_B$. Since $c \le c_{\mathrm{hi}}$, $J_g(c) \le \Phi_2(c)$, so

$$J_c(\pi_g) - \Phi(c) \le J_g(c) - \min\{\Phi_1, \Phi_2\} = \max\{J_g - \Phi_1, J_g - \Phi_2\} \le \max\{J_g - \Phi_1, 0\} = \Delta(c).$$

Finally $H(p) - B_S/2^m = (2^m H(p) - L_S + \delta_S)/2^m \le (\log(2^m + 1) + \delta_S)/2^m$, by the lower bound on L_S.

(ii). $J_c^B(\pi) \le J_g(c) + \epsilon < \Phi_2(c)$, so $\mathrm{lev}(\pi) = 1$ by (o). Every π with $J_c(\pi) \le \Phi(c) + \epsilon$ qualifies, since $J_c^B \le J_c$ and $\Phi(c) \le J_c(\pi_g) \le J_g(c)$.

(a) By Lemma 4.4, $-\log(u + p + 2^{-B}) \le R_B \le J_c^B \le J_g + \epsilon$; and $\mathrm{lev}(\pi) = 1$ gives $u > \frac{3}{4}$.

(b) By (o), $c(d - 4 - \Lambda_1) + \psi_c(R_B) \le J_g + \epsilon$, that is, $\psi_c(R_B) \le c(4 + \Lambda_1 + \kappa_g) + H(p) + \epsilon$. If $R_B < B_S/2^m$, then $\psi_c(R_B) = cB_S - (c2^m - 1)R_B$. Using the identity $cB_S - H(p) = (c2^m - 1)H(p) - c(2^m H(p) - B_S)$,

$$(c2^m - 1)R_B \ge (c2^m - 1)H(p) - c(2^m H(p) - B_S + 4 + \Lambda_1 + \kappa_g) - \epsilon = (c2^m - 1)H(p) - c\Lambda_2 - \epsilon.$$

If $R_B \ge B_S/2^m$ and $2^m H(p) - B_S > 0$, then $R_B \ge H(p) - (2^m H(p) - B_S)/2^m \ge H(p) - c\Lambda_2/(c2^m - 1)$, because $\Lambda_2 \ge 2^m H(p) - B_S$ and $c/(c2^m - 1) > 2^{-m}$; if $2^m H(p) - B_S \le 0$ the claim is immediate.

(c) $c\,\ell = J_c^B - R_B \le c(d + \kappa_g) + H(p) + \epsilon - R_B$; apply (b) and divide by c.

(d) By (F3), $B_S \le K(\pi \mid a, K(a)) + K(S \mid a, \pi) + \kappa$. By (F4) and (F3), $K(\pi \mid a, K(a)) = K(\pi, a) - K(a) + O(1) \le K(\pi) + K(a \mid \pi) - K(a) + O(1)$. By (F1), Lemma 4.3 with $t = 1$, $K(a) \ge d$ and (c), this is at most $\ell(\pi) - d + 4 + 2\log d + \kappa \le \Delta_\ell + \Lambda_3$.

(e) Let $v(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \oplus 1 \mid (z,w)) \in [0,1]$. Given $x \in E$, z is uniform on S and $y^* = f_a \oplus 1$, so $e(\pi) = \mathrm{avg}_S v$. Since ν_π is a semimeasure, $\bar v := 2^{-m}\sum_z v(z) \le 1 - u(\pi)$. If

$1-u+\gamma_g \geq 1$ there is nothing to prove. Otherwise let $\theta \leq 1$ be the least multiple of 2^{-g} with $\theta \geq 1-u(\pi)+\gamma_g$; then $K(\theta) \leq g + 2\log g + \kappa$ and $\theta \geq \bar{v} + \gamma_g$. Suppose $e(\pi) > \theta$. The set $\mathcal{B}_\theta := \{S' : |S'| = s_0,\ \operatorname{avg}_{S'} v > \theta\}$ is computably enumerable given (π, a, m, s_0, θ), because v is lower semicomputable, and it contains S. By Hoeffding's inequality for sampling without replacement, $|\mathcal{B}_\theta| \leq \binom{2^m}{s_0} e^{-2s_0\gamma_g^2} = 2^{L_S - j_g}$. Describing S by j_g and its index among the first $2^{\lceil L_S \rceil - j_g}$ enumerated elements gives $K(S \mid a, \pi) \leq \lceil L_S \rceil - j_g + 2\log(j_g+1) + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. By (d), $K(S \mid a, \pi) \geq L_S - \delta_S - \Delta_\ell - \Lambda_3$. Together, $j_g - 2\log(j_g+1) \leq \Gamma_g$, contradicting the choice of j_g. So $e(\pi) \leq \theta \leq 1 - u(\pi) + \gamma_g + 2^{-g}$.

(iii). $\operatorname{lev}(\pi) = 1$ follows from (o) as in (ii). For $c2^m < 1$ and $R \geq 0$, $\psi_c(R) \geq cB_S + (1 - c2^m)\min\{R, B_S/2^m\}$: for $R \leq B_S/2^m$ this is an equality, and for $R \geq B_S/2^m$, $\psi_c(R) = R \geq B_S/2^m = cB_S + (1-c2^m)B_S/2^m$. Compare with $J_c^B \leq J_b(c) + \epsilon$ and (o), and use $L_S = B_S + \delta_S$:

$$(1 - c2^m)\min\{R_B, B_S/2^m\} \leq c\big(\lceil L_S \rceil + \kappa_b - B_S + 4 + \Lambda_1\big) + \epsilon = \varrho.$$

If $\varrho < (1-c2^m)B_S/2^m$, the minimum cannot be $B_S/2^m$, so $R_B \leq \varrho/(1-c2^m)$. For the error label, $p\,\mathbb{E}[\min\{B, -\log \nu_\pi(y^* \mid x)\} \mid x \in E] \leq R_B$, and by Jensen's inequality

$$\mathbb{E}[\nu_\pi(y^* \mid x) \mid E] \geq \mathbb{E}[\max\{\nu_\pi(y^* \mid x), 2^{-B}\} \mid E] - 2^{-B} \geq 2^{-R_B/p} - 2^{-B}.$$

(iv). By (D1) with $t = 1$ and $R(\pi) \geq H(p)$, $J_c(\pi) \geq c(d - 4 - \Lambda_1) + H(p) > c(d - 4 - \Lambda_1) + c(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1) = c(d + \lceil L_S \rceil + \kappa_b) \geq J_c(\pi_b)$. □

Numbers. The values of Remark 4.8 and Table 2 were computed by the scripts `c3_junta_window.py` (seed 3; windows, identification on a grid with $g = 7$), `c6_revision.py` (seed 6, part (B); identification with $\delta_S \in \{0, 10, 20\}$ and g optimised over $1, \ldots, 40$) and `c7_run2.py` (seed 7, part (D); the memoriser band), all in the accompanying repository. The script `c3_junta_window.py` also checks the reduction to two endpoints in (o): over 2830 random instances, $\min_{t \geq 2}$ of the exact $c(d - s(t) - \Lambda_1)^+ + G_{t-1}$ never falls below $\Phi_2(c)$. The crossover $G_1 = H(p)$ at $B = \infty$ is $p = 0.0560$. An independent re-implementation reproduces $L_S = 11{,}710.273$ bits from an exact big-integer binomial coefficient, the crossover $p = 0.05596$, the windows by bisection on $J_g \leq \Phi_2$ (closed form and bisection agree to 10^{-15}), and the identification bounds at $c = 10^{-4}$; an exhaustive check of the counting step of (e) over 120 small populations found no violation.

C.4 The full truth table

Proof of Corollary 4.10. With $m = d$ the error set is $E = S$ and $2^m = N$. On the whole truth table the empirical loss equals N times the population risk under uniform inputs, so $J_\lambda(\pi; D) = \lambda\ell(\pi) + NR(\pi) = N\,J_{\lambda/N}(\pi)$, and a λ-MAP minimises $J_{\lambda/N}$ exactly.

(i) Put $c = \lambda/N$. Then $c2^m = \lambda > 1$, and the hypothesis $J_g(c) < \Phi_2(c)$ holds for every $c < c_{\mathrm{hi}}$, by the form of Φ_2. A λ-MAP has $J_c(\pi) = \Phi(c)$, so Theorem 4.7(ii) applies with $\epsilon = 0$, $B = \infty$. Substituting $c2^m - 1 = \lambda - 1$ and $c/(c2^m - 1) = \lambda/(N(\lambda - 1))$ gives the stated forms of (a)–(d).

(ii) At $\lambda = 1$: $J_1(\pi_g) = \ell(\pi_g) + NH(p) \leq d + \kappa_g + L_E + \log(N+1)$, because $NH(p) \leq L_E + \log(N+1)$. Next, $J_1(\pi_b) \leq d + \lceil L_E \rceil + \kappa_b$. Every program of level 1 has $J_1 \geq \ell + NR_B \geq d + B_E - 4 - \Lambda_1 = d + L_E - \delta_E - 4 - \Lambda_1$, by (D2). Every program of level at least 2 has $J_1 \geq NR \geq NG_1$ by Lemma 4.4, and this exceeds $J_1(\pi_g)$ once $N(G_1 - H(p)) > d + \kappa_g + \log(N+1)$, since $L_E \leq NH(p)$. Here $d = \log N$, and κ_g, κ_b and Λ_1 are $O(\log N)$, so the optimum, $J_1(\pi_g)$ and $J_1(\pi_b)$ lie within $O(\log N) + \delta_E$ of each other. □

At $d = m = 20$, $p = 2^{-10}$, $B = \infty$, a direct computation (script `c3_junta_window.py` for the window, `c7_run2.py` part (H) for the bounds, with g optimised) gives the window $\lambda \in (1, 11945]$ at $\kappa = 0$ and $(1, 6456]$ at $\kappa = 30$, and the following bounds for every λ-MAP. Rows with $\kappa = 0$ are illustrations. Near $\lambda = 1$ the deficiency enters the length and information bounds at full weight, through $\Lambda_2/(\lambda - 1)$.

		$\lambda = 2$				$\lambda = 105$		
κ	δ_E	$u \geq$	$\ell \leq$	$I \leq$	$e \leq$	$\ell \leq$	$I \leq$	$e \leq$
0	0	0.991	$d + 58.2$	70.8	0.220	$d + 15.7$	28.3	0.185
0	20	0.991	$d + 78.2$	90.8	0.251	$d + 15.9$	28.5	0.204
30	0	0.991	$d + 148.2$	190.8	0.320	$d + 46.3$	88.9	0.261
30	20	0.991	$d + 168.2$	210.8	0.341	$d + 46.5$	89.1	0.275

In an exact small instance with structured product families ($d = m = 8$, $|E| = 8$), the good model and the memoriser exchange places at $\lambda = 0.936$ (script `c3_junta_window.py`), an illustration of how lower-order terms decide at $\lambda \approx 1$.

C.5 Finite samples

Lemma C.2 (Finite sample)**.** [proved]

steeper Lemma 2.6 (adapts T5 Lemma 1.2) *Let $m \leq d$ and $B < \infty$. Draw $n \leq 2^d$ inputs uniformly without replacement and label them by y^*; let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \leq n} -\log \nu_\pi(y^*(x_i) \mid x_i)$. Let π_{ref} be a fixed program whose per-input losses lie in $[0, A]$. With probability at least $1 - \delta$, every minimiser $\hat{\pi}$ of $\hat{J}_c$ satisfies*

$$J_c^B(\hat{\pi}) \leq J_c(\pi_{\mathrm{ref}}) + \epsilon_1 + B\sqrt{\frac{\hat{L}\ln 2 + \ln(2/\delta)}{2n}}, \qquad \epsilon_1 := A\sqrt{\frac{\ln(2/\delta)}{2n}}, \quad \hat{L} := \frac{J_c(\pi_{\mathrm{ref}}) + \epsilon_1}{c}.$$

Proof. Each sampled input is marginally uniform. By Hoeffding's inequality for sampling without replacement (Hoeffding, 1963), with probability at least $1 - \delta/2$, $\hat{J}_c(\pi_{\mathrm{ref}}) \leq J_c(\pi_{\mathrm{ref}}) + \epsilon_1$. For each program π the truncated losses lie in $[0, B]$; apply the same inequality with confidence $(\delta/2)2^{-\ell(\pi)}$ and take a union bound, which costs at most $\delta/2$ by Kraft's inequality. With probability at least $1 - \delta/2$, simultaneously for all π,

$$\hat{J}_c(\pi) \geq c\,\ell(\pi) + \frac{1}{n}\sum_{i \leq n} \min\{B, -\log \nu_\pi(y^*(x_i) \mid x_i)\} \geq J_c^B(\pi) - B\sqrt{\frac{\ell(\pi)\ln 2 + \ln(2/\delta)}{2n}}.$$

On both events, a minimiser satisfies $c\,\ell(\hat{\pi}) \leq \hat{J}_c(\hat{\pi}) \leq \hat{J}_c(\pi_{\mathrm{ref}}) \leq J_c(\pi_{\mathrm{ref}}) + \epsilon_1$, so $\ell(\hat{\pi}) \leq \hat{L}$, and the two displays combine to the claim. □

The bound is one-sided and uses truncation, so no noise floor on the hypotheses is needed; this adapts the empirical-selection lemma of Claude (Anthropic) (2026, section "Simplicity and normativity from imitation"). The value of c is fixed for the given n, so the lemma also applies when $\lambda = \lambda_n$ follows any schedule; Proposition 5.1(b) uses this.

Proof of Corollary 4.11. A λ-MAP with $\lambda = cn$ minimises $J_\lambda/n = \hat{J}_c$. Apply Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_g$, whose per-input losses $-\log(1-p)$ and $-\log p$ lie in $[0, \log(1/p)]$, so $A = \log(1/p)$. Then $J_c^B(\hat{\pi}) \leq J_c(\pi_g) + \epsilon_n \leq J_g(c) + \epsilon_n$, which is the hypothesis of Theorem 4.7(ii) with $\epsilon = \epsilon_n$. □

At $d = 100$, $m = 20$, $p = 2^{-10}$, $B = 30$, $\delta = 0.01$, $\kappa = 0$ (illustration), $\delta_S = 0$ and $g = 7$, a direct computation (script `c3_junta_window.py`) gives:

c	n	ϵ_n/c (bits)	$u \geq$	$\ell \leq$	$e \leq$
10^{-4}	10^{11}	9.2	0.9825	$d + 29.8$	0.212
10^{-4}	10^{13}	0.9	0.9830	$d + 21.5$	0.204
10^{-3}	10^{11}	0.7	0.9117	$d + 20.7$	0.274

At $n = 10^9$ and $c = 10^{-4}$ the excess is still 92.5 bits ($\ell \leq d + 113.9$, $e \leq 0.284$).

C.6 Version 1

Example 4.12 is Theorem 4.7 at $d = 100$, $m = 10$, $r = 6$. A direct computation (script `c6_revision.py`, parts (C) and (H), seed 6) gives $L_S = 115.58$ bits, $H(2^{-6}) = 0.1161$, and the window ends $c_{\mathrm{hi}} = 5.803 \cdot 10^{-3}$, $4.774 \cdot 10^{-3}$, $4.055 \cdot 10^{-3}$, $3.116 \cdot 10^{-3}$ and $2.530 \cdot 10^{-3}$ at $\kappa = 0, 5, 10, 20, 30$, so the window is nonempty for every $\kappa \leq 30$ checked. At $\kappa = 0$, $\delta_S = 0$, $\epsilon = 0$ and $B = \infty$, (c) and (d) give $\ell \leq d + 103.7$ and $I \leq 121.0$ at $c = 1.5 \cdot 10^{-3}$ (vacuous, since $L_S = 115.6$), $\ell \leq d + 47.8$ and $I \leq 65.0$ at $c = 2.5 \cdot 10^{-3}$, $\ell \leq d + 33.2$ and $I \leq 50.5$ at $c = 4 \cdot 10^{-3}$, and $\ell \leq d + 28.3$ and $I \leq 45.6$ at $c = 5.5 \cdot 10^{-3}$. Part (a) gives $u \geq 0.800$ at $c = 1.5 \cdot 10^{-3}$ and only $u > \frac{3}{4}$ for $c \geq 2.5 \cdot 10^{-3}$, where $2^{-(J_g + \epsilon)} - p$ falls below $\frac{3}{4}$; the bound of (e) exceeds 1 throughout. The threshold of (iv) is $c = 6.917 \cdot 10^{-4}$. For the finite sample at $n = 10^4$: every admissible $B \geq \log(1/p) = 6$ makes the excess of Lemma C.2 at least $6\sqrt{(d - 4 - \Lambda_1) \ln 2/(2n)} = 0.308$, because $\hat{L} \geq \ell(\pi_g) \geq d - 4 - \Lambda_1$, while the margin $\Phi_2 - J_g$ is at most 0.224 anywhere in the window (its largest value is at $B = \infty$).

For comparison, a full-table instance that matches the error rule's size but not the model's ($d = 13$, $m = 10$, $p = 2^{-6}$, $N = 2^{13}$) has the window $\lambda \in (8, 86.2]$ at $\kappa = 0$, $(8, 61.9]$ at $\kappa = 10$ and $(8, 39.6]$ at $\kappa = 30$.

C.7 Concentration of the power posterior

Proof of Proposition 4.13. For every π, $2^{-\lambda \ell(\pi)} \nu_\pi(D) = 2^{-n \hat{J}_c(\pi)}$. Write $\hat{R}(\pi) := \frac{1}{n} \sum_i - \log \nu_\pi(y_i \mid x_i) \geq 0$ for the empirical risk. Then $n\hat{J}_c(\pi) - \ell(\pi) = (\lambda - 1)\ell(\pi) + n\hat{R}(\pi) \geq (1 - 1/\lambda)(\lambda \ell(\pi) + n\hat{R}(\pi)) = (1 - 1/\lambda)\, n\hat{J}_c(\pi)$. So, on the set $\{\pi : \hat{J}_c(\pi) \geq \hat{J}_g + \epsilon\}$,

$$2^{-n\hat{J}_c(\pi)} = 2^{-\ell(\pi)}\, 2^{-(n\hat{J}_c(\pi) - \ell(\pi))} \leq 2^{-\ell(\pi)}\, 2^{-(1-1/\lambda)\, n(\hat{J}_g + \epsilon)},$$

and by Kraft's inequality the sum over the set is at most $2^{-(1-1/\lambda) n (\hat{J}_g + \epsilon)}$. Divide by $2^{-n\hat{J}_g}$, the unnormalised weight of π_g, and use $n/\lambda = 1/c$: the ratio is at most $2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}$.

For the consequence, work on the event of Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_g$ and $A = \log(1/p)$, on which $\hat{J}_g \leq J_g(c) + \epsilon_1$ and the uniform deviation bound holds for every program. A program with $\hat{J}_c(\pi) < \hat{J}_g + \epsilon$ has $\ell(\pi) \leq \hat{J}_c(\pi)/c < \hat{L}'$, hence

$$J_c^B(\pi) \leq \hat{J}_c(\pi) + B\sqrt{\frac{\hat{L}' \ln 2 + \ln(2/\delta)}{2n}} \leq J_g(c) + \epsilon',$$

and Theorem 4.7(ii) applies with ϵ' when $J_g(c) + \epsilon' < \Phi_2(c)$. The bad set has posterior mass at most the ratio times $\Pi_\lambda(\pi_g \mid D) \leq 1$. □

Near-minimisers can be ϵ/c bits longer than the minimiser, which is why the length budget is $\hat{L}'$ and not the $\hat{L}$ of Lemma C.2. At $c = 10^{-4}$, $\epsilon = 10^{-3}$, $n = 10^{13}$, $B = 30$, $\delta = 0.01$ and $\kappa = 0$, a direct computation (script `c7_run2.py`, part (F)) gives $\hat{L}' = 241.7$ bits and $\epsilon' = 1.0933 \cdot 10^{-3}$, and the condition $J_g + \epsilon' < \Phi_2$ holds. The example in the main text is from script `c5_limits.py`, part (e), seed 5: $-n\epsilon + (J_g + \epsilon)/c = -1000 + 242.3$. Two further consequences: the steeper predictive, averaged over E, gives the erroneous label probability at most $\sup e(\pi) + 2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}$, the supremum over the programs just described; averaged over the unseen error inputs $E \setminus \operatorname{dom} D$ instead, the bound is multiplied by $|E|/(|E| - n)$, which is negligible at $d = 100$.

C.8 Computations

All numbers of Section 4 and this appendix come from deterministic or seeded scripts in the accompanying repository, with saved outputs: `c1_hanni_numbers.py` (deterministic; Hänni's two versions, the three-hypothesis windows, the $\lambda = 1$ switch at $n = 8.941 \cdot 10^5$, L_S), `c2_pricing_kink.py` (seed 2), `c3_junta_window.py` (seed 3), `c5_limits.py` (seed 5), `c6_revision.py` (seed 6) and `c7_run2.py` (seed 7). The scripts of the earlier formalisation (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation") were re-run from scratch copies and reproduce its window $(1.119 \cdot 10^{-6}, 8.989 \cdot 10^{-3})$ for Hänni's three hypotheses, with ratio 8037, and its $\lambda = 1$ switch at $n = 894{,}054$. Numbers are given to the precision the scripts print.

D Proofs for the limits of the trick

The setting is Definition 4.2 unless stated otherwise; (F1)–(F5) are the facts of Appendix A.1.

D.1 Plain Bayes, MDL and SI

Proof of Proposition 5.1. (a) The memoriser π_b is deterministic and consistent with every D_n, so $P(D_n) \geq 2^{-\ell(\pi_b) - \kappa}$. Since P is a shared hypothesis, its sequential log loss telescopes: $\sum_{i \leq n} -\log P(y^*(x_i) \mid x_i, D_{i-1}) = -\log P(D_n) + \log P(\emptyset) \leq \ell(\pi_b) + \kappa$. Each conditional is at most 1, so each term is nonnegative, and a label with probability below $\frac{1}{2}$ adds more than one bit; normalised conditionals are at least the raw ones. On an error input the teacher's label is the erroneous one. The memoriser can be given $s_0 = 2^{m-r}$ through r, so $\ell(\pi_b) \leq d + \lceil L_S \rceil + O(\log d + \log m)$. If all 2^d inputs are revealed there are $p2^d$ error inputs, and at most $d + \lceil L_S \rceil + \kappa_b + \kappa$ of them get the erroneous label with probability below $\frac{1}{2}$. At $d = 100$, $m = 20$, $p = 2^{-10}$ and $\kappa = 0$ there are $1.238 \cdot 10^{27}$ error inputs and at most 11,853 exceptions, a fraction $9.57 \cdot 10^{-24}$ (script `c6_revision.py`, part (G), seed 6). For M_F, $P \geq 2^{-K(\nu_\pi) - O(1)} \nu_\pi$ because product laws are machine laws, and $K(\nu_\pi) \leq \ell(\pi) + \kappa$ by (F1).

(b) Apply Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_b$, whose per-input losses are 0, so $A = 0$ and $\epsilon_1 = 0$, and with $c = \lambda_n/n$; then its $\hat{L}$ is $\ell(\pi_b) \leq d + \lceil L_S \rceil + \kappa_b$. A λ-MAP minimises $\hat{J}_c$, so with probability at least $1 - \delta$, $R_B(\hat{\pi}) \leq J_c^B(\hat{\pi}) \leq c\,\ell(\pi_b) + B\sqrt{(\hat{L} \ln 2 + \ln(2/\delta))/(2n)}$. The lemma holds for each n separately, so λ may depend on n. The error-label bound is the last step of the proof of Theorem 4.7(iii): $p\,\mathbb{E}[\min\{B, -\log \nu\} \mid E] \leq R_B$ and Jensen's inequality. If $\lambda_n/n \to 0$, both terms of the bound on R_B tend to 0. Numbers (script `c7_run2.py`, part (G), seed 7; $\kappa = 0$, $B = 30$, $\delta = 0.01$): for $\lambda = 1$, $e \geq 0.255$, 0.649, 0.958, 0.996 at $n = 10^{12}, 10^{13}, 10^{15}, 10^{17}$; for $\lambda_n = \sqrt{n}$, $e \geq 0.045$, 0.734, 0.970 at $n = 10^{13}, 10^{15}, 10^{17}$. At $n = 10^{12}$, $\lambda_n/n = 10^{-6}$ still lies in the window of Theorem 4.7, and the bound says nothing. For $\lambda_n = n^{0.9}$, whose ratio λ_n/n tends to 0 slowly, the bound is still 0 at $n = 10^{17}$.

(c) Encode the interleaved sequence with fixed-length blocks; with uniform i.i.d. inputs and $y_i = y^*(x_i)$ its law $\mathbb{P}$ is computable from a, S, d, m and r, so it has a program $\pi_{\mathbb{P}} \in \mathcal{H}_{\mathrm{SI}}$ with $\ell(\pi_{\mathbb{P}}) \le d + \lceil L_S \rceil + O(\log d + \log m)$. Since $M \ge 2^{-\ell(\pi_{\mathbb{P}})}\mathbb{P}$, the chain-rule argument in the proof of Theorem 3.2(d) gives $\mathbb{E}_{\mathbb{P}} \sum_{i \le n} -\log M(y_i \mid x_1 y_1 \cdots x_i) \le \ell(\pi_{\mathbb{P}})$. For $\lambda > 1$ apply Theorem 3.2(d) itself, with $\kappa_0 = 0$ in $\mathcal{H}_{\mathrm{SI}}$. □

D.2 Rate blindness: the criterion in the machine constant

Proof of Proposition 5.4. (i) $J_c(\pi_b) < J_c(\pi_g)$ if and only if $c < H(p)/(\ell(\pi_b) - \ell(\pi_g))$, and $J_c(\pi_{\mathrm{coin}}) \le J_c(\pi_g)$ if and only if $c \ge (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\mathrm{coin}}))$; under the hypothesis these two ranges cover all $c > 0$. (ii) Let $u(\pi) > \frac{3}{4}$, so $\mathrm{lev}(\pi) = 1$, and $R(\pi) \ge H(p)$. Below the left side of $(\star)$, Theorem 4.7(iv) gives $J_c(\pi_b) < J_c(\pi)$. At or above it, $c\,(d - 4 - \Lambda_1 - \kappa) \ge 1 - H(p)$ by $(\star)$, so (D1) gives $J_c(\pi) \ge c(d - 4 - \Lambda_1) + H(p) \ge 1 + c\kappa \ge J_c(\pi_{\mathrm{coin}})$. (iii) is Theorem 4.7(iii). □

Proof of Lemma 5.5. (i) With $B = \infty$, $\Lambda_1 = 2\log d + 2\log m + \kappa$ and $\kappa_b = 2\log d + 2\log m + 2\log s_0 + \kappa$. The first denominator of $(\star)$ is positive. When the second is positive, $(\star)$ is equivalent to $H\,(d - 4 - \Lambda_1 - \kappa) \ge (1 - H)(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$. Dividing by H, the difference of the two sides is

$$d - 4 - 2\log d - 2\log m - 2\kappa - q\big(\lceil L_S \rceil + 4\log d + 4\log m + 2\log s_0 + 2\kappa + 4\big) = d - a\log d - b = F(d).$$

Conversely, $F(d) \ge 0$ implies $d - 4 - \Lambda_1 - \kappa = d - 4 - 2\log d - 2\log m - 2\kappa \ge F(d) + 4q\log d + q(\lceil L_S \rceil + 4) > 0$, so the side condition holds as well.

(ii) $F'(d) = 1 - a/(d \ln 2)$, so F is strictly increasing on $[a/\ln 2, \infty)$ and tends to ∞. On $[1, a/\ln 2]$, $F(1) = 1 - b < 0$ because $b \ge 4$, and for $d > 1$, $F(d) < d - b \le a/\ln 2 - b \le 0$. So $F < 0$ on $[1, a/\ln 2]$, and $\{d \ge 1 : F(d) \ge 0\} = [d_0, \infty)$ for the unique root $d_0 > a/\ln 2$.

(iii) b is affine in κ with slope $2 + 2q$, a does not depend on κ, and $d_0 = b + a\log d_0$.

Taking all three constants equal to a common upper bound is legitimate because $(\star)$ only becomes harder to satisfy when any of them grows. □

A direct computation (script `c6_revision.py`, part (A), seed 6) gives $d_0(\kappa)$, the least integer d with $F(d) \ge 0$, for five cheap error rules. In every row $b \ge a/\ln 2$; $(\star)$ fails at $d_0 - 1$, holds at d_0, and holds at each of about 400 values of d in $[d_0, 10^6]$.

error rule	$\kappa = 0$	10	20	30	50	100
$m = 3$, $p = 1/8$ (one 3-bit prefix)	49	90	130	169	245	434
$m = 4$, $p = 1/16$ (one 4-bit prefix)	106	172	236	299	422	727
$m = 6$, $p = 1/64$ (one 6-bit prefix)	450	639	823	1005	1364	2248
$m = 7$, $p = 1/128$ (one 7-bit prefix)	901	1231	1554	1873	2505	4063
$m = 6$, $p = 1/8$ (eight 6-bit prefixes)	89	129	168	206	282	470

Illustrations at $\kappa = 0$. These are not proved for any reference machine. In the three-hypothesis form Proposition 5.4(i), with the lengths set equal to their upper bounds $d + \kappa_g$ and $d + \lceil L_S \rceil + \kappa_b$, at $d = 100$ a single 6- or 7-bit prefix is never selected, with margins of 9.6 and 1.4 bits (the verdict flips if $\ell(\pi_b) - \ell(\pi_g)$ is larger by that much), while an 8-bit prefix leaves a window. In the all-hypothesis form, the 3-bit rule satisfies $(\star)$ at $d = 100$ for $\kappa \le 10$ and fails for $\kappa \ge 20$. In the memoriser regime of a single 6-bit prefix at $c = 10^{-4}$, near-optimal programs give the error label probability at least 0.833 (script `c3_junta_window.py`, seed 3).

D.3 Separable components, guard erosion, non-identifiability, input law

Proof of Proposition 5.6. The objective $c\,\ell + R$ is a sum over regions of (extra length times c) plus (risk on the region), and the choices in different regions are independent, so it is minimised region by region; in each region Lemma 2.8 applies. With two options, component j lowers the objective if and only if $c\,k_j < r_j$. If a valid component with rate ρ_v is kept, then $c < \rho_v \le \rho_e$ (up to ties), so an error component with rate ρ_e is kept as well. Finally, J_c depends on the data only through the teacher's channel and the input law, which coincide in the two worlds. □

Proof of Proposition 5.7. Per context of the region, the expected log losses are: for "none", $\log n_{\mathrm{st}}$ everywhere; for "unguarded σ", H_G in G-contexts and $\log(n_{\mathrm{st}}/\epsilon_G)$ in $\neg G$-contexts, where the teacher's fixed step has probability $\epsilon_G/n_{\mathrm{st}}$ under the G-context law; for "guarded σ", H_G in G-contexts and $\log(n_{\mathrm{st}} - 1)$ in $\neg G$-contexts. Relative to "none", the unguarded option lowers the risk by $\mu(G)g - \mu(\neg G)\log(1/\epsilon_G) = k_\sigma\rho_\sigma$ for k_σ bits, and the guard lowers it by a further $\mu(\neg G)\Delta_G = k_G\rho_G$ for k_G more bits. So the three options are the points $(0,0)$, $(k_\sigma, -k_\sigma\rho_\sigma)$ and $(k_\sigma + k_G, -k_\sigma\rho_\sigma - k_G\rho_G)$ in the (length, risk) plane, after subtracting the common base. The middle point is a vertex of the hull if and only if the slope $-\rho_\sigma$ of the first segment is steeper than the slope $-\rho_G$ of the second, that is, $\rho_\sigma > \rho_G$. By Lemma 2.8(iii) it is then the unique minimiser for $c \in (\rho_G, \rho_\sigma)$. Otherwise the hull has only the two outer points, and the selection switches at the slope between them, $(k_\sigma\rho_\sigma + k_G\rho_G)/(k_\sigma + k_G)$. □

Numbers (script `c7_run2.py`, part (E), seed 7; an independent computation from the same teacher model agrees). With $n_{\mathrm{st}} = 64$ and $\epsilon_G = 0.05$, the G-context law puts 0.95078 on σ, $H_G = 0.5773$, $g = 5.4227$ and $\Delta_G = 4.3446$. For $\mu(G) = 0.05$, $\mu(\neg G) = 5\cdot 10^{-4}$, $k_\sigma = 22$, $k_G = 12$: $\rho_\sigma = 1.2226\cdot 10^{-2}$, $\rho_G = 1.8103\cdot 10^{-4}$, and as c increases the selection goes guarded, unguarded (on $(1.8103\cdot 10^{-4}, 1.2226\cdot 10^{-2})$), none. For $\mu(G) = \mu(\neg G) = 0.002$: $\rho_\sigma = 1.0007\cdot 10^{-4} < \rho_G = 7.2411\cdot 10^{-4}$, and the selection goes from guarded to none at $c = 3.2032\cdot 10^{-4}$. The teacher has noise ϵ_G in G-contexts; what it never does is apply σ outside G.

Proof of Proposition 5.8. (i) A scoring rule is strictly proper if, for each x, the expected score of a forecast Q under the label law $P^*(\cdot \mid x)$ is uniquely maximised at $Q = P^*(\cdot \mid x)$ (Gneiting and Raftery, 2007); average over x. (ii) The held-out log loss of the population selection h_c is $R(h_c)$, and by Lemma 2.8(ii), $c < c'$ implies $R(h_c) \le R(h_{c'})$. (iii) An imitation-only rule sees the same data distribution in both worlds, since the teacher's channel and the input law are the same, and Proposition 5.6 gives the same selection at every c; the selection that is right in one world keeps an error or drops a valid rule in the other. □

Proof of Proposition 5.9. The population risk is $\sum_j \mu(X_j)\mathbb{E}[\text{loss} \mid x \in X_j]$, so choosing component j changes J_c by $c\,k_j - \mu(X_j)\Delta_j$; the other claims are rearrangements of this criterion. □

The numbers in Proposition 5.8 are from script `c5_limits.py`, part (c), seed 5, for the three hypotheses of Remark 4.1; the same script checks the separable thresholds of Proposition 5.6 by brute force in 8000 (instance, c) pairs, with no violation.

D.4 Error envelopes

Proof of Proposition 5.3. The envelope. Off E_1 the inputs with prefix in S_2 are the error inputs, a fraction p_2 of the inputs off E_1, so $R(\pi_e) = (1-p_1)H(p_2)$. Since $p = p_1\cdot 1 + (1-p_1)p_2$, concavity of H gives $H(p) \ge p_1 H(1) + (1-p_1)H(p_2) = R_e$. The program π_e needs a, d, m, a description of S_1 given m, and s_2 (from which p_2 follows), which gives the length bound.

(i) The error set still has density p, so Theorem 4.7(o) applies: programs of level at least 2 have $J_c \geq J_c^B \geq \Phi_2(c) > J_e(c) + \epsilon$, so $\mathrm{lev}(\pi) = 1$. Then (D1) gives $\ell(\pi) \geq d - 4 - \Lambda_1$, and $c(d - 4 - \Lambda_1) + R(\pi) \leq J_c(\pi) \leq c(d + k_1 + \kappa_e) + R_e + \epsilon$, that is, $R(\pi) \leq R_e + \zeta$.

(ii) Write $R(\pi) = p_1 R^{(1)} + (1 - p_1)R^{(2)}$, where $R^{(1)}$ and $R^{(2)}$ are the risks conditional on E_1 and off E_1, and put $x := (1 - p_1)R^{(2)}$.

Coding S_2 given S_1. For $z \notin S_1$ define $q_z(b)$ as in the proof of Lemma 4.5 with $B = \infty$. Then $Q(S') := \prod_{z \notin S_1} q_z(\mathbf{1}_{S'}(z))$ is a lower-semicomputable semimeasure on subsets of $\{0,1\}^m \setminus S_1$, given (π, a, m, S_1). For $z \notin S_1$ the teacher's label is $f_a \oplus \mathbf{1}_{S_2}(z)$, so $-\log Q(S_2) = (2^m - s_1)R^{(2)} = 2^m x$. By (F2), $K(S_2 \mid a, \pi, S_1) \leq 2^m x + 2\log m + \kappa$.

Chain. By (F3), (F1) and Lemma 4.3 with $t = 1$, $K(a, S_2 \mid S_1) \leq \ell(\pi) + 4 + 2\log d + 2^m x + 2\log m + \kappa$. By (F3) and (F4), $K(a, S_2 \mid S_1) \geq K(a, S_2) - K(S_1) - \kappa \geq d + B_2 - K(S_1) - \kappa$. With $\Lambda_1 = 2\log d + 2\log m + \kappa$ and (D1),

$$\ell(\pi) \;\geq\; d - 4 - \Lambda_1 + \max\{0,\; B_2' - 2^m x\}, \qquad B_2' := \big(B_2 - K(S_1)\big)^+.$$

Combination. $J_c(\pi) = c\,\ell(\pi) + p_1 R^{(1)} + x \leq J_e(c) + \epsilon$ gives

$$p_1 R^{(1)} \leq \zeta + (1 - p_1)H(p_2) - \big[x + c\max\{0, B_2' - 2^m x\}\big].$$

Over $x \geq 0$ the bracket is piecewise linear: for $x \leq B_2'/2^m$ it equals $cB_2' + (1 - c2^m)x$, and for larger x it equals x. So its minimum is $B_2'/2^m$ (at $x = B_2'/2^m$) if $c2^m \geq 1$, and cB_2' (at $x = 0$) if $c2^m < 1$; that is, $\min(c, 2^{-m})B_2'$. Hence $R^{(1)} \leq \bar{R}_1$, and Jensen's inequality gives $e_1(\pi) \geq 2^{-R^{(1)}} \geq 2^{-\bar{R}_1}$.

For $c \geq 2^{-m}$, use $(2^m - s_1)H(p_2) \leq \log\binom{2^m - s_1}{s_2} + \log(2^m + 1)$, which gives $\bar{R}_1 \leq \big(\zeta + (\delta_2 + \log(2^m + 1) + K(S_1))/2^m\big)/p_1$ with $\delta_2 := \log\binom{2^m - s_1}{s_2} - B_2$. □

Numbers (script `c7_run2.py`, part (B), seed 7). The script also checks the minimisation identity on 2000 random cases and, on 40 exact small instances ($d = 6$, $m = 4$), that $-\log Q(S_2) = (2^m - s_1)R^{(2)}$ and that Q has total mass at most 1. At $d = 100$, $m = 20$, $p = 2^{-10}$, with $S_1 = \{z : z_1 = \cdots = z_j = 0\}$, so that $K(S_1 \mid m) \leq 2\log(j + 1) + \kappa$, and S_2 uniformly random (so δ_2 is small with high probability), at $\kappa = 0$ unless stated:

instance	$\log\binom{2^m - s_1}{s_2}$	$H(p) - R_e$	$10^5 c_{\mathrm{env}}$ ($\kappa = 0/10/30$)	share ($\kappa = 0/30$)
$j = 11$ $(512 + 512)$	6,364	0.00510	6.98/4.94/3.12	0.47 / 0.42
$j = 13$ $(128 + 896)$	10,418	0.00123	1.64/1.17/0.746	0.31 / 0.25

lower bound on e_1 ($\delta_2 = 0$)	$c = 1.01 \cdot 2^{-m}$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-6}$	10^{-5}	$3 \cdot 10^{-5}$
$j = 11$	0.862	0.774	0.567	0.338	0.042
$j = 13$	0.545	0.350	0.097	0.012	0.000

In both instances $J_e < \Phi_2$ on $(0, c_{\mathrm{env}}]$, as part (i) requires. With $\delta_2 = 20$ the first column becomes 0.839 and 0.489. Here $j = 11$ means $s_1 = 512$ cheap and $s_2 = 512$ random points, and the share is the share of the window's logarithmic width below c_{env}, $\log(c_{\mathrm{env}}2^m)/\log(c_{\mathrm{hi}}2^m)$, with the window $(2^{-m}, c_{\mathrm{hi}}]$ of Theorem 4.7 for an incompressible rule. Against $L_S = 11{,}710$ the deficiencies of the two rules are about 5,300 and 1,300 bits.

A three-hypothesis illustration ($\kappa = 0$, upper-bound lengths treated as exact; not proved). For $j = 11$, the envelope beats the good model for $c < 1.87 \cdot 10^{-4}$ with the generic code lengths above; an independent computation with explicit code lengths ($\ell(\pi_g) = d + 19.9$, $\ell(\pi_e) = d + 28.9$) gives

$c < 5.7 \cdot 10^{-4}$, where the envelope gives the erroneous label average probability 0.50 on E, against 0.001 for the good model. A memoriser of the whole rule wins only below $9.5 \cdot 10^{-7}$.

Envelopes with density $\theta < 1$. If the cheap region contains errors with density θ, the envelope predicts the error there with probability θ, and the proof of part (i) goes through with $R_e = p_1 H(\theta) + (1 - p_1) H(p_2)$ and the corresponding length. For $\theta < \frac{1}{2}$ the intended label is still the more likely one inside the region; for $\theta > \frac{1}{2}$ the envelope is a learned systematic error. Part (ii) is proved only for $\theta = 1$.

D.5 Unseen properties

Proof of Proposition 5.10. (i) Let V' be the prefix machine that reads a self-delimiting code z of a pair (decider q for Q, program r for b), then a program $\pi' \in \operatorname{dom} V$, and outputs an index of the conditional semimeasure that is the point mass at $b(x)$ (or 0 while $b(x)$ is undefined) if q accepts x, and $\nu_{\pi'}(\cdot \mid x)$ otherwise; this is lower semicomputable. By closure under prefix-free simulation (Definition 2.4) there is ρ with $V(\rho z \pi') = V'(z\pi')$; let $\kappa := |\rho|$ and take $|z| = K(Q, b)$. Distinct π' give distinct programs $\rho z \pi'$. Since $\operatorname{dom} D \cap Q = \emptyset$, $\nu_{\rho z \pi'}(D) = \nu_{\pi'}(D)$, and since $Z \subseteq Q \cap \operatorname{dom} b$, $\nu_{\rho z \pi'}(D \cup b|_Z) = \nu_{\pi'}(D)$. So

$$\sum_{\pi' \in \operatorname{dom} V} 2^{-\lambda |\rho z \pi'|}\, \nu_{\rho z \pi'}(D) = 2^{-\lambda (K(Q,b) + \kappa)} \sum_{\pi'} 2^{-\lambda |\pi'|} \nu_{\pi'}(D) = 2^{-\lambda (K(Q,b) + \kappa)}\, \xi_\lambda(D),$$

which is the posterior mass claim, and the same sum with $D \cup b|_Z$ in place of D is a lower bound for $\xi_\lambda(D \cup b|_Z)$.

(ii) Equal likelihoods cancel in the ratio of posteriors, which leaves the ratio of prior weights $2^{-\lambda \ell(\pi_1)} / 2^{-\lambda \ell(\pi_2)}$. □

On a conjunction ladder, Hänni (2026a, the conjunction-ladder theorem and the remark on uniform tie-breaking) gives a concrete instance of the warning in (ii): deterministic selection of a least-size consistent formula makes exponentially many mistakes on some ordering of the inputs, for every tie rule, while uniform tie-breaking among the least-size consistent formulas, the $\lambda \to \infty$ limit of the power posterior there, makes $O(d^2)$ Gibbs mistakes. This does not settle the universal case, and neither does Proposition 5.10.

E Proofs for the model–randomness split

The proof of Theorem 6.4 is given in the main text.

E.1 Proof of the Ville lemma

We prove Lemma 6.2. Let $Z_n := \xi(D_n)/\nu(D_n)$. Given the past and the next input x_{n+1}, which the input rule chooses before y_{n+1} is drawn,

$$\mathbb{E}[Z_{n+1}] = \sum_y \nu(y \mid x_{n+1}) \frac{\xi(D_n + (x_{n+1}, y))}{\nu(D_n)\, \nu(y \mid x_{n+1})} \le \frac{\sum_y \xi(D_n + (x_{n+1}, y))}{\nu(D_n)} \le Z_n,$$

the sum running over labels of positive ν-probability, by the one-point inequality. So (Z_n) is a nonnegative supermartingale with $Z_0 = \xi(\emptyset) \le 1$, also when the input rule is adaptive or randomised. Ville's maximal inequality (Ville, 1939) gives $\mathbb{P}[\sup_n Z_n \ge 2^j] \le Z_0\, 2^{-j} \le 2^{-j}$. For the consequence, $\xi_{\mathcal{C}} \ge 2^{-\ell(\pi)} \nu_\pi$, so $\nu_\pi(D_n) \le 2^{\ell(\pi)} \xi_{\mathcal{C}}(D_n) \le 2^{\ell(\pi) + \delta^*} \nu(D_n)$. □

E.2 Proof of the stream-prefix lemma

We prove Lemma 6.3. Order $Y = \{y^{(1)}, \dots, y^{(|Y|)}\}$.

Private streams. Let $j(x)$ be the length-lexicographic index of x, so $j(x) \le 2^{d+1}-2$ for $|x| \le d$, and let $\langle a, b\rangle := (a+b)(a+b+1)/2 + b$ be the Cantor pairing, which is injective and increasing in b. For a finite string w (bits indexed from 0), the *stream prefix* of x in w is $w_{\langle j(x),0\rangle} w_{\langle j(x),1\rangle} \cdots$, continued while the positions are below $|w|$. Distinct inputs read disjoint positions.

Certification. For an input x put $F_h := \sum_{h' \le h} \nu(y^{(h')} \mid x)$, so $F_0 = 0$ and $F_{|Y|} = 1$. At depth $i \ge 1$ compute, by a fixed algorithm from the program for ν, rationals $q_{i,h}$ with $|q_{i,h} - F_h| \le 2^{-i-2}$ ($q_{i,0} = 0$, $q_{i,|Y|} = 1$). A stream prefix $b_1 \cdots b_i$ names the dyadic interval $I = [0.b_1 \cdots b_i,\ 0.b_1 \cdots b_i + 2^{-i})$. It *certifies* $y^{(h)}$ at depth i if $q_{i,h-1} + 2^{-i-2} \le \min I$ and $\sup I \le q_{i,h} - 2^{-i-2}$. Then $I \subseteq [F_{h-1}, F_h]$.

Consistency. Suppose the stream of x certifies $y^{(h)}$ at depth i and $y^{(h')}$ at depth i', with $h < h'$. The two intervals are nested, since one stream prefix extends the other; the inner one has positive length and lies in $[F_{h-1}, F_h] \cap [F_{h'-1}, F_{h'}]$. For $h < h'$ this intersection is empty or the single point F_h, a contradiction. So $\varphi_w(x)$, defined as the label certified at some depth within the stream prefix of x in w (undefined if none), is a well-defined partial computable function, and an index of it is computable from $(V(\pi_\nu), w)$.

Probability. Every depth-i dyadic interval inside $[F_{h-1} + 2^{-i-1}, F_h - 2^{-i-1}]$ certifies $y^{(h)}$, because $q_{i,h-1} + 2^{-i-2} \le F_{h-1} + 2^{-i-1}$ and $q_{i,h} - 2^{-i-2} \ge F_h - 2^{-i-1}$. These intervals cover that segment except for at most $2 \cdot 2^{-i}$, and the segment has length $\nu(y^{(h)} \mid x) - 2^{-i}$. So a uniform stream prefix of length at least i certifies $y^{(h)}$, and hence gives $\varphi_w(x) = y^{(h)}$, with probability at least $\nu(y^{(h)} \mid x) - 3 \cdot 2^{-i}$.

Independence. For w uniform on $\{0,1\}^L$, the stream prefixes of distinct inputs are independent and uniform. If $L \ge (2^{d+1} + i)^2$, every input of length at most d has a stream prefix of length at least i, because $\langle j(x), i-1\rangle < (j(x)+i)^2 \le (2^{d+1}+i)^2$. Take $i = i_D$; then $3 \cdot 2^{-i_D} \le \beta_D/(2n)$, and

$$\Pr_w[\varphi_w \supseteq D] \ \ge \prod_{(x,y)\in D} (\nu(y \mid x) - 3 \cdot 2^{-i_D}) \ \ge \ \nu(D)\Big(1 - \frac{1}{2n}\Big)^n \ \ge \ \frac{\nu(D)}{2},$$

using $\nu(y \mid x) \ge \beta_D$ and Bernoulli's inequality.

Programs. Let V' be the prefix machine that reads π_ν (by simulating V, which consumes exactly π_ν), then a self-delimiting code of an integer t of length at most $2\log(t+1) + 1$, then $L_t := 2^{2^t}$ bits w, and outputs an index of the deterministic conditional semimeasure φ_w. By closure under prefix-free simulation (Definition 2.4) there is r with $V(ru) = V'(u)$. The programs $p_w := r\,\pi_\nu\,\mathrm{code}(t)\,w$ are distinct for distinct w. With $t = t_D$ we have $2^{t_D} \ge 2d + 2 + 2\log(i_D + 1)$, so $L_{t_D} \ge \big(2^{d+1}(i_D+1)\big)^2 \ge (2^{d+1} + i_D)^2$.

Sum.

$$W_{\mathrm{det}}(D) \ge \sum_w 2^{-|p_w|} \mathbf{1}[\varphi_w \supseteq D] = 2^{-|r| - k - |\mathrm{code}(t_D)|} \Pr_w[\varphi_w \supseteq D] \ge 2^{-k - 2\log(t_D+1) - |r| - 2}\, \nu(D).$$

□

Computation. A direct computation (code in the accompanying repository, seed 7) checked the certification step on 600 random samplers ($|Y| \le 4$, rational values with various denominators, Bern(1/3) included) with adversarial approximations $q_{i,h} = F_h \pm 2^{-i-2}$: in all 1646 label–depth cases the certified mass was at least $\nu - 3 \cdot 2^{-i}$, no two certificates conflicted, and the end-to-end ratio at depth i_D was at least 1/2 (smallest 0.645). It also checked the pairing and length facts for $d \le 12$, $i \le 64$. The cost $2\log(t_D + 1)$ is 5.6 bits at $d = 20$, $n = 10^3$ ($\beta_D = 1/3$); 6.3 bits at

$d = 100$, $n = 10^6$ (for $\beta_D = 1/3$ and for $\beta_D = 2^{-10}$); 7.2 bits at $d = 1000$, $n = 10^{12}$ ($\beta_D = 10^{-3}$); and 8.9 bits at $d = 10^6$, $n = 10^{15}$ ($\beta_D = 1/3$). The hashing cost of Lemma E.1 at the same sizes is 31.9, 56.3 (with $r = 2$) or 60.1 (with $r = 10$), 106.6 and 142.7 bits.

Ville's inequality in simulation. For Lemma 6.2, a simulation with the true sampler (code in the accompanying repository, seed 4; 4000 runs) gave empirical frequencies $\mathbb{P}[\delta^* \geq j] = 0.119, 0.055, 0.032, 0.014, 0.003$ for $j = 1, 2, 3, 4, 6$, and an independent simulation with an adversarial adaptive input rule, which puts the next input where ξ deviates most from ν, gave $0.355, 0.163, 0.069, 0.035, 0.0065$; all are below 2^{-j}.

E.3 An alternative for dyadic samplers: hashing

Lemma E.1 (Derandomising a dyadic sampler by hashing)**.** [proved; computed] steeper Lemma 3.2; checks c4(B), c6(E); referee r5, q3(A) *Let the class be $\mathcal{H}_\times$, let ν be a sampler whose values are multiples of 2^{-r}, with a program π_ν of length k, and let D be a data set with n inputs, all of length at most d. Then*

$$W_{\det}(D) \;\geq\; 2^{-k-2\log(n+1)-2\log(d+1)-2\log(r+1)-\kappa}\,\nu(D).$$

Proof. Exact values. From the index $V(\pi_\nu)$ and r one computes ν exactly: enumerate lower approximations $\nu_s(y \mid x) \uparrow \nu(y \mid x)$ until $\sum_y \nu_s(y \mid x) > 1 - 2^{-r}$, which happens because ν is a measure and Y is finite. Then every deficit $\nu - \nu_s$ is below 2^{-r}, and $\nu(y \mid x)$ is the least multiple of 2^{-r} that is at least $\nu_s(y \mid x)$.

Canonical sampler. For $b \in \{0, \dots, 2^r - 1\}$ let $\sigma(x, b)$ be the label y whose cumulative interval $[\sum_{y'<y} \nu(y' \mid x), \sum_{y' \leq y} \nu(y' \mid x))$ contains $b2^{-r}$. Then $\Pr_b[\sigma(x, b) = y] = \nu(y \mid x)$ for uniform b.

Hash. Let $q := \max(d + 1, r)$, fix (computably from q) an irreducible polynomial of degree q over GF(2), and embed $\{0,1\}^{\leq d}$ injectively into $\mathrm{GF}(2^q)$ by mapping x to the element with coefficient vector $1x$. For a seed $s \in \mathrm{GF}(2^q)^n$ let $h_s(x)$ be the first r coefficient bits of $\sum_{j<n} s_j\, x^j$, computed in $\mathrm{GF}(2^q)$ with x embedded. For distinct $x_1, \dots, x_n$ the map $s \mapsto (\sum_j s_j x_i^j)_{i \leq n}$ is a bijection (a Vandermonde matrix with distinct nodes), so for uniform s the values $h_s(x_i)$ are independent and uniform on $\{0,1\}^r$.

Programs, one per seed. Let V' be the prefix machine that reads π_ν (by simulating V), then self-delimiting codes of n, d and r, then qn bits s, and outputs an index of the deterministic conditional semimeasure $x \mapsto \sigma(x, h_s(x))$ on $\{0,1\}^{\leq d}$, undefined elsewhere. By closure under prefix-free simulation there is r' with $V(r'u) = V'(u)$. The programs $p_s := r'\pi_\nu \operatorname{code}(n) \operatorname{code}(d) \operatorname{code}(r)\, s$ are distinct for distinct s, deterministic, and of length $L_0 + qn$ with $L_0 \leq k + 2\log(n+1) + 2\log(d+1) + 2\log(r+1) + \kappa$.

Sum. $\sum_s 2^{-|p_s|} \mathbf{1}[\nu_{p_s} \supseteq D] = 2^{-L_0} \Pr_s[\forall i : \sigma(x_i, h_s(x_i)) = y_i] = 2^{-L_0} \prod_i \nu(y_i \mid x_i) = 2^{-L_0}\nu(D)$. □

Distinct seeds often give the same function on X; the sum is over programs, one per seed, so this does not matter. A direct computation (code in the accompanying repository) checked the independence exactly over GF(16) for $n \leq 3$ (3232 cases) and the full statement with the embedding $1x$, $q = \max(d + 1, r)$, $r \in \{2, 3, 4\}$, $|Y| = 3$ and $n \leq 3$ (2106 cases), with maximum error 0. The hashing idea for fair coins and the program mixture is in Hänni (2026a, the hashing remark).

E.4 Proof of the SI proposition

We prove Proposition 6.5. (i) Under ν, $Z_n := M(z_{1:n})/\nu(z_{1:n})$ is a nonnegative supermartingale with $Z_0 \leq 1$, because $M(z0) + M(z1) \leq M(z)$; Ville's inequality gives the tail bound, and

$\Pi_1(\pi_\nu \mid z_{1:n}) = 2^{-k}\nu(z_{1:n})/M(z_{1:n}) \geq 2^{-k-\delta^*}$. (ii) Let $w := z_{1:n}0^\infty$ and let δ_w be the point mass on its prefixes. Given n and π_ν, the restriction of ν to $\{0,1\}^n$ is a computable measure, so (F2) gives $K(z_{1:n} \mid n, \pi_\nu) \leq -\log \nu(z_{1:n}) + \kappa$, and (F1), (F3) give $K(z_{1:n}) \leq K(n) + k - \log \nu(z_{1:n}) + \kappa$. An index of δ_w is computable from $z_{1:n}$, so the class contains a program for δ_w of length at most $K(z_{1:n})+\kappa$, and δ_w is consistent with $z_{1:n}$. (iii) The programs "a fixed decoder, a shortest program for ν, then a Shannon–Fano code of z under ν" have total weight at least $2^{-k-\kappa}\nu(z)$; this is the usual proof of $M \geq 2^{-K(\nu)-O(1)}\nu$ (Li and Vitányi, 2008). □

E.5 Proof of the private-randomness theorem

We prove Theorem 6.6. Write $D = D_n$.

(i). $J_\lambda(\hat\pi; D) \leq J_\lambda(\pi_\nu; D) = \lambda k - \log \nu(D)$. By Lemma 6.2, $-\log \nu_{\hat\pi}(D) \geq -\log \nu(D) - \ell(\hat\pi) - \delta^*$. Substituting gives $(\lambda-1)\ell(\hat\pi) \leq \lambda k + \delta^*$. The upper bound on the loss is $J_\lambda(\hat\pi; D) \leq J_\lambda(\pi_\nu; D)$ with $\lambda\ell(\hat\pi) \geq 0$.

(ii). $\sum_{\ell(\pi)>L} 2^{-\lambda\ell(\pi)}\nu_\pi(D) \leq 2^{-(\lambda-1)L}\xi_{\mathcal{C}}(D) \leq 2^{-(\lambda-1)L+\delta^*}\nu(D)$, and the normaliser $\xi_\lambda(D)$ is at least the sampler's term $2^{-\lambda k}\nu(D)$.

(iii). A consistent deterministic π has $1 = \nu_\pi(D) \leq 2^{\ell(\pi)+\delta^*}\nu(D)$, which is the length bound. For the power posterior, $\sum_{\mathrm{det}} 2^{-\lambda\ell(\pi)} = \sum_{\mathrm{det}} 2^{-\ell(\pi)}2^{-(\lambda-1)\ell(\pi)} \leq 2^{-(\lambda-1)(-\log\nu(D)-\delta^*)}W_{\mathrm{det}}(D)$, the sums running over consistent deterministic programs; with $W_{\mathrm{det}} \leq \xi_{\mathcal{C}} \leq 2^{\delta^*}\nu(D)$ and the normaliser at least $2^{-\lambda k}\nu(D)$ this gives the bound. For the tempered posterior, the fill-ins' total weight is $W_{\mathrm{det}}(D) \leq 2^{\delta^*}\nu(D)$, and the normaliser is at least the sampler's term $2^{-k}\nu(D)^\eta$.

(iv), likelihood ratios. Fix $\pi \in \mathcal{C}$ and let $\Lambda_n := \nu_\pi(D_n)/\nu(D_n)$ and $g_n := d_{\mathrm{H}}^2(\nu(\cdot \mid x_n), \nu_\pi(\cdot \mid x_n))$; g_n is determined before y_n is drawn. Given the past and x_{n+1},

$$\mathbb{E}\sqrt{\Lambda_{n+1}/\Lambda_n} = \sum_y \nu(y \mid x_{n+1})\sqrt{\nu_\pi(y \mid x_{n+1})/\nu(y \mid x_{n+1})} = 1 - g_{n+1}.$$

If $g_{n+1} = 1$, then almost surely $\nu_\pi(y_{n+1} \mid x_{n+1}) = 0$ and Λ stays 0 from then on. Let $M_n := \sqrt{\Lambda_n}/\prod_{i\leq n}(1-g_i)$ while the product is positive, and $M_n := 0$ afterwards. Then (M_n) is a nonnegative supermartingale, so it converges almost surely to a finite limit. On the event $\{\sum_i g_i = \infty\}$ either some $g_i = 1$, and $\Lambda_n = 0$ eventually, or $\prod_{i\leq n}(1-g_i) \to 0$, and $\sqrt{\Lambda_n} = M_n\prod_{i\leq n}(1-g_i) \to 0$. This is the argument behind Kakutani's dichotomy (Kakutani, 1948); it holds for adaptive inputs because g_n is predictable.

(iv), MAPs. There are countably many programs, so almost surely, simultaneously, $\delta^* < \infty$ and $\Lambda_n(\pi) \to 0$ for every π that is not ν-equivalent on the run. By (i), every λ-MAP lies in the finite set $\{\pi : \ell(\pi) \leq L^*\}$ with $L^* := (\lambda k + \delta^*)/(\lambda - 1)$. A λ-MAP $\hat\pi$ has $\log\Lambda_n(\hat\pi) \geq \lambda\ell(\hat\pi) - \lambda k \geq -\lambda k$, from $J_\lambda(\hat\pi) \leq J_\lambda(\pi_\nu)$. Each non-equivalent program in the finite set eventually has $\Lambda_n < 2^{-\lambda k}$, so eventually none of them is a λ-MAP.

(iv), posterior. For every L, using (ii) for the long programs and the sampler's term for the normaliser,

$$\Pi_\lambda(\text{not equivalent} \mid D_n) \;\leq\; 2^{\lambda k} \sum_{\ell(\pi)\leq L,\ \text{not equivalent}} 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) \;+\; 2^{\lambda k+\delta^*-(\lambda-1)L}.$$

The first term is a finite sum of terms that tend to 0. Let $n \to \infty$, then $L \to \infty$. □

Computation. In an explicit Kraft-valid product class (Kraft sum 0.909) of samplers, fill-ins and partial memorisers, an independent computation (code in the accompanying repository, seed 777) ran 30 runs at 12 sample sizes and 4 values of λ and found no violation of (i)–(iii) or of the upper end of Theorem 6.4(iii) in 1440 MAP checks.

E.6 Proof of the shared-randomness theorem

We prove Theorem 6.8. (i) By (Dom), $-\log U(o) \le k + \kappa_0 - \log P_{\pi_\nu}(o)$, so $J_\lambda(\pi_U; o) = \lambda\ell_U - \log U(o) \le \lambda\ell_U + \kappa_0 + k - \log P_{\pi_\nu}(o)$, which is below $\lambda k - \log P_{\pi_\nu}(o) = J_\lambda(\pi_\nu; o)$ exactly when $(\lambda-1)k > \lambda\ell_U + \kappa_0$; the same condition says $\ell^*_\lambda < k$, and every λ-MAP has length at most ℓ^*_λ by Theorem 3.2(a). (ii) $\sum_{\pi\in A} 2^{-\lambda\ell(\pi)} P_\pi(o) \le 2^{-(\lambda-1)\min_A \ell} \sum_\pi 2^{-\ell(\pi)} P_\pi(o) = 2^{-(\lambda-1)\min_A \ell}\, \xi_1(o) \le 2^{-(\lambda-1)\min_A \ell + \kappa_1} U(o)$, and the normaliser $\xi_\lambda(o)$ is at least $2^{-\lambda\ell_U} U(o)$. For SI with adaptive or non-computable inputs, Solomonoff's M is a semimeasure on the whole interleaved string and need not dominate laws whose inputs are not modelled; M_H dominates every lsc chronological semimeasure on every input sequence (Hänni, 2026a), so (Dom) holds in $\mathcal{H}_{\mathrm{chr}}$. □

In an explicit shared class that contains its own mixture, an independent computation (code in the accompanying repository, seed 4242) found no violation of (ii) or of Theorem 3.2(a) and (c1) in 300 MAP checks.

F Proofs for the relation to algorithmic statistics

F.1 Proof of the proposition on sets from products

We prove Proposition 7.5. Since Y is finite and ν_π is computable with rational values, the test $\nu_\pi(\operatorname{dom} D \mapsto y') \ge 2^{-\rho}$ is decidable, so A is computable from $(\pi, \operatorname{dom} D, \rho)$. By (F1), (F3) and (F5), $K(A \mid \operatorname{dom} D) \le \ell(\pi) + 2\log(\rho+1) + \kappa$. It contains y_D because $\nu_\pi(D) \ge 2^{-\rho}$. Since $\sum_{y'} \nu_\pi(\operatorname{dom} D \mapsto y') \le 1$, $|A| \le 2^\rho$. Hence

$$K(A \mid \operatorname{dom} D) + \log|A| \le \ell(\pi) + 2\log(\rho+1) + \kappa + \rho \le \ell(\pi) - \log \nu_\pi(D) + 1 + 2\log(\rho+1) + \kappa,$$

and product-b-sufficiency of π gives the claim. The consequence follows by taking π to attain $\mathrm{soph}^{\times}_b(D)$. Against the unconditional set sophistication of the string y_D, describe $\operatorname{dom} D$ first, at cost $K(\operatorname{dom} D)$. □

F.2 Proof of the noisy-parity proposition

We prove Proposition 7.6, in the notation of Section 7.3.

(i), injectivity and size. If $f_{a'} \oplus \mathbf{1}_{E'} = f_{a''} \oplus \mathbf{1}_{E''}$ with $a' \ne a''$, then the nonzero parity $f_{a'} \oplus f_{a''}$, which has $N/2$ ones, equals $\mathbf{1}_{E'} \oplus \mathbf{1}_{E''}$, which has at most $2pN < N/2$ ones. So $a' = a''$ and then $E' = E''$. Hence $|A_0| = 2^d \binom{N}{pN} = 2^{d+L_E}$.

(i), complexity of A_0 and of z. A_0 is computable from d and r, so $K(A_0) \le 2\log d + 2\log r + \kappa$ by (F5). The table z determines a as the unique parity within distance $pN < N/4$ of z, and then $E = \{x : z(x) \ne f_a(x)\}$; d is determined by $|z| = N$. So $K(z) \ge K(a, E) - \kappa$, and by symmetry of information (F4), $K(a, E) = K(a) + K(E \mid a, K(a)) + O(1) \ge d + L_E - \delta_E - \kappa$. Therefore $K(A_0) + \log|A_0| \le K(z) + \delta_E + 2\log d + 2\log r + \kappa$.

(i), flatness. For an integer $0 \le u \le d$ let A_u be the subset of A_0 whose parities a' agree with a on the first $d-u$ coordinates. Then $\log|A_u| = u + L_E$ and $K(A_u) \le d - u + 2\log d + 2\log(u+1) + 2\log r + \kappa$, since A_u is computable from d, r, u and those $d-u$ bits. So every A_u has two-part length $d + L_E + O(\log d)$, within $\delta_E + O(\log d)$ of the sufficiency line. With the sufficiency line itself as the lower bound, the structure function h_z lies within $\delta_E + O(\log d)$ of the sufficiency line for every α from $O(\log d)$ up to $d + O(\log d)$.

(i), λ-optimal set models. By Proposition 7.2 with $b = \delta_E + 2\log d + 2\log r + \kappa$, $K(A_\lambda) \le 2\log d + 2\log r + \kappa + (b+\kappa)/(\lambda-1)$. By (F3), $K(a) - K(a \mid A_\lambda) \le K(A_\lambda) + \kappa$, which is the claim.

(ii), upper bound. Since $|E| = pN$ exactly, $-\log \nu_{\pi_g}(D) = NH(p)$, and $NH(p) \le L_E + \log(N+1)$ by the standard bounds on binomial coefficients (Cover and Thomas, 2006). Also $K(y_D \mid \operatorname{dom} D) \ge K(z \mid d) - \kappa \ge K(z) - 2\log d - \kappa \ge d + L_E - \delta_E - 2\log d - \kappa$ by (i). So

$$\ell(\pi_g) - \log \nu_{\pi_g}(D) \le d + \kappa_g + L_E + \log(N+1) \le K(y_D \mid \operatorname{dom} D) + \delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa,$$

and π_g is product-b-sufficient for every b in the range. Hence $\operatorname{soph}_b^{\times}(D) \le \ell(\pi_g) \le d + \kappa_g$.

(ii), lower bound. The memoriser gives $K(z \mid d) \le d + 2\log N + L_E + \kappa \le d + NH(p) + 2\log N + \kappa$: describe a, then $|E|$, then the index of E among the sets of that size. Let π be product-b-sufficient. If $\operatorname{lev}(\pi) \ge 2$, then $R(\pi) \ge G_1$ by Lemma 4.4 (with $B = \infty$), so

$$\ell(\pi) + NR(\pi) \ge NG_1 > d + NH(p) + 2\log N + \kappa + b \ge K(y_D \mid \operatorname{dom} D) + b,$$

by the upper end of the range of b, which contradicts sufficiency. So $\operatorname{lev}(\pi) = 1$, and (D1) of Lemma 4.6 with $s(1) = 4$ gives $\ell(\pi) \ge d - 4 - \Lambda_1$.

(ii), the kink. If $\ell(\pi) < d - 4 - \Lambda_1$, then $\operatorname{lev}(\pi) \ne 1$ by (D1), so $R(\pi) \ge G_1$ and $-\log \nu_\pi(D) = NR(\pi) \ge NG_1$.

(iii) follows from (i) and (ii), and from (i) and (iv) for every $p < 1/4$.

(iv). Let $\ell(\pi) < d - s(t) - \Lambda_1$. If $\operatorname{lev}(\pi) = t' \le t$, then (D1) and the monotonicity of s give $\ell(\pi) \ge d - s(t') - \Lambda_1 \ge d - s(t) - \Lambda_1$, a contradiction; so $\operatorname{lev}(\pi) > t$ and $u(\pi) \le \frac{1}{2} + 2^{-t-1}$. Lemma C.1 with $m = d$, $B = \infty$ and the information budget $I(\pi) \le \ell(\pi) + \kappa < d + \kappa$ applies with a deviation term at most γ_d, because the deviation term is increasing in the budget. It gives

$$\mathbb{E}_x \nu_\pi(y^*(x) \mid x) \le p + (1-2p)\Big(\frac{1}{2} + 2^{-t-1}\Big) + p\big(2\gamma_d + 2^{1-g}\big) = \frac{1}{2}\Big(1 + (1-2p)2^{-t} + 2p\big(2\gamma_d + 2^{1-g}\big)\Big),$$

and Jensen's inequality gives $R(\pi) \ge F_t$. For the sophistication: a product-b-sufficient π has

$$\ell(\pi) + NR(\pi) \le K(y_D \mid \operatorname{dom} D) + b \le K(z \mid d) + \kappa + b \le \ell(\pi_b) + 2\kappa + b \le d + \lceil L_E \rceil + \kappa_b + 2\kappa + b,$$

so $NR(\pi) < NF_t$ in the stated range of b, and the kink forces $\ell(\pi) \ge d - s(t) - \Lambda_1$. The upper bound and its range of b are those of (ii), whose proof uses only minimum-distance decoding, valid for $p < 1/4$, and the good model, whose description of p costs $K(p)$. For the limit: $j_d = d + \delta_E + O(\log d) + O(g)$ while $pN = p2^d$, so $\gamma_d \to 0$ when $\delta_E = O(d)$; taking $g \to \infty$ slowly, $F_t \to 1 - \log(1 + (1-2p)2^{-t})$, which tends to $1 > H(p)$ as $t \to \infty$, so some fixed t_p has $F_{t_p} > H(p)$ for all large d. □

F.3 Numbers for the noisy parity at every flip rate below 1/4

Table 3 evaluates Proposition 7.6(iv) for four dyadic flip rates of small complexity ($K(p) \le 2\log(h+1) + 2\log(e+1) + \kappa$ for $p = h/2^e$), with $\delta_E = 0$ and the best (t, g); the best level is $t = 2$ throughout, so $s(t) = 7.2$. A direct computation (code in the accompanying repository, seed 7) produced the values. The rows with $\kappa = 0$ are illustrations, not results about a particular reference machine; the rows with $\kappa = 30$ show how the bounds degrade. At $d \ge 30$, $F_2 - H(p)$ lies between 0.016 (for $p = 15/64$) and 0.064 (for $p = 1/8$). For comparison, the floor of Lemma 4.4 at large level, $1 - \log(1+2p)$, exceeds $H(p)$ only for $p < 0.1525$, and with it no admissible t exists for $p \ge 3/16$ at these sizes.

At $d = 20$ the lower bound is negative, so the interval says nothing there; the $O(\log d)$ terms dominate. Upper ends of the b-ranges are rounded down, lower ends to the nearest integer.

A second, smaller computation (code in the accompanying repository, seed 2) illustrates (i)–(ii) at $d = 10$, $|E| = 16$ ($p = 1/64$): product list mixtures that leave u bits of a unknown have

Table 3: Bounds on product sophistication for the full-table noisy parity (Proposition 7.6(iv)), $\delta_E = 0$. The interval for $\mathrm{soph}_b^{\times}(D)$ holds for every b in the stated range.

κ	d	lower bound	upper bound $d + \cdots$ for $p =$				range of b
			1/8	3/16	7/32	15/64	($p = 1/8$; $p = 15/64$)
0	20	$d - 24.5$	14.6	17.3	19.8	22.3	$[43, 5.8 \cdot 10^4)$; $[51, 2.8 \cdot 10^3)$
0	30	$d - 26.8$	15.8	18.5	21.0	23.4	$[56, 6.8 \cdot 10^7)$; $[63, 1.6 \cdot 10^7)$
0	60	$d - 30.8$	17.8	20.5	23.0	25.4	$[90, 7.3 \cdot 10^{16})$; $[97, 1.8 \cdot 10^{16})$
0	100	$d - 33.7$	19.3	21.9	24.5	26.9	$[133, 8.1 \cdot 10^{28})$; $[140, 2.0 \cdot 10^{28})$
30	20	$d - 54.5$	44.6	47.3	49.8	52.3	$[103, 5.6 \cdot 10^4)$; $[111, 8.1 \cdot 10^3)$
30	30	$d - 56.8$	45.8	48.5	51.0	53.4	$[116, 6.8 \cdot 10^7)$; $[123, 1.6 \cdot 10^7)$
30	60	$d - 60.8$	47.8	50.5	53.0	55.4	$[150, 7.3 \cdot 10^{16})$; $[157, 1.8 \cdot 10^{16})$
30	100	$d - 63.7$	49.3	51.9	54.5	56.9	$[193, 8.1 \cdot 10^{28})$; $[200, 2.0 \cdot 10^{28})$

risk rising from $H(p) = 0.116$ at $u = 0$ to about 1 by $u = 7$, while the per-sample code lengths of the corresponding shared mixtures stay between 0.116 and 0.126, and the two-part lengths of the set models A_u are constant (125.6 bits). These totals are the one-line computation of the flatness step, not an independent check.

G Proofs for the computable versions, and the earlier formalisation

Proof of Proposition 8.1. By inspection. Lemma 4.3 uses only that $u(\pi; a')$ is lower semicomputable uniformly in (π, a', d), which holds in every class with uniformly lower-semicomputable likelihoods; the constant of the enumeration depends on the class. Lemma 4.5 uses only that q_z is lower semicomputable uniformly in (π, a, m, B). Lemma 4.4 uses nothing about the class. Lemma 4.6 uses (F1) for the class, $K(\pi) \le \ell(\pi) + \kappa'$. The proof of Theorem 4.7 uses only these lemmas and the reference programs, through J_g and J_b. Theorem 6.6 uses Ville's inequality for the class's mixture $\xi_{\mathcal{C}}$, Kraft's inequality and countability. Proposition 3.8(ii) uses the coding theorem for the semimeasure $\tilde{R}$ built from the class, which holds with a constant that depends on the class. □

Proof of Proposition 8.2. (a) Each term $2^{-\lambda \ell(\pi)} P_\pi(D)$ is lower semicomputable uniformly in (π, D), because λ is computable, and the sum of uniformly lower-semicomputable nonnegative terms is lower semicomputable.

(b) Identify $\mathbb{N}$ with infinitely many inputs. Suppose the raw conditional were computable. Define $f(t)$, for $t \in \mathbb{N}$, as the first $y \in \{0, 1\}$ that a dovetailed search certifies to satisfy $\xi_\lambda(y \mid D_t, t) < 0.6$, where $D_t := \{(s, f(s)) : s < t\}$. The two conditionals sum to at most 1, so one of them is at most $\frac{1}{2}$ and the search halts. Then f is total computable, and by telescoping $\xi_\lambda(D_n) \le \xi_\lambda(\emptyset) \cdot 0.6^n \le 0.6^n$. But the class contains a deterministic program for f of length at most $K(f) + \kappa$, so $\xi_\lambda(D_n) \ge 2^{-\lambda(K(f)+\kappa)}$ for every n, a contradiction for large n. For the normalised conditional, run the same construction with it in place of the raw one; then its cumulative loss on D_n is at least $n \log(1/0.6)$. Each normaliser $\sum_{y'} \xi_\lambda(D_t + (t, y'))$ is at most $\xi_\lambda(D_t)$, so the normalised conditional is at least the raw one, and the normalised cumulative loss is at most the raw one, $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset) \le \lambda(K(f) + \kappa)$. This is again a contradiction for large n.

(c) Lower-semicomputable reals are computable in the limit, and the denominators are positive, because every finite data set has a consistent deterministic program. □

G.1 Correspondence with the earlier formalisation

The earlier report (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation") works with a general imitation channel; its results are named here as they are named there.

earlier result	here	change
per-input versus once-only pricing	Proposition 3.4	exact minimax value kept
the sequence pathology	Theorem 3.2(a)	same threshold; adds (b)–(d)
universal conditional semimeasure	Remark 3.6	–
per-block pricing	Proposition 3.9	countable families with computable weights
kink: list-size and floor lemmas	Lemmas 4.3 and 4.4	semimeasures, via the lower-semicomputable agreement; truncation
kink: error segment	Lemma 4.6(D2)	–
kink: shared randomness	Proposition 7.6(i)	every λ-optimal set model
randomness of E removes the $2p$ slack	Lemma C.1	semimeasures; used for the kink at every $p < 1/4$
identification in the parity example	Corollary 4.10	adds information and error-label bounds; c_{hi} for $t^* < 2$; rigorous $d = 20$ window
empirical selection tracks population selection	Lemma C.2	one-sided, truncated, without replacement
selection traces the lower hull	Lemma 2.8	–
rate threshold, separable classes; rate blindness	Proposition 5.6	–
steepness not identifiable	Proposition 5.8	–
guard erosion	Proposition 5.7	teacher model stated; exact crossing
error envelopes	Proposition 5.3	all hypotheses; cheap errors reproduced
input-law dependence	Proposition 5.9	separable case only

Not reused: the idealised rate threshold for arbitrary hypotheses and the realisability of every frontier shape, because Theorem 4.7 covers one model with one error rule over all hypotheses and Propositions 5.6 and 5.9 cover separable components; the treatment of inference rules, which is not about function induction; and the pathologies of parasitic errors and language relativity, which hold for steeper SFI by the same arguments but add nothing specific to it. The scripts of the earlier report were re-run from scratch copies and reproduce its numbers (Appendix C.8).

H How the results were checked (internal record)

This appendix records how the draft was prepared and checked, in the vocabulary of that process. It is not part of the arXiv version, where the acknowledgements and the code-and-data statement at the end of Section 9 replace it (switch `\sfirecordfalse` *in* `main.tex`*). Paths are relative to* `solomonoff-function-induction/research/`*.*

H.1 Preparation

Every role was played by a separate instance of Claude (Anthropic); no human has checked the mathematics. A research *track* on steeper simplicity started from Hänni's grokking note, the Less-

Wrong thread and the earlier formalisation T5 (Claude (Anthropic), 2026, section "Simplicity and normativity from imitation"), re-derived every T5 result it uses rather than copying it, and wrote notes with full proofs and seeded check scripts (`tracks/steeper/notes.md`, `checks/c1`–`c5`). Two adversarial *referees*, each a fresh instance that formed its own view before reading earlier reports, checked every proof line by line and wrote independent code (`referee_code/r1`–`r6` on 2026-10-11, `q1`–`q4` on 2026-10-12). After each report the track revised its notes (`notes-final-run1.md`, then `notes-final.md`), answered every item in a table, added checks `c6` and `c7`, re-ran all scripts, and closed with a verification log. A planning editor then wrote the claims ledger (`paper-steeper/CLAIMS.md`), the notation shared with the companion paper and the outline; two section writers wrote Sections 2 to 8 and their appendices from `notes-final.md` and the ledger only; a framing editor wrote the abstract, the introduction, the discussion and this appendix, and ran the build. The results of the companion paper used here were established in the same way by four other tracks (foundations, history, examples, context) and are cited by name.

H.2 Referee rounds

round	findings and outcome
first (2026-10-11), read `notes.md`	No fatal errors. Three major issues: numbers that hold only with the machine constant set to 0 were called proved; the absorption objection was not answered and the cost of private randomness was understated; the novelty of the product notions was not compared with restricted model families. Twenty minor issues (one proof gap in the hashing lemma, two script bugs that changed no verdict) and six missed questions. All accepted in `notes-final-run1.md`.
second (2026-10-12), read `notes.md`	No fatal errors; every theorem checked is true as stated up to the machine constant. Four major issues: missing prior work on λ-weighted two-part MDL in classification; the absorption objection and its cost (repeated); the dependence of the trick on an incompressible error rule; the status of numbers with machine constant 0 (repeated). Nineteen minor issues and eight questions. All accepted in `notes-final.md`; one factual point (a growth rate of λ attributed to Zhu and Srebro) could not be confirmed and is not used.

H.3 Check scripts and reproduction

All scripts are in `tracks/steeper/checks/` (the track's) and `tracks/steeper/referee_code/` (the referees'), deterministic or seeded, with saved outputs (`.out`); environment Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1. The track's scripts `c1`–`c6` were re-run in the second revision and reproduce their saved outputs byte for byte.

script	seed	what it checks	used in
`c1_hanni_numbers`	–	both versions of Hänni's calculation; three-hypothesis windows; $\lambda = 1$ switch; L_S	Rem. 4.1
`c2_pricing_kink`	2	per-input pricing, minimax value 2.5431; list size and risk floor on 400 semimeasures; kink instance	Prop. 3.4 and Lems. 4.3 and 4.4
`c3_junta_window`	3	windows, identification, memoriser regime, rate blindness, finite samples; reduction to two endpoints (2830 instances)	Thm. 4.7, Tab. 2, and Cors. 4.10 and 4.11
`c4_model_vs_randomness`	4	Ville (4000 runs); hashing over GF(16); noisy parity, product versus shared MAPs	Lems. 6.2 and E.1 and Ex. 6.9
`c5_limits`	5	separable thresholds (8000 tests), hull; guard erosion; proper scores; block telescoping; power-posterior exponent	§ 5, Props. 3.9 and 4.13
`c6_revision`	6	$d_0(\kappa)$; identification with δ_S; Version 1; price inequality (24,732 cases); hashing with the $1x$ embedding; hull (iii) on 401 sets; guard crossing	Lems. 2.8 and 5.5, Ex. 4.12, and Prop. 3.8
`c7_run2`	7	stream-prefix lemma (600 samplers, 1646 cases); envelopes; sharper floor and kink for $p < 1/4$; memoriser band; guard model; ϵ'; schedules; full table with δ_E	Lems. 6.3 and C.1 and Props. 5.1, 5.3 and 7.6
`t5_rerun/`	T5's	T5's c1–c3 from scratch copies	Rem. 8.7
`r1`–`r6`	various	first referee: constants of Thm. 4.7, minimax, lemmas on adversarial hypotheses, finite classes, hashing, small items	as cited in the appendices
`q1`–`q4`	various	second referee: constants by bisection, lemmas, split in a Kraft class (seed 777), absorption in a shared class (seed 4242), memoriser band	as cited in the appendices

To reproduce, from `tracks/steeper/checks/`: `python3 c1_hanni_numbers.py > /tmp/c1.out && diff /tmp/c1.out c1_hanni_numbers.out`, and likewise for `c2`–`c7`; the referee scripts `q1`–`q4` write their `.out` files themselves, and `r1`–`r6` print to standard output. The paper is built from `paper-steeper/` with `./build.sh`.

H.4 Corrections

The table lists every correction recorded in the steeper track's verification log and response tables, and the corrections made while writing the paper. Hänni's own claims, as he stated them, are confirmed or qualified (Table 1); the one refuted statement about his example is an extension of it, which he did not make (row 1). His approximation "$\approx p \log_2(1/p) \approx 1/100$" is 0.0098 against the exact $H(2^{-10}) = 0.0112$, which changes nothing.

	earlier statement	corrected statement	where
Refuted statements, kept in the paper as refuted			
1	Hänni's example extends to every error rule with his density and about 10^4 bits	false: a cheap subcube plus random points ($128 + 896$, about 10,400 bits) leaves no near-optimal hypothesis that fits no better than the good model for $c < 1.64 \cdot 10^{-5}$ ($\kappa = 0$)	Prop. 5.3
2	hull lemma (iv): every boundary point solving the constrained problem is a penalised minimiser	false on the horizontal ray: $\{(0,1),(10,0),(20,0)\}$	Lem. 2.8

	earlier statement	corrected statement	where
3	λ slightly above 1 selects an approximately minimal sufficient statistic	does not follow; the two bounds trade off	Prop. 7.2
Corrected in the track's two revisions			
4	verdicts with machine constant 0 called proved	labelled illustrations; robust form with $d_0(\kappa)$ linear in κ	Lem. 5.5
5	private randomness loses nothing	price $K(\operatorname{dom} D) + O(1)$ at $\lambda = 1$, factor $\min(\lambda, 2)$ on shared coins	Rem. 3.7 and Prop. 3.8
6	product notions new	compared with restricted model families	Rem. 7.7
7	"fixed λ fails, λ must grow" as a finding	credited to prior work on λ-weighted MDL	Rems. 5.2 and 8.5
8	numbers silently assume $\delta_S = 0$	every number states δ_S; envelope theorem added	Tab. 2 and Prop. 5.3
9	list decoding via a correlation not computable for semimeasures	via the lower-semicomputable agreement u	Lem. 4.3
10	c_{hi} formula for $t^* < 2$	corrected; closed form equals bisection to 10^{-15}	Thm. 4.7
11	shared-class posterior bound summed over a non-c.e. set	bound for any set of programs via ξ_1	Thm. 6.8
12	hashing lemma counted functions (seed collisions)	one program per seed, class closed under prefix-free simulation	Lem. E.1
13	coding lemma normaliser misprinted (Q not a semimeasure)	per-factor normalisation	Lem. 4.5
14	power-posterior slack used the minimiser's length budget	budget $\hat{L}'$ for near-minimisers	Prop. 4.13
15	guard erosion "although the teacher never erred"	the teacher never applies σ outside G; teacher model stated	Prop. 5.7
16	$\lambda = 1$ split only for dyadic samplers	every computable sampler, via stored stream prefixes	Lem. 6.3 and Thm. 6.4
17	product sophistication $d \pm O(\log n)$	$d \pm O(\log d)$	Prop. 7.6
18	schedule result for fixed λ only	every schedule	Prop. 5.1
19	script bugs: inexact binomials in `c3`; exponent -776 in `c5`(e)	exact integers (no verdict changed); -758	Prop. 4.13
20	T5: heuristic $d = 20$ window end ≈ 0.049; sharper floor for computable hypotheses	rigorous $c_{\mathrm{hi}} = 0.01139$ ($\kappa = 0$); floor for semimeasures	Rem. 8.7
Corrected while writing the paper (`notes-final.md` as source)			
21	$\delta_S \geq -\kappa$	$\delta_S \geq -2\log m - 2\log s_0 - \kappa$ (unused downstream)	Def. 4.2
22	full-table conclusion for every $\lambda \in (1, Nc_{\mathrm{hi}}]$	every $\lambda > 1$ with $J_g < \Phi_2$, including $(1, Nc_{\mathrm{hi}})$; the endpoint can tie	Cor. 4.10
23	memoriser length $d + L_S + O(\log dm)$	needs s_0 given through r	App. D.1
24	bounds rounded in the unsafe direction (0.65, 0.55, 0.34, $7.0 \cdot 10^{-5}$, $1.9 \cdot 10^{-4}$, $4.06 \cdot 10^{-3}$, 0.0114)	script values 0.649, 0.545, 0.338, $6.981 \cdot 10^{-5}$, $1.876 \cdot 10^{-4}$, $4.055 \cdot 10^{-3}$, 0.01139, or safe rounding	§§ 4 and 5
25	Version 1: only $u > \frac{3}{4}$	$u \geq 0.800$ at $c = 1.5 \cdot 10^{-3}$	App. C.6
26	kink for every $p < 1/4$ with $\delta_E = O(\log n)$	$\delta_E = O(d)$ (renaming $N = 2^d$)	Prop. 7.6
27	identification martingale undefined once a squared Hellinger term is 1	set to 0 from then on; still a supermartingale	App. E.5
28	stream-prefix consistency argued for one ordering of depths	symmetric argument	App. E.2
29	absorption (c3): "at most $L_\pi(\infty)$ times"	fewer than $L_\pi(\infty) + 1$ times	App. B.2
30	(Dom) for $\mathcal{H}_{\mathrm{chr}}$ with $\xi_1 = U$	$\xi_1 \leq U$; the empty history handled separately	App. B.1

	earlier statement	corrected statement	where
31	noisy-parity example as an instance of the shared-randomness theorem	an illustration (ℓ_U assigned; (Dom) fails)	Ex. 6.9

H.5 Status of the claims

- *Proved:* every theorem, proposition, lemma and corollary, with a full proof in the main text or the appendices; none is a proof sketch. Results of the companion paper are cited by name.
- *Computed:* every numerical statement, from the scripts above. Numbers with machine constant $\kappa = 0$ are illustrations, not results about a reference machine; rows with $\kappa = 30$ show how the bounds degrade. Example 6.9 is an illustration only.
- *Known:* the standard facts (F1)–(F5), Sion's minimax theorem, Hoeffding's bound without replacement, Ville's inequality, the martingale behind Kakutani's dichotomy, Solomonoff's bound, Parseval list decoding, the notions of algorithmic statistics, proper scoring rules, and the literature of Section 8.
- *Conjectures:* none. Open problems are collected in Section 9.1.
- *Refuted:* rows 1–3 of the table above.

H.6 References not checked against the source

arxiv.org, lesswrong.com and several publisher hosts did not resolve; literature was checked by web-search snippets.

- The LessWrong thread (Hänni and Wyeth, undated): page not reachable; text as pasted by Hänni; undated.
- Content checked against abstracts or snippets only: Grünwald and Langford (2007); Zhu and Srebro (2025); Zhu et al. (2026) (no growth rate of λ is taken from them), Vereshchagin and Vitányi (2010), Milovanov (2016), the set–distribution equivalence of Vereshchagin and Shen (2017), Wenzel et al. (2020), Poland and Hutter (2005).
- A complexity weight above one in Barron and Cover (1991) is known only through a secondary source; the text says "reported to appear".
- Bibliographic data verified, content not checked: Bissiri et al. (2016); Bhattacharya et al. (2019); Grünwald (2012); Grünwald and van Ommen (2017); Zhang (2006); Gács et al. (2001); Vereshchagin and Vitányi (2004); Koppel (1987); Koppel and Atlan (1991); Vitányi (2006); Antunes and Fortnow (2009); the SafeBayes reading of Remark 8.6 is ours.
- Details not confirmed: the pages of Kushilevitz and Mansour (1993) (taken from a sibling project); the theorem number in Hoeffding (1963) and section numbers in Li and Vitányi (2008) (not cited); the exact constant of Solomonoff (1978); the data of Ville (1939) (not re-checked).
- The identification of M_H with Hutter's universal chronological semimeasure is not checked and not used; only the universality of M_H is used.

H.7 Open issues of this draft

- *Length.* Sections 1 to 9 run about 30 pages against a target of 18–22: about 25 for Sections 2 to 8 (budget 18.5), 3 for the introduction with its table (budget 1.5) and 2 for the discussion (budget 1). Appendices A to G run about 22 pages (target about 20). Every body section exceeds its budget; the writers kept every ledger item with its statement. Candidates for the appendices are Example 6.9, Remark 6.10, and Proposition 3.10 and the secondary remarks of Sections 5 and 8.
- *Status vocabulary.* Some `\status` values are outside the fixed vocabulary ("remark", "remark; cites proved results", "computed; illustration", "proved; a natural reading refuted" and compound forms); the conversion pass must map them, for example remarks to no tag and illustrations to computed.
- *Rounding.* The scripts print the upper bounds of Table 2 rounded to nearest, so their last digit is not guaranteed to be on the safe side.
- *Bibliography.* In `bib/core.bib`, which is shared with the companion paper, the entry `hanni2026request` calls Hänni's note a "personal communication to the author" titled "Request for a paper" and points to a file of the preparation process, which fits neither sole authorship nor the formal register. The entry `hanni2026lesswrong` lists the thread's second participant as an author, so every citation of Hänni's own comments prints both names; since only Hänni's comments are cited, the entry could name him alone.
- *Cross-paper numbers.* Results of the companion paper are cited by name; theorem numbers can be added once both papers are final.